True or false?

A temperature that is 7°C at t = 0, 23°C at t = 12 h, and repeats every 24 hours is modeled by $15 - 8\cos(\pi t/12)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q3/5

1601

QUESTION ID

S43-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Minimum 7 at t = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A blood pressure that starts at 100 mm Hg and rises, varies between 80 and 120, and repeats every 0.8 s is modeled by

A. 100 + 20sin(0.8t)

B. 20 + 100sin(2.5πt)

C. 120sin(2.5πt)

D. 100 + 20sin(2.5πt)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q4/5

1602

QUESTION ID
S43-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D.
100 + 20sin(2.5πt)

GRADING NOTE Period 0.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A piston's position starts at 0 cm and increases, varies between −3 and 3 cm, and repeats every 1 s. Write it in the form p = A·sin(Bt) with A > 0 and B > 0. Give B exactly, in terms of π.

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q5/5

1603

QUESTION ID

S43-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3; 2π

GRADING NOTE Form required: both. Amplitude 3, period 1 so B = 2π; sine starts at 0 and increases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

State domains of radical, rational, logarithmic, and polynomial models

MAIN	Algebra II main question	**HSF-IF.B.5**
BACKWARD 1	Domains of linear, quadratic, and exponential models	**HSF-IF.B.5**
BACKWARD 2	Division by zero	**7.NS.A.2.b**
BACKWARD 3	Solve linear inequalities	**HSA-REI.B.3**
FORWARD 1	Domains defined by nonlinear conditions	**HSF-IF.B.5**
FORWARD 2	Domains and ranges of inverse functions	**HSF-BF.B.4.a**

SET 44 • HSF-IF.B.5

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

A car's speed is $s(d) = \sqrt{(22.5d)}$ mph for a skid of d feet. Which gives the domain of the function and a domain that makes sense in context?

A. Function: $d > 22.5$; context: $d > 22.5$, up to the longest skid that can occur

B. Function: $d \geq 0$; context: $d \geq 0$, up to the longest skid that can actually occur

C. Function: $d \geq 0$; context: all real numbers, since any skid length is possible

D. Function: all real numbers; context: $d \geq 0$, up to the longest skid that can occur

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q1/5

1606

QUESTION ID
S44-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Function: $d \geq 0$; context: $d \geq 0$, up to the longest skid that can actually occur

GRADING NOTE $\sqrt{(22.5d)}$ is real only for $d \geq 0$, and real skid lengths are nonnegative and bounded. The others ignore the square root, invent a bound, or ignore the context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The average cost of x shirts is A(x) = (500 + 4x)/x dollars. Which gives the domain of the function and the domain in context?

A. Function: x > 0; context: all real numbers

B. Function: x ≠ 500; context: x ≥ 500

C. Function: all real numbers; context: whole numbers x ≥ 0

D. Function: all real x ≠ 0; context: whole numbers x ≥ 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q2/5

1607

QUESTION ID
S44-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
D. Function: all real x ≠ 0; context: whole numbers x ≥ 1

GRADING NOTE Division by x excludes 0; a number of shirts is a positive whole number. The others include x = 0 or confuse the fixed cost with an exclusion.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q2/5

pH = −log[H⁺], where [H⁺] is the hydrogen ion concentration. What is the domain?

A. [H⁺] > 0

B. [H⁺] ≥ 0

C. All real numbers

D. 0 < [H⁺] < 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q3/5

1608

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. [H⁺] > 0

GRADING NOTE A logarithm needs a positive input, so 0 is excluded. The range of pH values, not the domain, is often near 0 to 14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q3/5

A box made from a 12-in square sheet has volume $V(x) = x(12 - 2x)^2$. Which gives the domain of the polynomial and the domain in context?

A. Polynomial: all real numbers; context: $0 < x < 12$

B. Polynomial: all real numbers; context: $0 < x < 6$

C. Polynomial: $x \geq 0$; context: $x \geq 6$

D. Polynomial: $0 < x < 6$; context: all real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q4/5

1609

QUESTION ID

S44-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Polynomial: all real numbers; context: $0 < x < 6$

GRADING NOTE A polynomial is defined everywhere; a cut x must be positive and less than half of 12. The others swap the domains or use the full side length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q4/5

Which is the domain of $f(x) = \sqrt{(x - 2)/(x - 5)}$?

A. $x > 5$

B. $x \geq 2$

C. $x \geq 2$ and $x \neq 5$

D. All real numbers except 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q5/5

1610

QUESTION ID

S44-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $x \geq 2$ and $x \neq 5$

GRADING NOTE The square root needs $x - 2 \geq 0$, and the denominator needs $x \neq 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • MAIN • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 1

Domains of linear, quadratic, and exponential models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 64t$ models height until the ball lands. Give the domain in context.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q1/5

1612

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$0 \le t \le 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q1/5

C(n) = 2n + 5 is the cost in dollars of n tickets. Which domain makes sense in context?

A. Negative numbers

B. All real numbers

C. All real numbers n ≥ 5

D. Whole numbers n ≥ 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q2/5

1613

QUESTION ID

S44-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Whole numbers n ≥ 0

GRADING NOTE A number of tickets is a whole number and cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q2/5

True or false?

The domain of $f(x) = 2^x$ is $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q3/5

1614

QUESTION ID

S44-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2^x is defined for every real x; $y > 0$ is its range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q3/5

The input n of a model counts people. Which domain fits?

A. Fractions between 0 and 1

B. All real numbers

C. Nonnegative whole numbers

D. Negative numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q4/5

1615

QUESTION ID

S44-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Nonnegative whole numbers

GRADING NOTE A count of people is 0, 1, 2, ...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q4/5

Why might the domain that makes sense for a model be smaller than the domain of its formula?

A. The context restricts the inputs, for example to nonnegative numbers or whole numbers.

B. The formula is always wrong outside small inputs, so only small inputs make sense.

C. The domain of a model must always be all real numbers, as for its formula.

D. Formulas never have restricted domains, so only the context can restrict them.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q5/5

1616

QUESTION ID

S44-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. The context restricts the inputs, for example to nonnegative numbers or whole numbers.

GRADING NOTE For example, $h(t) = -16t^2 + 64t$ is defined for every t, but the ball is in the air only for $0 \le t \le 4$. The others make false general claims: formulas are not wrong outside small inputs, and many formulas do have restricted domains.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 2

Division by zero • Grade 7 • The Number System • 7.NS.A.2.b

ANSWER KEY Reduced copy of the question page, with its answer

For which x is 1/(x − 5) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q1/5

1618

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is (500 + 4x)/x undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q2/5

1619

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q2/5

True or false?

3/0 = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q3/5

1620

QUESTION ID

S44-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(x + 1)/(x^2 - 9)$ is undefined at

A. $x = 0$

B. $x = 9$

C. $x = -1$

D. $x = \pm 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q4/5

1621

QUESTION ID

S44-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x = \pm 3$

GRADING NOTE $x^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the value of 0/5.

0/5 = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q5/5

1622

QUESTION ID

S44-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE 0/5 = 0 because 5 · 0 = 0 (dividing 0 by a nonzero number is defined).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.B.5

BACKWARD 3

Solve linear inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x - 2 \geq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q1/5

1624

QUESTION ID

S44-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $12 - 2x > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q2/5

1625

QUESTION ID

S44-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$x < 6$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

22.5d ≥ 0 means d ≥ 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q3/5

1626

QUESTION ID

S44-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide by a positive number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3 - x \geq 0$ means

A. $x \geq 3$

B. $x \leq 3$

C. $x \geq -3$

D. $x \leq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q4/5

1627

QUESTION ID

S44-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x \leq 3$

GRADING NOTE Add x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 8 > 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q5/5

1628

QUESTION ID

S44-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x > −4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 1

Domains defined by nonlinear conditions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $f(x) = \sqrt{(9 - x^2)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q1/5

1630

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$-3 \leq x \leq 3$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the domain of $g(x) = \log(x^2 - 4)$?

A. $x \geq 2$

B. $-2 < x < 2$

C. $x > 2$ only

D. $x < -2$ or $x > 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q2/5

1631

QUESTION ID

S44-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $x < -2$ or $x > 2$

GRADING NOTE A logarithm needs a positive input: $x^2 - 4 > 0$ when $|x| > 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q2/5

$h(x) = 1/(x^2 - x - 6)$. List the real numbers excluded from the domain.

Excluded values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q3/5

1632

QUESTION ID

S44-F1-Q3

TYPE

Short answer

CORRECT ANSWER

{−2, 3}

GRADING NOTE Form required: both, in any order. $x^2 - x - 6 = (x - 3)(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q3/5

True or false?

The domain of $\sqrt{(x^2)}$ is $x \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q4/5

1633

QUESTION ID
S44-F1-Q4

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x^2 \geq 0$ for every x, so $\sqrt{(x^2)}$ is defined for all real x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $\log(x - 1) + \sqrt{(5 - x)}$ is

A. $1 < x \leq 5$

B. $x \leq 5$

C. $1 \leq x \leq 5$

D. $x > 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q5/5

1634

QUESTION ID

S44-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $1 < x \leq 5$

GRADING NOTE Both conditions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.5 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.B.5

FORWARD 2

Domains and ranges of inverse functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{(x - 2)}$ has domain $x \geq 2$ and range $y \geq 0$. Write $f^{-1}(x)$ and give its domain.

$f^{-1}(x) =$ _____

Domain of f^{-1}: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q1/5

1636

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2; x \geq 0$

GRADING NOTE Form required: both. $y = \sqrt{(x - 2)}$ gives $x = y^2 + 2$; the domain of f^{-1} is the range of f.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2^x$ has range $y > 0$. What is the domain of $f^{-1}(x) = \log_2 x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q2/5

1637

QUESTION ID
S44-F2-Q2

TYPE
Short answer

CORRECT ANSWER
x > 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q2/5

True or false?

The domain of an inverse function is the range of the original function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q3/5

1638

QUESTION ID
S44-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Inputs and outputs swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 1/x + 3$ $(x \neq 0)$ has an inverse with domain

A. $x \neq 0$

B. $x > 3$

C. all real x

D. $x \neq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q4/5

1639

QUESTION ID

S44-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x \neq 3$

GRADING NOTE The range of f excludes 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q4/5

$s(d) = \sqrt{(22.5d)}$ gives a car's speed s from its skid length d. Why does its inverse $d = s^2/22.5$ need the restriction $s \geq 0$?

A. Because $s^2/22.5$ is undefined for $s < 0$, so the formula cannot be used for negative speeds.

B. Because d must be a whole number of feet, and $s \geq 0$ makes $s^2/22.5$ a whole number.

C. The range of s(d) is $s \geq 0$, so $d = s^2/22.5$ undoes s only for $s \geq 0$; without the restriction it is not one-to-one.

D. Because 22.5 is positive, so s must also be positive for the quotient $s^2/22.5$ to make sense.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q5/5

1640

QUESTION ID

S44-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The range of s(d) is $s \geq 0$, so $d = s^2/22.5$ undoes s only for $s \geq 0$; without the restriction it is not one-to-one.

GRADING NOTE The inverse uses the outputs of s as inputs. The others are false ($s^2/22.5$ is defined for all s) or irrelevant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 44 • FORWARD 2 • Q5/5

SET 45 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate and interpret average rates of change of nonlinear models

MAIN	Algebra II main question	**HSF-IF.B.6**
BACKWARD 1	Average rate of change of simpler functions	**HSF-IF.B.6**
BACKWARD 2	Slope between two points	**8.F.B.4**
BACKWARD 3	Evaluate nonlinear functions	**HSF-IF.A.2**
FORWARD 1	Compare average rates on different intervals	**HSF-IF.B.6**
FORWARD 2	Compare rates of change across representations	**HSF-IF.C.9**

SET 45 • HSF-IF.B.6

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

A culture has $N(t) = 200 \cdot 2^t$ bacteria after t hours. Find the average rate of change from t = 1 to t = 3.

Average rate (bacteria per hour): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q1/5

1643

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

600

GRADING NOTE Form required: the value. (1600 − 400)/2 = 600: on average the culture grew by 600 bacteria each hour from hour 1 to hour 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A stone's height is $h(t) = 64 - 16t^2$ ft after t seconds. Find the average rate of change from t = 0.5 to t = 1.5 s.

Average rate (ft per second): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q2/5

1644

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

−32

GRADING NOTE Form required: the value. (28 − 60)/1 = −32: the stone fell an average of 32 feet per second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q2/5

The table shows a car's value. Find the average rate of change from t = 2 to t = 4 years, to the nearest cent.

t (years)	0	1	2	3	4	5
V ($)	20000	17000	14450	12282.50	10440.06	8874.11

Average rate (dollars per year): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q3/5

1645

QUESTION ID

S45-M-Q3

TYPE

Short answer

CORRECT ANSWER

−2004.97

GRADING NOTE Form required: the value. (10440.06 − 14450)/2 = −2004.97: the value dropped about $2,005 per year on average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Speed from skid length is $s(d) = \sqrt{(24d)}$ mph. Find the average rate of change from $d = 24$ to $d = 96$ ft.

Average rate (mph per foot): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q4/5

1646

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Form required: the value. (48 − 24)/72 = 1/3: on average each extra foot of skid adds 1/3 mph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Temperature is T(t) = 15 − 8cos(πt/12) °C after t hours.
Find the average rate of change from t = 0 to t = 6.

Average rate (°C per hour): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q5/5

1647

QUESTION ID
S45-M-Q5

TYPE
Short answer

CORRECT ANSWER
4/3

GRADING NOTE Form required: the value. (15 − 7)/6 = 4/3: on average the temperature rose about 1.33°C per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • MAIN • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 1

Average rate of change of simpler functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$. Find the average rate of change from $x = 1$ to $x = 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q1/5

1649

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A table gives f(2) = 10 and f(5) = 1. Find the average rate of change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q2/5

1650

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q2/5

True or false?

The average rate of change from a to b is (f(b) − f(a))/(b − a).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q3/5

1651

QUESTION ID

S45-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q3/5

The average rate of change of f(x) = 3x + 1 on any interval is

A. it varies

B. 1

C. 3

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q4/5

1652

QUESTION ID

S45-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE Linear.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q4/5

f is in dollars and x is in years. What are the units of the average rate of change of f?

A. Years per dollar

B. Dollars per year

C. Dollar-years

D. Dollars

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q5/5

1653

QUESTION ID

S45-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Dollars per year

GRADING NOTE Change in output units divided by change in input units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 2

Slope between two points • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope between (1, 400) and (3, 1600).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q1/5

1655

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

600

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope between (0.5, 60) and (1.5, 28).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q2/5

1656

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q2/5

True or false?

A negative slope means the output increases as the input increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q3/5

1657

QUESTION ID

S45-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A negative slope means the output decreases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The slope between (24, 24) and (96, 48) is

A. 72/24

B. 1/3

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q4/5

1658

QUESTION ID

S45-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1/3

GRADING NOTE 24/72 = 1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope between (0, 7) and (6, 15).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q5/5

1659

QUESTION ID

S45-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.B.6

BACKWARD 3

Evaluate nonlinear functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find N(3) for $N(t) = 200 \cdot 2^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q1/5

1661

QUESTION ID

S45-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1600

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find h(1.5) for $h(t) = 64 - 16t^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q2/5

1662

QUESTION ID

S45-B3-Q2

TYPE

Short answer

CORRECT ANSWER

28

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{(24 \cdot 96)} = 2304$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q3/5

1663

QUESTION ID

S45-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $24 \cdot 96 = 2304$ and $\sqrt{2304} = 48$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

15 − 8cos(π/2) equals

A. 15

B. 7

C. 23

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q4/5

1664

QUESTION ID

S45-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 15

GRADING NOTE cos(π/2) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find T(0) for T(t) = 15 − 8cos(πt/12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q5/5

1665

QUESTION ID

S45-B3-Q5

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 45 • BACKWARD 3 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 1

Compare average rates on different intervals • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

For $N(t) = 200 \cdot 2^t$ bacteria after t hours, find the average rates of change on [0, 1] and on [3, 4].

Rate on [0, 1] (bacteria per hour): _____

Rate on [3, 4] (bacteria per hour): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q1/5

1667

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

200; 1600

GRADING NOTE Form required: both. (400 − 200)/1 and (3200 − 1600)/1: the growth speeds up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For s(d) = √(24d), find the average rates of change on [0, 24] and on [96, 150].

Rate on [0, 24]: _____

Rate on [96, 150]: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q2/5

1668

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1; 2/9

GRADING NOTE Form required: both (2/9 ≈ 0.22). (24 − 0)/24 and (60 − 48)/54: the rate decreases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q2/5

True or false?

For a function with period 24, the average rate of change over [0, 24] is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q3/5

1669

QUESTION ID

S45-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(24) = f(0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = x^3, the average rate on [−1, 1] is

A. 3

B. 0

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q4/5

1670

QUESTION ID

S45-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE (1 − (−1))/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q4/5

$T(t) = 15 - 8\cos(\pi t/12)$. On which interval of length 6 is the average rate of change $-4/3$?

A. [6, 12]

B. [3, 9]

C. [12, 18]

D. [0, 6]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q5/5

1671

QUESTION ID

S45-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. [12, 18]

GRADING NOTE $T(12) = 23$ and $T(18) = 15$, so $(15 - 23)/6 = -4/3$. [0, 6] and [6, 12] give $+4/3$; [3, 9] gives about 1.89.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.6 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSF-IF.B.6

FORWARD 2

Compare rates of change across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

$f(t) = 200 \cdot 2^t$, and g is given by g(1) = 500 and g(3) = 1500. Find the average rate of change of each on [1, 3].

Rate of f: _____

Rate of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q1/5

1673

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

600; 500

GRADING NOTE Form required: both. f: (1600 − 400)/2; g: (1500 − 500)/2. f has the greater average rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph of a cooling liquid falls from 90°C at t = 0 to 50°C at t = 20 min. Find the average rate of change of the graph and of T(t) = 90 − 1.5t on [0, 20].

Graph (°C per min): _____

Formula (°C per min): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q2/5

1674

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−2; −1.5

GRADING NOTE Form required: both. (50 − 90)/20 and −1.5: the graphed liquid cools faster on average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q2/5

True or false?

A function with a greater average rate on one interval must be greater everywhere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q3/5

1675

QUESTION ID

S45-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rates and values are different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q3/5

A table shows h(0) = 0, h(2) = 40; the formula $k(t) = 5t^2$ gives

A. no rate for k

B. a greater rate for k

C. a greater rate for h

D. equal average rates on [0, 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q4/5

1676

QUESTION ID

S45-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a greater rate for h

GRADING NOTE h: 40/2 = 20; k: (20 − 0)/2 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q4/5

One function is given by a graph and another by a formula. How can their average rates of change on the same interval be compared?

A. Compare the two functions at the right endpoint only, since the larger final value means the larger average rate of change.

B. Read the two endpoint values from the graph, evaluate the formula at the same inputs, and compute both quotients (change in output)/(change in input).

C. Compare the steepest part of the graph with the leading coefficient of the formula, since both measure the rate of change on the interval.

D. Compare the y-intercepts of the graph and the formula, since the larger starting value has the larger average rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q5/5

1677

QUESTION ID

S45-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Read the two endpoint values from the graph, evaluate the formula at the same inputs, and compute both quotients (change in output)/(change in input).

GRADING NOTE An average rate uses both endpoints of the same interval.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root and cube root functions with transformations

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Graph $y = \sqrt{x}$ and $y = \sqrt[3]{x}$ from a table	**HSF-IF.C.7.b**
BACKWARD 2	Translations and reflections of graphs	**HSF-BF.B.3**
BACKWARD 3	Square and cube roots of numbers	**8.EE.A.2**
FORWARD 1	Graph radical functions with horizontal stretches and reflections	**HSF-IF.C.7.b**
FORWARD 2	Graph a radical function and its inverse	**HSF-BF.B.4.a**

SET 46 • HSF-IF.C.7.b

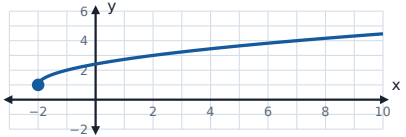
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

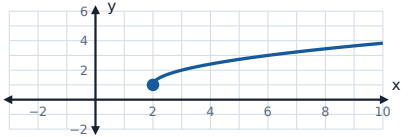
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \sqrt{(x - 2)} + 1$?

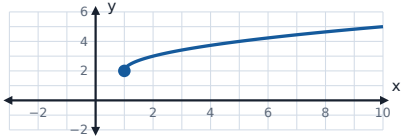
A



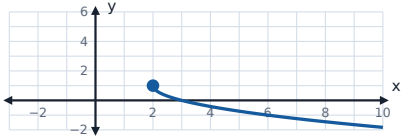
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q1/5

1680

QUESTION ID

S46-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE The graph starts at (2, 1) (right 2 and up 1 from $y = \sqrt{x}$) and rises through (3, 2) and (6, 3); domain $x \geq 2$, range $y \geq 1$. The others show $\sqrt{(x + 2)} + 1$ (shifted the wrong way), $\sqrt{(x - 1)} + 2$ (shifts swapped), and $-\sqrt{(x - 2)} + 1$ (reflected).

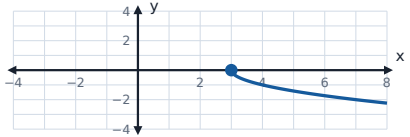
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q1/5

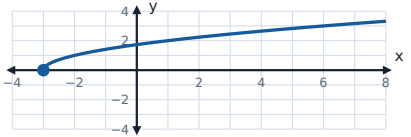
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -\sqrt{(x + 3)}$?

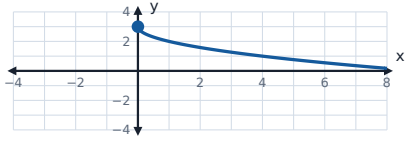
A



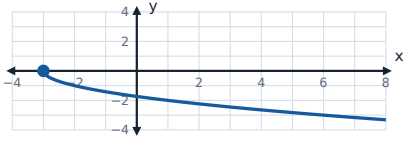
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q2/5

1681

QUESTION ID

S46-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Left 3, then reflected over the x-axis: starts at (−3, 0) and falls through (−2, −1) and (1, −2); domain $x \geq -3$, range $y \leq 0$. The others show $-\sqrt{(x - 3)}$, $\sqrt{(x + 3)}$, and $-\sqrt{x} + 3$.

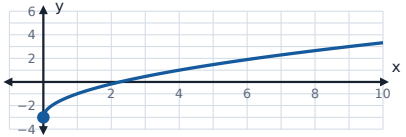
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q2/5

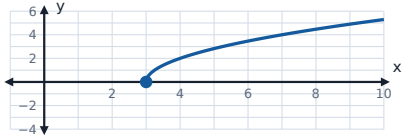
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 2\sqrt{x} - 3$?

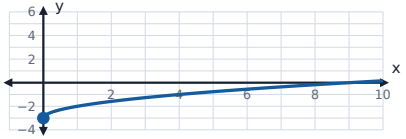
A



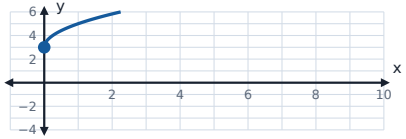
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q3/5

1682

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Vertical stretch by 2, then down 3: starts at (0, −3) and passes (1, −1), (4, 1), (9, 3); domain $x \geq 0$, range $y \geq -3$. The others show $2\sqrt{(x-3)}$, $\sqrt{x} - 3$ (no stretch), and $2\sqrt{x} + 3$.

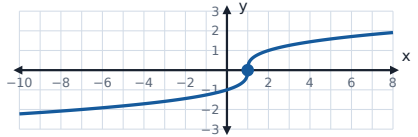
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q3/5

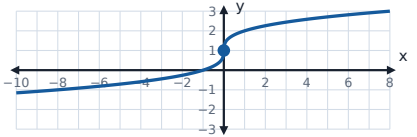
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \sqrt[3]{(x + 1)}$?

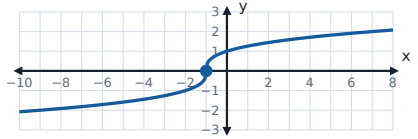
A



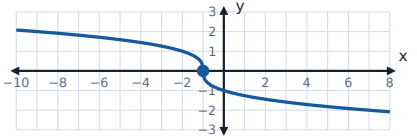
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q4/5

1683

QUESTION ID

S46-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Left 1: increasing, centered at (−1, 0), through (−9, −2), (−2, −1), (0, 1), (7, 2); domain and range all real numbers. The others show $\sqrt[3]{(x - 1)}$, $\sqrt[3]{x} + 1$, and $-\sqrt[3]{(x + 1)}$.

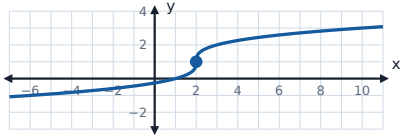
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q4/5

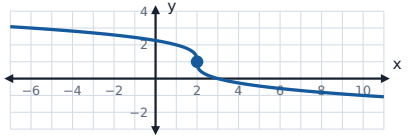
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -\sqrt[3]{(x - 2)} + 1$?

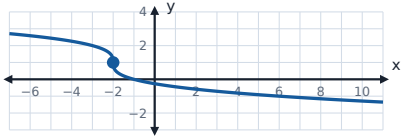
A



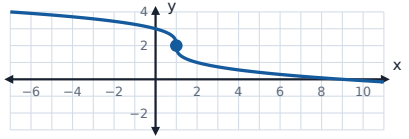
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q5/5

1684

QUESTION ID

S46-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Right 2, reflected, up 1: decreasing, centered at (2, 1), through (1, 2), (3, 0), (10, −1), (−6, 3); domain and range all real numbers. The others show $\sqrt[3]{(x - 2)} + 1$, $-\sqrt[3]{(x + 2)} + 1$, and $-\sqrt[3]{(x - 1)} + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • MAIN • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 1

Graph $y = \sqrt{x}$ and $y = \sqrt[3]{x}$ from a table • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

Find y = √x at x = 0, 1, 4, and 9, in that order.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q1/5

1686

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 1, 2, 3

GRADING NOTE Form required: all four, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $y = \sqrt[3]{x}$ at $x = -8, -1, 0, 1,$ and 8, in that order.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q2/5

1687

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−2, −1, 0, 1, 2

GRADING NOTE Form required: all five, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q2/5

True or false?

The graph of $y = \sqrt{x}$ starts at the origin and decreases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q3/5

1688

QUESTION ID

S46-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{x}$ increases as x increases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $y = \sqrt[3]{x}$ is

A. $x > 0$

B. all real numbers

C. $x \leq 0$

D. $x \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q4/5

1689

QUESTION ID
S46-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. all real numbers

GRADING NOTE Cube roots of negatives exist.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $y = \sqrt{x}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q5/5

1690

QUESTION ID

S46-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$y \geq 0$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 2

Translations and reflections of graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = \sqrt{(x - 2)} + 1$ obtained from the graph of $y = \sqrt{x}$?

A. Shift right 2 and up 1

B. Shift up 2 and right 1

C. Shift left 2 and up 1

D. Shift right 2 and down 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q1/5

1692

QUESTION ID

S46-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Shift right 2 and up 1

GRADING NOTE $x - 2$ shifts right 2; $+ 1$ shifts up 1. The starting point moves from (0, 0) to (2, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How is the graph of $y = -\sqrt{(x + 3)}$ obtained from the graph of $y = \sqrt{x}$?

A. Shift right 3, then reflect over the x-axis

B. Shift down 3, then reflect over the x-axis

C. Shift left 3, then reflect over the y-axis

D. Shift left 3, then reflect over the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q2/5

1693

QUESTION ID

S46-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Shift left 3, then reflect over the x-axis

GRADING NOTE x + 3 shifts left 3; the minus sign outside reflects over the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 2\sqrt{x}$ is a vertical stretch of $y = \sqrt{x}$ by 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q3/5

1694

QUESTION ID

S46-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outputs doubled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q3/5

The point (0, 0) of $y = \sqrt{x}$ moves to which point for $y = \sqrt{(x - 2)} + 1$?

A. (2, 1)

B. (1, 2)

C. (−2, 1)

D. (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q4/5

1695

QUESTION ID

S46-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 1)

GRADING NOTE Right 2, up 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Where does (1, 1) of $y = \sqrt[3]{x}$ go under $y = \sqrt[3]{(x + 1)}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q5/5

1696

QUESTION ID

S46-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSF-IF.C.7.b

BACKWARD 3

Square and cube roots of numbers • Grade 8 • Expressions & Equations • 8.EE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find $\sqrt{9}$ and $\sqrt[3]{(-8)}$.

$\sqrt{9}$ = _____

$\sqrt[3]{(-8)}$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q1/5

1698

QUESTION ID

S46-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3; −2

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $\sqrt[3]{27}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q2/5

1699

QUESTION ID

S46-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

√(−4) is a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q3/5

1700

QUESTION ID

S46-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No real square root of a negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\sqrt[3]{(-1)}$ equals

A. i

B. undefined

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q4/5

1701

QUESTION ID

S46-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −1

GRADING NOTE $(-1)^3 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find √16 − 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q5/5

1702

QUESTION ID

S46-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • HSF-IF.C.7.b

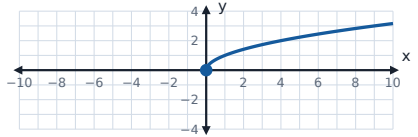
FORWARD 1

Graph radical functions with horizontal stretches and reflections • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

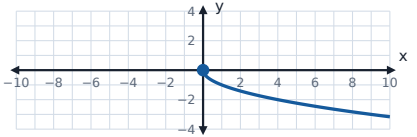
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = √(−x)?

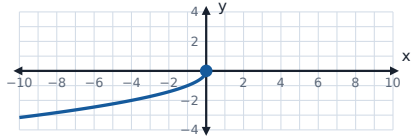
A



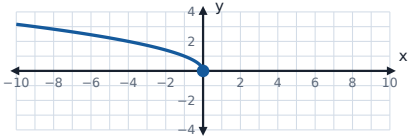
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q1/5

1704

QUESTION ID

S46-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Reflection of y = √x over the y-axis: starts at (0, 0) and rises to the left through (−1, 1), (−4, 2), (−9, 3); domain x ≤ 0. The others show √x, −√x, and −√(−x).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q1/5

For $x \geq 0$, $\sqrt{(4x)} = 2\sqrt{x}$. What does this say about the graphs of $y = \sqrt{(4x)}$ and $y = 2\sqrt{x}$?

A. They are the same graph: a horizontal compression of $y = \sqrt{x}$ by a factor of 1/4 gives the same result as a vertical stretch by 2.

B. They are reflections of each other over the y-axis, because one changes the input and the other the output.

C. They are different graphs that meet only at the origin, because a horizontal change never equals a vertical one.

D. They are different graphs: $y = \sqrt{(4x)}$ is $y = \sqrt{x}$ shifted right 4, and $y = 2\sqrt{x}$ is $y = \sqrt{x}$ stretched up by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q2/5

1705

QUESTION ID

S46-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. They are the same graph: a horizontal compression of $y = \sqrt{x}$ by a factor of 1/4 gives the same result as a vertical stretch by 2.

GRADING NOTE $\sqrt{(4x)} = \sqrt{4} \cdot \sqrt{x} = 2\sqrt{x}$ for $x \geq 0$. The others contradict the identity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = \sqrt[3]{(-x)}$ is the same as $y = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q3/5

1706

QUESTION ID

S46-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt[3]{(-x)} = -\sqrt[3]{x}$; at $x = 8$ they give -2 and 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $y = \sqrt{(3 - x)}$ is

A. all reals

B. $x \geq 3$

C. $x \leq 3$

D. $x \geq -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q4/5

1707

QUESTION ID

S46-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $x \leq 3$

GRADING NOTE $3 - x \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which square root function has a graph that starts at (4, −2) and goes up to the left?

A. $y = \sqrt{(4 - x)} + 2$

B. $y = -\sqrt{(4 - x)} - 2$

C. $y = \sqrt{(x - 4)} - 2$

D. $y = \sqrt{(4 - x)} - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q5/5

1708

QUESTION ID

S46-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $y = \sqrt{(4 - x)} - 2$

GRADING NOTE $\sqrt{(4 - x)}$ is defined for $x \leq 4$ and grows as x decreases; − 2 starts it at height −2. The others go right, go down, or start at (4, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSF-IF.C.7.b

FORWARD 2

Graph a radical function and its inverse • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{(x - 2)} + 1$. Write $f^{-1}(x)$ and give its domain.

$f^{-1}(x) =$ _____

Domain of f^{-1}: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q1/5

1710

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 1)^2 + 2; x \geq 1$

GRADING NOTE Form required: both. $y - 1 = \sqrt{(x - 2)}$, so $x = (y - 1)^2 + 2$; the domain of f^{-1} is the range of f, $y \geq 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $g(x) = \sqrt[3]{(x + 1)}$.

$g^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q2/5

1711

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x^3 - 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q2/5

True or false?

The graph of an inverse is the reflection of the graph over y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q3/5

1712

QUESTION ID

S46-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Inputs and outputs swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (6, 3) on $f(x) = \sqrt{(x - 2)} + 1$ gives which point on f^{-1}?

A. (6, 3)

B. (3, 6)

C. (−6, −3)

D. (3, −6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q4/5

1713

QUESTION ID

S46-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 6)

GRADING NOTE Swap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q4/5

Why does the inverse of $f(x) = \sqrt{(x - 2)} + 1$ need the domain $x \geq 1$?

A. Because $f(x) = \sqrt{(x - 2)} + 1$ is undefined for $x < 1$, so its inverse is undefined there too.

B. Because every inverse of a square root function has domain $x \geq 1$, whatever the shifts are.

C. The range of f is $y \geq 1$, so f^{-1} has domain $x \geq 1$; otherwise $(x - 1)^2 + 2$ is not one-to-one.

D. Because the inverse of f is $\sqrt{(x - 1)} + 2$, and $\sqrt{(x - 1)}$ is undefined for $x < 1$, so the inverse needs $x \geq 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q5/5

1714

QUESTION ID
S46-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
C. The range of f is $y \geq 1$, so f^{-1} has domain $x \geq 1$; otherwise $(x - 1)^2 + 2$ is not one-to-one.

GRADING NOTE Domain of f^{-1} = range of f. The others name an expression not in f^{-1}, misstate the domain of f ($x \geq 2$), or overgeneralize.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 46 • FORWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph piecewise-defined and absolute value functions

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate and graph simple piecewise functions	**HSF-IF.C.7.b**
BACKWARD 2	Absolute value as distance	**6.NS.C.7**
BACKWARD 3	Open and closed endpoints	**6.EE.B.8**
FORWARD 1	Write piecewise functions from graphs and contexts	**HSF-IF.C.7.b**
FORWARD 2	Describe absolute value graphs as transformations, and identify…	**HSF-BF.B.3**

SET 47 • HSF-IF.C.7.b

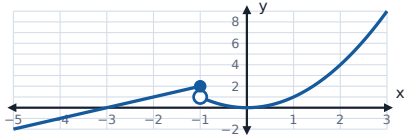
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

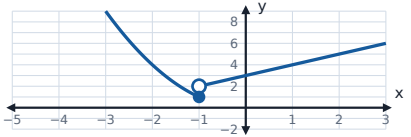
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows f(x) = x + 3 for x < −1, and f(x) = x² for x ≥ −1?

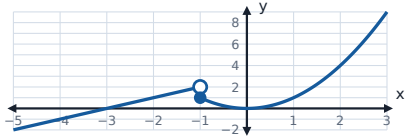
A



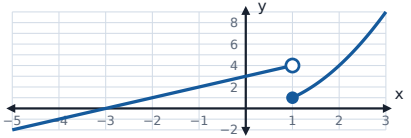
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q1/5

1717

QUESTION ID

S47-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE The line y = x + 3 is used to the left of −1 and ends with an open circle at (−1, 2) (x = −1 is not included); the parabola y = x² starts with a closed dot at (−1, 1). The others swap the circles, swap the pieces, or break at x = 1.

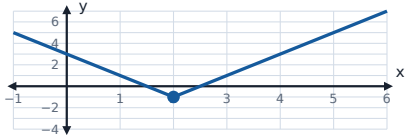
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q1/5

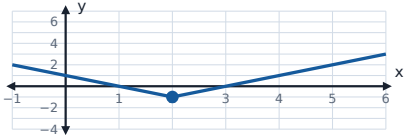
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = |2x - 4| - 1$?

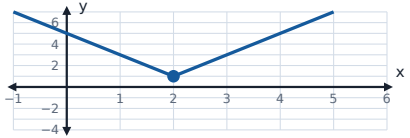
A



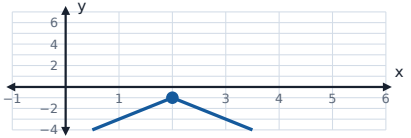
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q2/5

1718

QUESTION ID

S47-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE $|2x - 4| = 2|x - 2|$: vertex $(2, -1)$, slopes ± 2, through $(0, 3)$, $(1, 1)$, $(3, 1)$, $(4, 3)$. The others show $|x - 2| - 1$ (slopes ± 1), $|2x - 4| + 1$, and $-|2x - 4| - 1$.

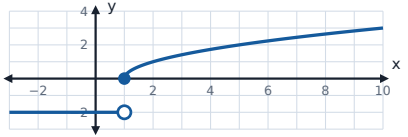
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q2/5

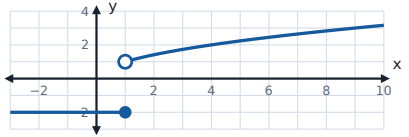
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows f(x) = −2 for x ≤ 1, and f(x) = √(x − 1) for x > 1?

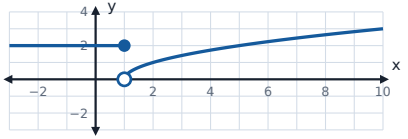
A



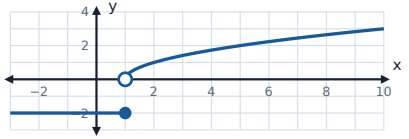
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q3/5

1719

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE The constant piece includes x = 1 (closed dot at (1, −2)); the square-root piece starts at x = 1 but does not include it (open circle at (1, 0)) and passes (2, 1), (5, 2), (10, 3). The others swap the circles, use √x, or use y = 2.

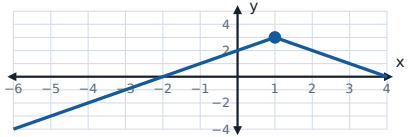
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q3/5

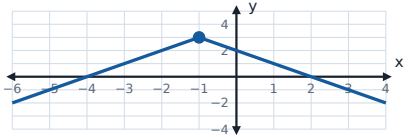
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -|x + 1| + 3$?

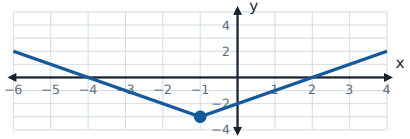
A



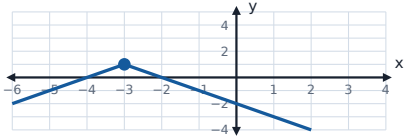
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q4/5

1720

QUESTION ID

S47-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Left 1, reflected, up 3: vertex (−1, 3), opening down, through (−4, 0), (0, 2), (2, 0). The others show $-|x - 1| + 3$, $|x + 1| - 3$, and $-|x + 3| + 1$.

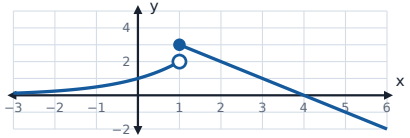
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q4/5

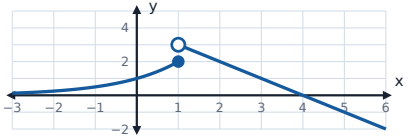
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $f(x) = 2^x$ for $x < 1$, and $f(x) = 4 - x$ for $x \geq 1$?

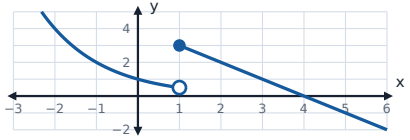
A



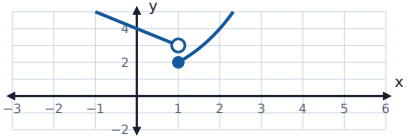
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q5/5

1721

QUESTION ID

S47-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE For $x < 1$ the graph is $y = 2^x$ (through (0, 1), approaching $y = 0$ to the left) with an open circle at (1, 2); for $x \geq 1$ it is $y = 4 - x$ from a closed dot at (1, 3) through (4, 0). The others swap the circles, use $(1/2)^x$, or swap the pieces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 1

Evaluate and graph simple piecewise functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(-3)$ and $f(5)$.

$f(-3) =$ _____

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q1/5

1723

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2; 5

GRADING NOTE Form required: both. $-3 < 0$ uses the first rule; $5 \geq 0$ uses the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2$ for $x < 0$ and $f(x) = x$ for $x \geq 0$. Find $f(0)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q2/5

1724

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q2/5

True or false?

A piecewise function uses different rules on different parts of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q3/5

1725

QUESTION ID

S47-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x + 3$ $(x < -1)$, x^2 $(x \geq -1)$, $f(-1)$ is

A. 3

B. 2

C. -1

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q4/5

1726

QUESTION ID

S47-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1

GRADING NOTE Use the second rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x + 3$ for $x < -1$, and $f(x) = x^2$ for $x \geq -1$. Find $f(-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q5/5

1727

QUESTION ID

S47-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 2

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Find |−5| and |3|.

|−5| = _____

|3| = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q1/5

1729

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5; 3

GRADING NOTE Form required: both. Absolute value is distance from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $|2(1) - 4| - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q2/5

1730

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q2/5

True or false?

|x| can be negative for negative x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q3/5

1731

QUESTION ID

S47-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE |x| is a distance, so it is never negative: |−3| = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q3/5

|x + 1| = 0 when

A. x = 0

B. x = −1

C. x = 1

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q4/5

1732

QUESTION ID

S47-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. x = −1

GRADING NOTE Vertex input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find $-|0 + 1| + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q5/5

1733

QUESTION ID

S47-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.7.b

BACKWARD 3

Open and closed endpoints • Grade 6 • Expressions & Equations • 6.EE.B.8

On a number line, is $x < -1$ drawn with an open or a closed circle at -1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q1/5

1735

QUESTION ID

S47-B3-Q1

TYPE

Short answer

CORRECT ANSWER

Open

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is x ≥ 1 drawn with an open or a closed circle at 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q2/5

1736

QUESTION ID

S47-B3-Q2

TYPE

Short answer

CORRECT ANSWER

Closed

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q2/5

True or false?

An open circle means the endpoint is included.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q3/5

1737

QUESTION ID

S47-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

On a number line, $x \leq 1$ is drawn with

A. an open circle and shading right

B. a closed circle at 1 and shading right

C. an open circle at 1 and shading left

D. a closed circle at 1 and shading left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q4/5

1738

QUESTION ID

S47-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. a closed circle at 1 and shading left

GRADING NOTE Included endpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inequality for "all numbers greater than 1".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q5/5

1739

QUESTION ID

S47-B3-Q5

TYPE

Short answer

CORRECT ANSWER

x > 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 47 • BACKWARD 3 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 1

Write piecewise functions from graphs and contexts • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

A shipping company charges $5 for packages up to 2 lb and $5 plus $1.50 per pound over 2 lb. Which gives the cost C(w) for a package of w pounds?

A. C(w) = 5 for $0 < w \le 2$, and C(w) = 1.5(w − 2) for w > 2

B. C(w) = 5 for $0 < w \le 2$, and C(w) = 5 + 1.5(w − 2) for w > 2

C. C(w) = 5 for $0 < w < 2$, and C(w) = 5 + 1.5w for $w \ge 2$

D. C(w) = 5 + 1.5w for every w > 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q1/5

1741

QUESTION ID
S47-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. C(w) = 5 for $0 < w \le 2$, and C(w) = 5 + 1.5(w − 2) for w > 2

GRADING NOTE "Up to 2 lb" includes 2; only the weight over 2 lb is charged $1.50 per pound, on top of the $5. The others charge every pound, drop the $5, or misplace the endpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q1/5

Which writes |x − 3| as a piecewise function without absolute value bars?

A. 3 − x for x < 3, and x − 3 for x ≥ 3

B. x − 3 for every x

C. 3 − x for x < 0, and x − 3 for x ≥ 0

D. x − 3 for x < 3, and 3 − x for x ≥ 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q2/5

1742

QUESTION ID

S47-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3 − x for x < 3, and x − 3 for x ≥ 3

GRADING NOTE |u| = −u when u < 0. The others swap the pieces, ignore the sign, or split at 0 instead of 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A piecewise function can have a jump and still be a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q3/5

1743

QUESTION ID
S47-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each input still has one output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which piecewise function is continuous at $x = 1$?

A. 2^x for $x < 1$, $4 - x$ for $x \geq 1$

B. x^2 for $x < 1$, $2x - 1$ for $x \geq 1$

C. 2 for $x < 1$, 3 for $x \geq 1$

D. x for $x < 1$, $x + 1$ for $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q4/5

1744

QUESTION ID
S47-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. x^2 for $x < 1$, $2x - 1$ for $x \geq 1$

GRADING NOTE Both pieces give 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find k so that $f(x) = kx$ for $x < 2$ and $f(x) = x^2$ for $x \geq 2$ has no jump at 2.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q5/5

1745

QUESTION ID

S47-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. 2k = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSF-IF.C.7.b

FORWARD 2

Describe absolute value graphs as transformations, and identify symmetry • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Which describes $y = |2x - 4| - 1$ as a transformation of $y = |x|$?

A. $y = 2|x - 4| - 1$: shift right 4, stretch vertically by 2, shift down 1

B. $y = 2|x - 2| - 1$: shift right 2, stretch vertically by 2, shift down 1

C. $y = 2|x + 2| - 1$: shift left 2, stretch vertically by 2, shift down 1

D. $y = |x - 2| - 1$: shift right 2, shift down 1, no vertical stretch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q1/5

1747

QUESTION ID

S47-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 2|x - 2| - 1$: shift right 2, stretch vertically by 2, shift down 1

GRADING NOTE $|2x - 4| = 2|x - 2|$. The others shift by 4, shift the wrong way, or miss the factor 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q1/5

An absolute value function $y = a|x + 1| + 3$ has vertex $(-1, 3)$, opens down, and passes through $(2, 0)$. Find a.

$a =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q2/5

1748

QUESTION ID
S47-F2-Q2

TYPE
Short answer

CORRECT ANSWER
−1

GRADING NOTE Form required: the value. $0 = a \cdot 3 + 3$ gives $a = -1$ (negative, so it opens down).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q2/5

True or false?

y = |x| is an odd function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q3/5

1749

QUESTION ID

S47-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE |−x| = |x|, so it is even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = |x - 2|$ is symmetric about

A. the origin

B. $y = 2$

C. $x = 0$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q4/5

1750

QUESTION ID

S47-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. x = 2

GRADING NOTE Vertex at 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = -|x + 1| + 3$. Find $f(1)$ and $f(-1)$, and decide whether f is even.

$f(1) =$ _____

$f(-1) =$ _____

Is f even? (Yes or No) _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q5/5

1751

QUESTION ID

S47-F2-Q5

TYPE

Short answer

CORRECT ANSWER

1; 3; No

GRADING NOTE Form required: all three. An even function has $f(-x) = f(x)$ for every x; $f(1) \neq f(-1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 47 • FORWARD 2 • Q5/5

SET 85 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph step functions

MAIN	Algebra II main question	**HSF-IF.C.7.b**
BACKWARD 1	Evaluate step functions	**HSF-IF.C.7.b**
BACKWARD 2	Included and excluded endpoints of intervals	**6.EE.B.8**
BACKWARD 3	Integers just below a number	**6.NS.C.7**
FORWARD 1	Write and interpret step-function models	**HSF-IF.C.7.b**
FORWARD 2	Transform step functions	**HSF-BF.B.3**

SET 85 • HSF-IF.C.7.b

MAIN

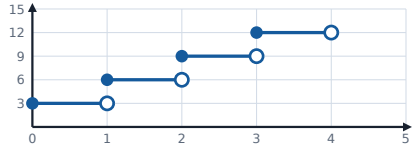
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY

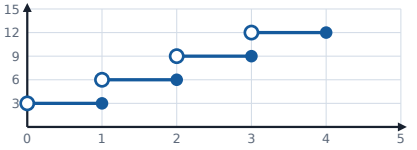
Reduced copy of the question page, with its answer

A parking garage charges $3 for each hour or part of an hour, so C(t) = 3⌈t⌉ dollars for 0 < t ≤ 4 hours, where ⌈t⌉ is the least integer greater than or equal to t. Which graph shows C?

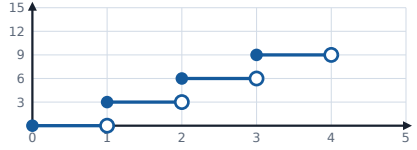
A



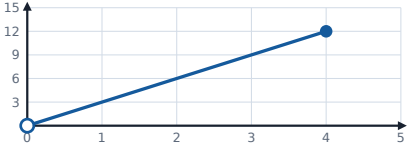
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q1/5

1754

QUESTION ID

S85-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Any part of an hour costs a full $3: 0 < t ≤ 1 costs 3, so the first step is open at t = 0 and closed at t = 1 (exactly 1 hour costs $3, not $6). The others close the wrong ends, use rounding down (3⌊t⌋), or charge for exact fractions of an hour (3t).

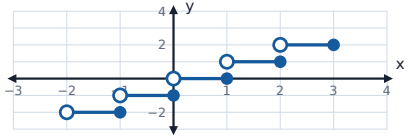
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q1/5

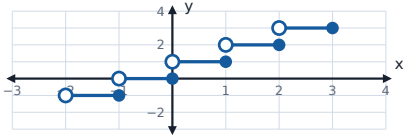
ANSWER KEY Reduced copy of the question page, with its answer

f(x) = ⌊x⌋, the greatest integer less than or equal to x, for −2 ≤ x < 3. Which graph shows f?

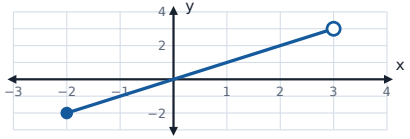
A



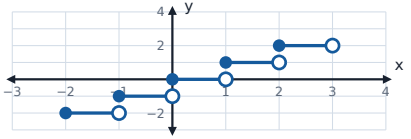
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q2/5

1755

QUESTION ID

S85-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE ⌊x⌋ = k for k ≤ x < k + 1, so each step includes its left end (closed dot) and not its right end (open circle); for example ⌊1⌋ = 1 and ⌊1.9⌋ = 1. The others close the wrong ends, show the least integer ⌈x⌉, or show y = x.

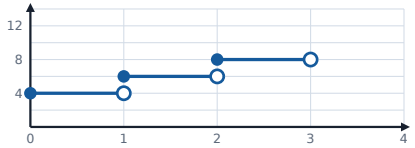
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q2/5

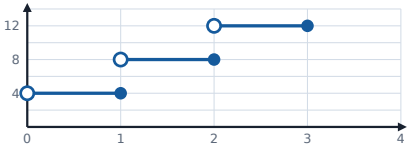
ANSWER KEY Reduced copy of the question page, with its answer

A courier charges $4 for a package up to 1 lb, $6 for more than 1 lb up to 2 lb, and $8 for more than 2 lb up to 3 lb. Which graph shows the cost for weights $0 < w \leq 3$?

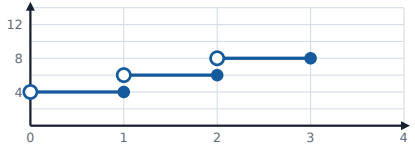
A



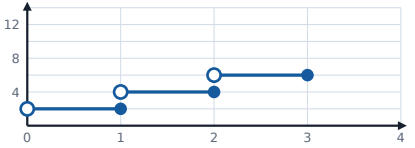
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q3/5

1756

QUESTION ID

S85-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE "Up to 1 lb" includes 1 lb, so the $4 step is closed at w = 1 and the $6 step is open there; the steps are 4, 6, 8. The others close the wrong ends, add $4 per pound, or use $2 per pound.

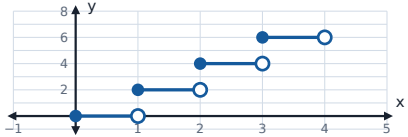
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q3/5

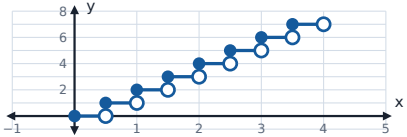
ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 2\lfloor x \rfloor$ for $0 \le x < 4$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x. Which graph shows g?

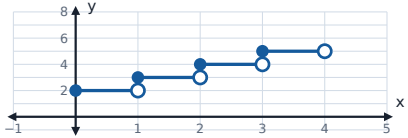
A



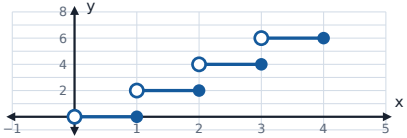
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q4/5

1757

QUESTION ID

S85-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Multiplying the output by 2 makes each step twice as high (0, 2, 4, 6) and keeps the width 1 and the closed left ends. The others show $\lfloor 2x \rfloor$ (half-width steps), $\lfloor x \rfloor + 2$, and the wrong ends closed.

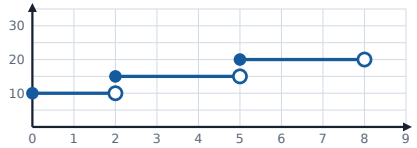
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q4/5

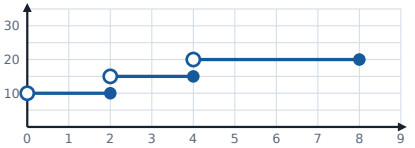
ANSWER KEY Reduced copy of the question page, with its answer

A phone plan costs $10 for up to 2 GB of data, $15 for more than 2 GB up to 5 GB, and $20 for more than 5 GB up to 8 GB. Which graph shows the monthly cost for data use $0 < d \leq 8$?

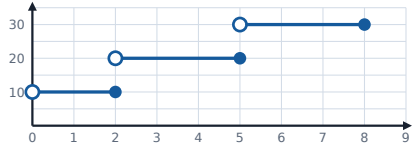
A



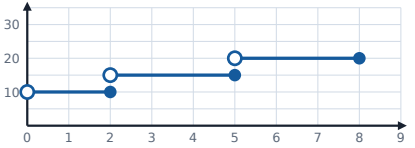
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q5/5

1758

QUESTION ID
S85-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. Graph D

GRADING NOTE "Up to 2 GB" includes 2, so the $10 step is closed at d = 2; the steps change at 2 and 5 GB and cost 10, 15, 20. The others close the wrong ends, break at 4 GB, or use the wrong prices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • MAIN • Q5/5

SET 85 • HSF-IF.C.7.b

BACKWARD 1

Evaluate step functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

$\lfloor x \rfloor$ is the greatest integer less than or equal to x. Find $\lfloor 2.7 \rfloor$ and $\lfloor -1.3 \rfloor$.

$\lfloor 2.7 \rfloor =$ _____

$\lfloor -1.3 \rfloor =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q1/5

1760

QUESTION ID

S85-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2; −2

GRADING NOTE Form required: both. −2 is the greatest integer that is at most −1.3 (not −1, which is greater than −1.3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A parking garage charges $3 for each hour or part of an hour. How many dollars does it charge for 2.5 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q2/5

1761

QUESTION ID

S85-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE 2.5 hours starts 3 hours: 3 × 3 = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q2/5

True or false?

$\lfloor x \rfloor$, the greatest integer less than or equal to x, equals x exactly when x is an integer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q3/5

1762

QUESTION ID

S85-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example $\lfloor 4 \rfloor = 4$ but $\lfloor 4.5 \rfloor = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q3/5

$\lfloor x \rfloor$ is the greatest integer less than or equal to x. What is $\lfloor -0.5 \rfloor$?

A. 1

B. 0

C. -1

D. -0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q4/5

1763

QUESTION ID

S85-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −1

GRADING NOTE $-1 \le -0.5 < 0$. Rounding toward 0 gives 0, which is too large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A parking garage charges $3 for each hour or part of an hour. How many dollars does it charge for exactly 4 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q5/5

1764

QUESTION ID
S85-B1-Q5

TYPE
Short answer

CORRECT ANSWER
12

GRADING NOTE Exactly 4 hours is 4 hours, not part of a 5th: 3 × 4 = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • BACKWARD 1 • Q5/5

SET 85 • HSF-IF.C.7.b

BACKWARD 2

Included and excluded endpoints of intervals • Grade 6 • Expressions & Equations • 6.EE.B.8

ANSWER KEY Reduced copy of the question page, with its answer

Is 2 included in the interval $1 < t \leq 2$? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q1/5

1766

QUESTION ID

S85-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE ≤ includes the endpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is 1 included in the interval $1 < t \leq 2$? Answer Yes or No.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q2/5

1767

QUESTION ID

S85-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE < excludes the endpoint.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q2/5

True or false?

On a graph, a closed dot at the end of a segment means that endpoint is included.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q3/5

1768

QUESTION ID

S85-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE An open circle means excluded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q3/5

Which inequality describes all t greater than 2 and at most 3?

A. $2 < t < 3$

B. $2 \leq t < 3$

C. $2 \leq t \leq 3$

D. $2 < t \leq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q4/5

1769

QUESTION ID

S85-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $2 < t \leq 3$

GRADING NOTE "Greater than" excludes 2; "at most" includes 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A horizontal segment from (1, 6) to (2, 6) has an open circle at (1, 6) and a closed dot at (2, 6). Which inputs does it represent?

A. $1 < x \leq 2$

B. $1 \leq x < 2$

C. $1 < x < 2$

D. $1 \leq x \leq 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q5/5

1770

QUESTION ID

S85-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $1 < x \leq 2$

GRADING NOTE Open circle: excluded; closed dot: included.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 85 • BACKWARD 2 • Q5/5

SET 85 • HSF-IF.C.7.b

BACKWARD 3

Integers just below a number • Grade 6 • The Number System • 6.NS.C.7

What is the greatest integer that is less than or equal to 3.2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q1/5

1772

QUESTION ID

S85-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE $3 \leq 3.2 < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q1

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q1/5

What is the greatest integer that is less than or equal to -3.2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q2/5

1773

QUESTION ID

S85-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−4

GRADING NOTE $-4 \le -3.2 < -3$; -3 is greater than -3.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q2

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q2/5

True or false?

On a number line, −4 is to the left of −3.2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q3/5

1774

QUESTION ID

S85-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −4 < −3.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q3

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which integer is the greatest integer less than or equal to −0.1?

A. −0.1

B. 1

C. −1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q4/5

1775

QUESTION ID

S85-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −1

GRADING NOTE 0 is greater than −0.1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q4

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q4/5

What is the least integer that is greater than or equal to 2.1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q5/5

1776

QUESTION ID

S85-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE $2 < 2.1 \leq 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-B3-Q5

Grade 6 • The Number System • 6.NS.C.7 • Set 85 • BACKWARD 3 • Q5/5

SET 85 • HSF-IF.C.7.b

FORWARD 1

Write and interpret step-function models • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

A parking garage charges $3 for each hour or part of an hour, up to a maximum of $15 per day. [t] is the least integer greater than or equal to t. Which rule gives the cost C(t) for $0 < t \le 24$ hours?

A. C(t) = 3[t] for $0 < t \le 5$, and C(t) = 15 for $5 < t \le 24$

B. C(t) = 3[t] for every t with $0 < t \le 24$

C. C(t) = 3t for $0 < t \le 5$, and C(t) = 15 for $5 < t \le 24$

D. C(t) = 3⌊t⌋ for $0 < t \le 5$, and C(t) = 15 for $5 < t \le 24$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q1/5

1778

QUESTION ID
S85-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. C(t) = 3[t] for $0 < t \le 5$, and C(t) = 15 for $5 < t \le 24$

GRADING NOTE Each started hour costs $3 until the $15 maximum at 5 hours. The others round down, charge for exact time, or ignore the maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function has a jump of 2 at every integer and steps of width 1?

A. $y = \lfloor 2x \rfloor$

B. $y = \lfloor x \rfloor + 2$

C. $y = \lfloor x/2 \rfloor$

D. $y = 2\lfloor x \rfloor$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q2/5

1779

QUESTION ID

S85-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 2\lfloor x \rfloor$

GRADING NOTE $2\lfloor x \rfloor$ rises by 2 at each integer. $\lfloor 2x \rfloor$ jumps by 1 every 0.5; $\lfloor x \rfloor + 2$ jumps by 1; $\lfloor x/2 \rfloor$ has steps of width 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q2/5

True or false?

For $f(x) = \lfloor x \rfloor$, the average rate of change over any interval inside one step, such as $1.2 \le x \le 1.8$, is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q3/5

1780

QUESTION ID

S85-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $f(1.8) - f(1.2) = 1 - 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = ⌊x⌋, how many solutions does f(x) = 3 have?

A. none

B. infinitely many: every x with 3 ≤ x < 4

C. exactly one: x = 3

D. exactly two: x = 3 and x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q4/5

1781

QUESTION ID

S85-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. infinitely many: every x with 3 ≤ x < 4

GRADING NOTE The whole step 3 ≤ x < 4 has output 3; ⌊4⌋ = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Postage is $0.70 for a letter up to 1 oz plus $0.25 for each additional ounce or part of an ounce. What is the postage, in dollars, for a 2.3-oz letter? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q5/5

1782

QUESTION ID

S85-F1-Q5

TYPE

Short answer

CORRECT ANSWER

1.2

GRADING NOTE 2.3 oz is 1.3 oz over the first ounce, which starts 2 additional ounces: 0.70 + 2(0.25) = 1.20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 85 • FORWARD 1 • Q5/5

SET 85 • HSF-IF.C.7.b

FORWARD 2

Transform step functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How is the graph of $y = 3\lfloor x \rfloor$ related to the graph of $y = \lfloor x \rfloor$?

A. shifted right 3: each step moves 3 units to the right

B. compressed horizontally by 3: each step is 1/3 as wide

C. stretched vertically by 3: each step is 3 times as high

D. shifted up 3: each step is 3 units higher than before

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q1/5

1784

QUESTION ID

S85-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. stretched vertically by 3: each step is 3 times as high

GRADING NOTE The outputs are multiplied by 3. The others change the inputs or add a constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How is the graph of $y = \lfloor x/2 \rfloor$ related to the graph of $y = \lfloor x \rfloor$?

A. stretched horizontally by 2: each step is 2 units wide

B. shifted right 2: each step moves 2 units to the right

C. stretched vertically by 2: each step is 2 times as high

D. compressed horizontally by 2: each step is 1/2 unit wide

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q2/5

1785

QUESTION ID
S85-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. stretched horizontally by 2: each step is 2 units wide

GRADING NOTE Dividing the input by 2 doubles the width of each step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q2/5

True or false?

The graphs of $y = \lfloor x \rfloor + 1$ and $y = \lfloor x + 1 \rfloor$ are the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q3/5

1786

QUESTION ID

S85-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Adding 1 to the input moves x up past exactly one more integer: $\lfloor x + 1 \rfloor = \lfloor x \rfloor + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function has steps of width 1 at heights ..., −2, 0, 2, 4, ..., with a closed dot at the left end of each step?

A. $y = 2\lceil x \rceil$

B. $y = \lfloor 2x \rfloor$

C. $y = \lfloor x \rfloor + 2$

D. $y = 2\lfloor x \rfloor$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q4/5

1787

QUESTION ID

S85-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 2\lfloor x \rfloor$

GRADING NOTE $\lceil x \rceil$ steps are closed at the right end; $\lfloor 2x \rfloor$ has steps of width 0.5 at every integer height; $\lfloor x \rfloor + 2$ rises by 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q4/5

$g(x) = 2\lfloor x - 1 \rfloor + 3$, where $\lfloor x \rfloor$ is the greatest integer less than or equal to x. Find g(2.5).

g(2.5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S85-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q5/5

1788

QUESTION ID

S85-F2-Q5

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE $\lfloor 1.5 \rfloor = 1$, so $2(1) + 3 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S85-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 85 • FORWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c • INTERPRETING FUNCTIONS

Graph polynomial functions showing zeros and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.c**
BACKWARD 1	Factor out common factors and special products	**HSA-SSE.A.2**
BACKWARD 2	Crossing and touching from multiplicity	**HSA-APR.B.3**
BACKWARD 3	Evaluate polynomials at test points	**HSF-IF.A.2**
FORWARD 1	Locate relative extrema of polynomial graphs	**HSF-IF.C.7.c**
FORWARD 2	Interpret polynomial graphs in context	**HSF-IF.B.4**

SET 48 • HSF-IF.C.7.c

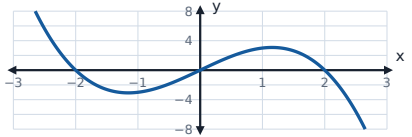
MAIN

Algebra II main question • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

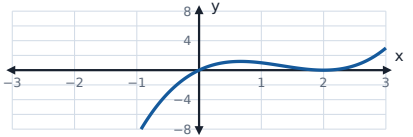
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = x^3 - 4x$?

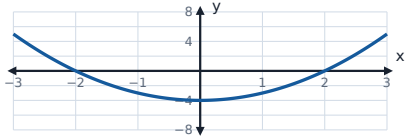
A



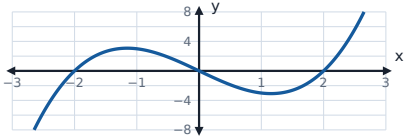
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q1/5

1791

QUESTION ID

S48-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE $x^3 - 4x = x(x - 2)(x + 2)$: crosses at -2, 0, 2; $(-1, 3)$ and $(1, -3)$; falls left, rises right (odd degree, positive leading coefficient). The others show $-x^3 + 4x$, $x(x - 2)^2$, and $x^2 - 4$.

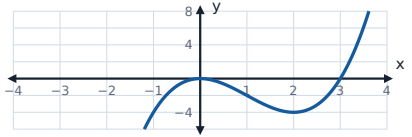
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q1/5

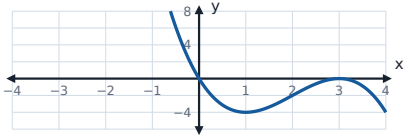
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -x^3 + 3x^2$?

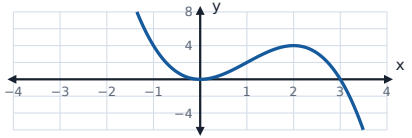
A



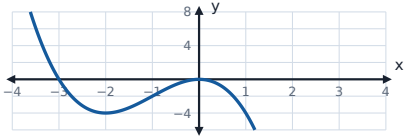
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q2/5

1792

QUESTION ID

S48-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE $-x^3 + 3x^2 = -x^2(x - 3)$: touches at 0 (double zero), crosses at 3, (2, 4), rises left and falls right. The others show $x^3 - 3x^2$, $-x(x - 3)^2$, and $-x^2(x + 3)$.

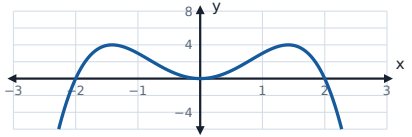
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q2/5

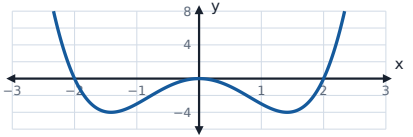
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = x^4 - 4x^2$?

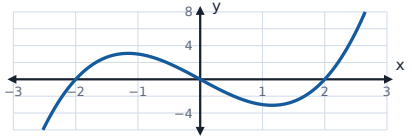
A



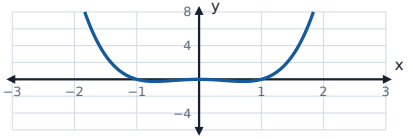
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q3/5

1793

QUESTION ID

S48-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE $x^4 - 4x^2 = x^2(x - 2)(x + 2)$: touches at 0, crosses at ±2, (±1, −3), rises at both ends (even degree, positive leading coefficient). The others show $-x^4 + 4x^2$, $x^3 - 4x$, and $x^4 - x^2$.

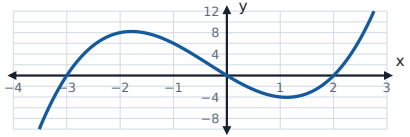
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q3/5

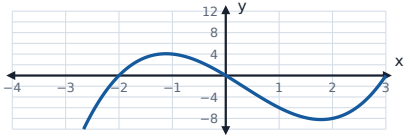
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = x^3 + x^2 - 6x$?

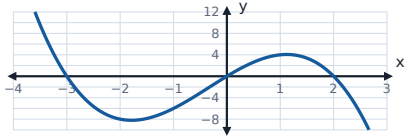
A



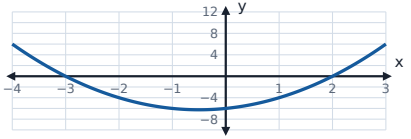
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q4/5

1794

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE $x^3 + x^2 - 6x = x(x + 3)(x - 2)$: crosses at -3, 0, 2; $(-2, 8)$ and $(1, -4)$; falls left, rises right. The others show $x(x - 3)(x + 2)$, its reflection $-x(x + 3)(x - 2)$, and $x^2 + x - 6$.

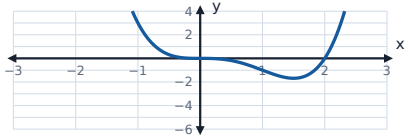
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q4/5

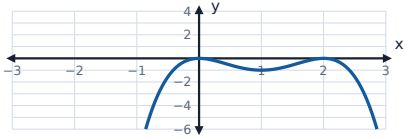
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -x^4 + 2x^3$?

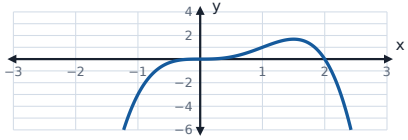
A



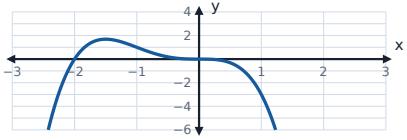
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q5/5

1795

QUESTION ID

S48-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE $-x^4 + 2x^3 = -x^3(x - 2)$: crosses at 0 with a flattened shape (multiplicity 3) and at 2; (1, 1); falls at both ends (even degree, negative leading coefficient). The others show $x^4 - 2x^3$, $-x^2(x - 2)^2$, and $-x^3(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • MAIN • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 1

Factor out common factors and special products • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q1/5

1797

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x(x − 2)(x + 2)

GRADING NOTE Form required: complete.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $-x^3 + 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q2/5

1798

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$-x^2(x - 3)$

GRADING NOTE Any equivalent complete factorization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^4 - 4x^2 = x^2(x^2 - 4) = x^2(x - 2)(x + 2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q3/5

1799

QUESTION ID

S48-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Common factor, then difference of squares.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^3 + x^2 - 6x$ equals

A. $(x + 3)(x - 2)$

B. $x(x + 3)(x - 2)$

C. $x(x - 3)(x + 2)$

D. $x(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q4/5

1800

QUESTION ID

S48-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $x(x + 3)(x - 2)$

GRADING NOTE $x^2 + x - 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Factor $-x^4 + 2x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q5/5

1801

QUESTION ID

S48-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$-x^3(x - 2)$

GRADING NOTE Any equivalent complete factorization.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 2

Crossing and touching from multiplicity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Does $y = x^2(x - 3)$ cross or touch the x-axis at 0?
Answer Crosses or Touches.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q1/5

1803

QUESTION ID
S48-B2-Q1

TYPE
Short answer

CORRECT ANSWER
Touches

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q1/5

Does the graph of $y = x^3(x - 2)$ cross the x-axis at $x = 0$?

A. Yes, as a straight line (multiplicity 1).

B. Yes, and it flattens there (odd multiplicity 3).

C. No, $x = 0$ is not a zero.

D. No, it only touches there (even multiplicity).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q2/5

1804

QUESTION ID
S48-B2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. Yes, and it flattens there (odd multiplicity 3).

GRADING NOTE The factor x^3 has odd multiplicity 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q2/5

True or false?

A zero of odd multiplicity is a crossing point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q3/5

1805

QUESTION ID

S48-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sign change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $y = (x + 2)^2(x - 1)$ at $x = -2$

A. crosses

B. touches

C. has an asymptote

D. is undefined

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q4/5

1806

QUESTION ID

S48-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. touches

GRADING NOTE Even multiplicity.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which describes the end behavior of $y = -x^4 + 2x^3$?

A. Rises at both ends

B. Rises to the left and falls to the right

C. Falls to the left and rises to the right

D. Falls at both ends

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q5/5

1807

QUESTION ID

S48-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Falls at both ends

GRADING NOTE The leading term $-x^4$ has even degree and a negative coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-IF.C.7.c

BACKWARD 3

Evaluate polynomials at test points • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $x^3 - 4x$ at $x = -1$ and at $x = 1$.

At $x = -1$: _____

At $x = 1$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q1/5

1809

QUESTION ID

S48-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3; −3

GRADING NOTE Form required: both. $-1 + 4 = 3$ and $1 - 4 = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-x^3 + 3x^2$ at $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q2/5

1810

QUESTION ID

S48-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q2/5

True or false?

$x^4 - 4x^2$ at $x = 1$ equals 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S48-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q3/5

1811

QUESTION ID

S48-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1 - 4 = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S48-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$x^3 + x^2 - 6x$ at $x = -2$ equals

A. 8

B. 0

C. -8

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q4/5

1812

QUESTION ID

S48-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8

GRADING NOTE $-8 + 4 + 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-x^4 + 2x^3$ at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q5/5

1813

QUESTION ID

S48-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 1

Locate relative extrema of polynomial graphs • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

Use a table with steps of 0.1 (or technology) to estimate the relative maximum of $y = x^3 - 4x$. Give the x-coordinate to the nearest tenth and the y-value to the nearest hundredth. A calculator may be used.

$x \approx$ _____

$y \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q1/5

1815

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−1.2; 3.08

GRADING NOTE Form required: both. Exact $x = -2/\sqrt{3} \approx -1.155$, $y = 16/(3\sqrt{3}) \approx 3.079$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q1/5

Use technology or a table to estimate the relative minimum of $y = x^3 + x^2 - 6x$ on $0 < x < 2$. Give the x-coordinate to the nearest tenth and the y-value to the nearest hundredth. A calculator may be used.

$x \approx$ _____

$y \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q2/5

1816

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.1; −4.06

GRADING NOTE Form required: both. Exact $x = (-1 + \sqrt{19})/3 \approx 1.120$; $y \approx -4.061$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 4 always has exactly 3 turning points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q3/5

1817

QUESTION ID

S48-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has at most 3; $y = x^4$ has only one turning point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

How many turning points can $y = x^4 - 4x^2$ have?

A. 1

B. 4

C. 3

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q4/5

1818

QUESTION ID

S48-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

GRADING NOTE It has 3: two minima and one maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why does the graph of $y = x^3 - 4x$ have a turning point between $x = 0$ and $x = 2$?

A. Because y is positive between 0 and 2 (for example y(1) = 3), so the graph must come back down to 0 at x = 2.

B. Because the graph crosses the x-axis at x = 1, halfway between 0 and 2, and a graph turns at every crossing.

C. y is 0 at x = 0 and at x = 2 and negative between them (for example y(1) = −3), so the graph must turn back up somewhere between them.

D. Because every cubic with a positive leading coefficient turns at x = 1, which lies between 0 and 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q5/5

1819

QUESTION ID

S48-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. y is 0 at x = 0 and at x = 2 and negative between them (for example y(1) = −3), so the graph must turn back up somewhere between them.

GRADING NOTE A continuous graph that leaves 0, goes negative, and returns to 0 has a lowest point in between. The others state false facts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-IF.C.7.c

FORWARD 2

Interpret polynomial graphs in context • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

A box's volume is $V(x) = x(12 - 2x)^2$ in³, where x is the side (in) of the squares cut from a 12-in square sheet. Which interval of x is meaningful, and what is the maximum volume there (it occurs at x = 2)?

Meaningful interval: _____

Maximum volume (in³): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q1/5

1821

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0 < x < 6; 128

GRADING NOTE Form required: both. A cut must be positive and less than 6; V(2) = 128.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A company's profit (thousands of dollars) is $P(x) = -x^3 + 9x^2 - 15x$ for x hundred items, $0 \le x \le 8$. For which x is the profit positive? Give the endpoints to the nearest hundredth.

Profit positive for: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q2/5

1822

QUESTION ID
S48-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$2.21 < x < 6.79$

GRADING NOTE Form required: the compound inequality, endpoints rounded to 0.01. $P(x) = -x(x^2 - 9x + 15)$, with zeros $(9 \pm \sqrt{21})/2 \approx 2.2087$ and 6.7913.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q2/5

True or false?

A zero of a profit model is a break-even point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q3/5

1823

QUESTION ID

S48-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Profit 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A population model $P(t) = -t^3 + 6t^2 + 15t + 100$ rises then falls. The relative maximum means

A. the largest population before decline

B. the starting population

C. the end behavior

D. the time the population is 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q4/5

1824

QUESTION ID
S48-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. the largest population before decline

GRADING NOTE Turning point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q4/5

Why does the end behavior of $V(x) = x(12 - 2x)^2$ not matter for the box made from a 12-in sheet?

A. The context uses only $0 < x < 6$, so the behavior for large $|x|$ is never used.

B. Because $V(x) = x(12 - 2x)^2$ has no end behavior at all.

C. Because the volume of the box is always 128 cubic inches.

D. Because x must be negative for the box, so large positive x is never used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q5/5

1825

QUESTION ID
S48-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. The context uses only 0 < x < 6, so the behavior for large |x| is never used.

GRADING NOTE End behavior describes inputs outside the meaningful interval. The others are false: V does have end behavior, the volume is not constant, and x cannot be negative for a box.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 48 • FORWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions with intercepts, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph basic exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Zero and negative exponents	**8.EE.A.1**
BACKWARD 3	Vertical and horizontal shifts and reflections	**HSF-BF.B.3**
FORWARD 1	Compare and describe exponential graphs	**HSF-IF.C.7.e**
FORWARD 2	Graph the inverse of an exponential function	**HSF-BF.B.4.a**

SET 49 • HSF-IF.C.7.e

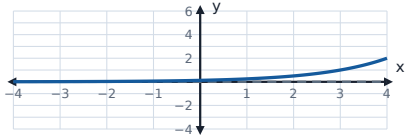
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

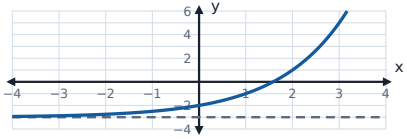
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 2^x - 3$?

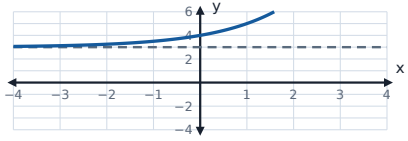
A



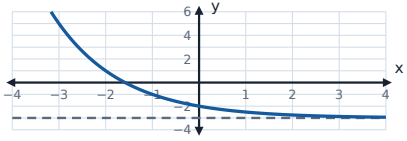
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q1/5

1828

QUESTION ID

S49-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Down 3 from $y = 2^x$: asymptote $y = -3$, y-intercept $(0, -2)$, x-intercept $\log_2 3 \approx 1.58$, increasing. The others show 2^{x-3}, $2^x + 3$, and $(1/2)^x - 3$.

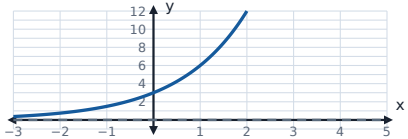
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q1/5

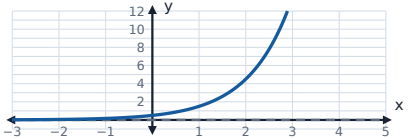
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 3(0.5)^x$?

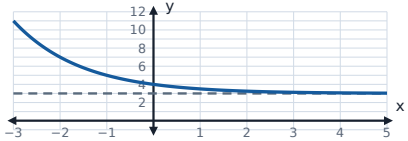
A



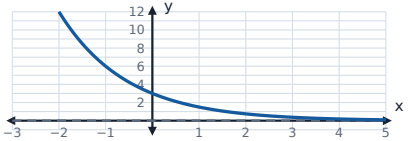
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q2/5

1829

QUESTION ID

S49-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Initial value 3, factor 0.5: y-intercept (0, 3), (1, 1.5), (−1, 6); decreasing; asymptote $y = 0$ ($y \to 0$ as $x \to \infty$). The others show $3(2)^x$, $0.5(3)^x$, and $(0.5)^x + 3$.

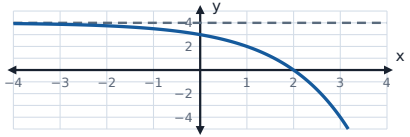
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q2/5

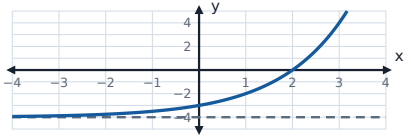
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -2^x + 4$?

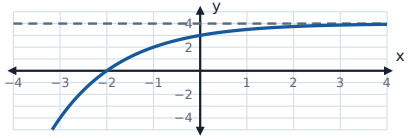
A



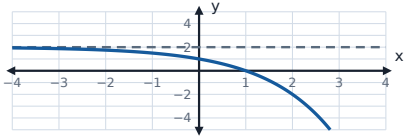
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q3/5

1830

QUESTION ID

S49-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Reflect $y = 2^x$ over the x-axis and move up 4: asymptote $y = 4$ (approached to the left), y-intercept (0, 3), x-intercept (2, 0), decreasing. The others show $2^x - 4$, $4 - 2^{-x}$, and $-2^x + 2$.

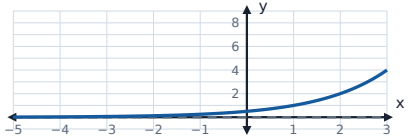
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q3/5

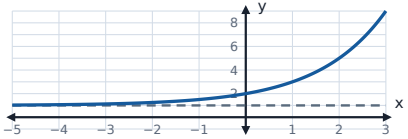
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = 2^{x+1}$?

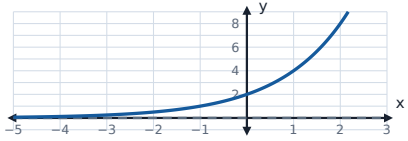
A



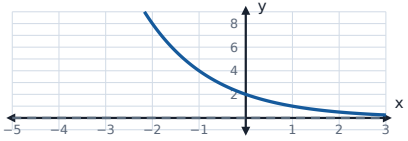
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q4/5

1831

QUESTION ID

S49-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Left 1 from $y = 2^x$ (or $2 \cdot 2^x$): asymptote $y = 0$, y-intercept $(0, 2)$, through $(-1, 1)$, $(1, 4)$, $(2, 8)$. The others show 2^{x-1}, $2^x + 1$ (same y-intercept, asymptote $y = 1$), and 2^{1-x}.

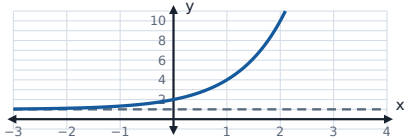
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q4/5

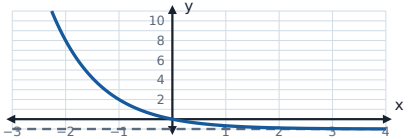
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = (1/3)^x + 1$?

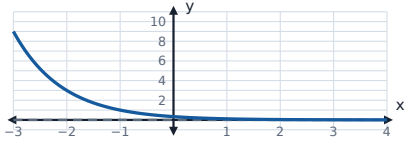
A



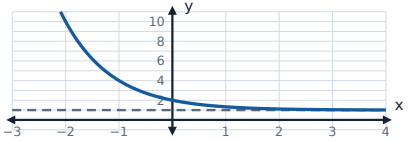
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q5/5

1832

QUESTION ID

S49-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Up 1 from $y = (1/3)^x$: asymptote $y = 1$, y-intercept $(0, 2)$, through $(-1, 4)$ and $(-2, 10)$, decreasing toward 1. The others show $3^x + 1$, $(1/3)^x - 1$, and $(1/3)^{x+1}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • MAIN • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 1

Graph basic exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Find $y = 2^x$ at $x = -2, -1, 0, 1$, and 2, in that order.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q1/5

1834

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4, 1/2, 1, 2, 4

GRADING NOTE Form required: all five, in order (0.25 and 0.5 are equivalent).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the y-intercept of $y = 3(0.5)^x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q2/5

1835

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q2/5

True or false?

$y = 2^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q3/5

1836

QUESTION ID

S49-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2^x \to 0$ as $x \to -\infty$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S49-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = (0.5)^x$ is

A. linear

B. increasing

C. constant

D. decreasing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q4/5

1837

QUESTION ID

S49-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. decreasing

GRADING NOTE Base less than 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q4/5

Does the graph of $y = 2^x$ have an x-intercept?

A. Yes: at x = 0

B. No: $2^x > 0$ for every x, so y is never 0

C. Yes: at x = −1

D. No: the graph does not cross the y-axis either

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q5/5

1838

QUESTION ID

S49-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. No: $2^x > 0$ for every x, so y is never 0

GRADING NOTE Every power of 2 is positive. At x = 0, y = 1 (the y-intercept).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 2

Zero and negative exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2^{-2}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q1/5

1840

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $(1/3)^{-2}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q2/5

1841

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$b^0 = 0$ for $b \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q3/5

1842

QUESTION ID

S49-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Any nonzero base to the 0 power is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3(0.5)^{-1}$ equals

A. 6

B. 3

C. -1.5

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q4/5

1843

QUESTION ID

S49-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6

GRADING NOTE $0.5^{-1} = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $2^{-1} - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q5/5

1844

QUESTION ID

S49-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−2.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-IF.C.7.e

BACKWARD 3

Vertical and horizontal shifts and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How is the graph of $y = 2^x - 3$ related to the graph of $y = 2^x$?

A. Shifted left 3

B. Shifted up 3

C. Shifted right 3

D. Shifted down 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q1/5

1846

QUESTION ID

S49-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Shifted down 3

GRADING NOTE Subtracting 3 from the output shifts down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

How is the graph of $y = 2^{x+1}$ related to the graph of $y = 2^x$?

A. Shifted left 1

B. Shifted up 1

C. Shifted down 1

D. Shifted right 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q2/5

1847

QUESTION ID

S49-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Shifted left 1

GRADING NOTE Replacing x by x + 1 shifts left.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A vertical shift moves the horizontal asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q3/5

1848

QUESTION ID

S49-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE y = 0 becomes y = k.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = -2^x + 4$ has asymptote

A. y = 4

B. y = 0

C. y = −4

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q4/5

1849

QUESTION ID

S49-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. y = 4

GRADING NOTE Reflect, then up 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the asymptote of $y = (1/3)^x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q5/5

1850

QUESTION ID

S49-B3-Q5

TYPE

Short answer

CORRECT ANSWER

y = 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 1

Compare and describe exponential graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

Why are the graphs of $y = 2^x$ and $y = (1/2)^x$ reflections of each other over the y-axis?

A. Because 1/2 is less than 1, so $y = (1/2)^x$ decreases while $y = 2^x$ increases.

B. $(1/2)^x = 2^{-x}$, so $y = (1/2)^x$ is $y = 2^x$ with x replaced by $-x$.

C. $(1/2)^x = -2^x$, so $y = (1/2)^x$ is $y = 2^x$ with every output made negative.

D. Because both graphs pass through (0, 1), which lies on the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q1/5

1852

QUESTION ID
S49-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. $(1/2)^x = 2^{-x}$, so $y = (1/2)^x$ is $y = 2^x$ with x replaced by $-x$.

GRADING NOTE Replacing x by $-x$ reflects over the y-axis. The others give incomplete or false reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q1/5

$2^{x+1} = 2 \cdot 2^x$. What does this say about two transformations of $y = 2^x$?

A. A shift right by 1 gives the same graph as a vertical stretch by 2.

B. A shift left by 1 gives the same graph as a shift up by 2.

C. A shift left by 1 gives the same graph as a vertical stretch by 2.

D. They are never the same graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q2/5

1853

QUESTION ID
S49-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. A shift left by 1 gives the same graph as a vertical stretch by 2.

GRADING NOTE 2^{x+1} is the left shift; $2 \cdot 2^x$ is the vertical stretch; they are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q2/5

True or false?

$y = 5 \cdot 3^x$ and $y = 3^x$ have different horizontal asymptotes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q3/5

1854

QUESTION ID

S49-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A vertical stretch keeps the asymptote y = 0 for both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which exponential graph passes through (0, 4) and (1, 2)?

A. $y = 2^x + 3$

B. $y = 4 \cdot 2^x$

C. $y = 0.5 \cdot 4^x$

D. $y = 4(0.5)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q4/5

1855

QUESTION ID
S49-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $y = 4(0.5)^x$

GRADING NOTE a = 4, b = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which exponential function is increasing, has asymptote $y = -1$, has y-intercept 2, and passes through (1, 5)?

A. $y = 3(1/2)^x - 1$

B. $y = 3 \cdot 3^x - 1$

C. $y = 3 \cdot 2^x - 1$

D. $y = 2 \cdot 2^x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q5/5

1856

QUESTION ID

S49-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 3 \cdot 2^x - 1$

GRADING NOTE $3 \cdot 2^0 - 1 = 2$ and $3 \cdot 2 - 1 = 5$. $3 \cdot 3^x - 1$ passes (1, 8); $2 \cdot 2^x - 1$ has y-intercept 1; $3(1/2)^x - 1$ decreases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-IF.C.7.e

FORWARD 2

Graph the inverse of an exponential function • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Which is the inverse of $f(x) = 2^x$?

A. $f^{-1}(x) = \log_2 x$

B. $f^{-1}(x) = x^2$

C. $f^{-1}(x) = (1/2)^x$

D. $f^{-1}(x) = x/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q1/5

1858

QUESTION ID

S49-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $f^{-1}(x) = \log_2 x$

GRADING NOTE $y = 2^x$ means $x = \log_2 y$. The others confuse inverse with a power, a reciprocal base, or division.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (3, 8) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q2/5

1859

QUESTION ID
S49-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(8, 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q2/5

True or false?

The inverse of $y = 2^x$ has the vertical asymptote $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q3/5

1860

QUESTION ID
S49-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Reflection of y = 0 over y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q3/5

The inverse of $y = 2^x - 3$ is

A. $y = \log_2(x - 3)$

B. $y = \log_2 x - 3$

C. $y = 2^{x+3}$

D. $y = \log_2(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q4/5

1861

QUESTION ID

S49-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = \log_2(x + 3)$

GRADING NOTE $x = 2^y - 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which gives the domain and range of $y = \log_2 x$?

A. Domain: all real numbers; range: $y > 0$

B. Domain: $x \geq 0$; range: $y \geq 0$

C. Domain: $x > 1$; range: $y > 0$

D. Domain: $x > 0$; range: all real numbers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q5/5

1862

QUESTION ID

S49-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Domain: $x > 0$; range: all real numbers

GRADING NOTE They are the range and domain of 2^x, swapped.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 49 • FORWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph logarithmic functions with intercept, asymptote, and end behavior

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Graph exponential functions	**HSF-IF.C.7.e**
BACKWARD 2	Find inverses by swapping inputs and outputs	**HSF-BF.B.4.a**
BACKWARD 3	Shift and reflect graphs	**HSF-BF.B.3**
FORWARD 1	Graph logarithms with reflections that change the domain	**HSF-IF.C.7.e**
FORWARD 2	Solve equations with logarithms	**HSF-LE.A.4**

SET 50 • HSF-IF.C.7.e

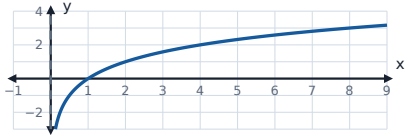
MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

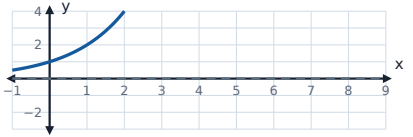
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_2 x$?

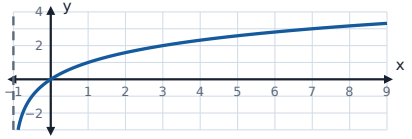
A



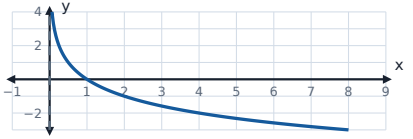
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q1/5

1865

QUESTION ID

S50-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Asymptote $x = 0$, x-intercept (1, 0), through (2, 1), (4, 2), (8, 3); domain $x > 0$; increases slowly. The others show 2^x (the inverse), $\log_2(x + 1)$, and $-\log_2 x$.

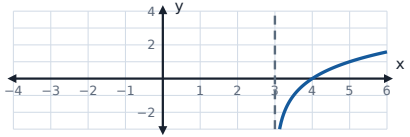
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q1/5

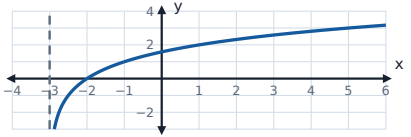
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_2(x + 3)$?

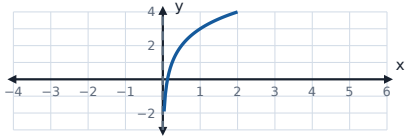
A



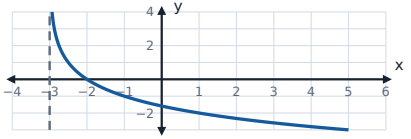
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q2/5

1866

QUESTION ID

S50-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Left 3 from $y = \log_2 x$: asymptote $x = -3$, x-intercept $(-2, 0)$, through $(-1, 1)$, $(1, 2)$, $(5, 3)$; domain $x > -3$. The others show $\log_2(x - 3)$, $\log_2 x + 3$, and $-\log_2(x + 3)$.

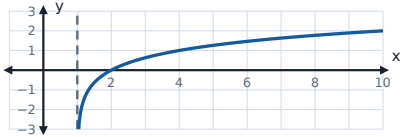
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q2/5

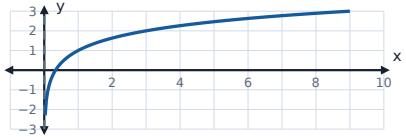
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_3 x - 1$?

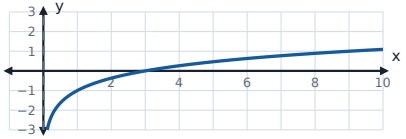
A



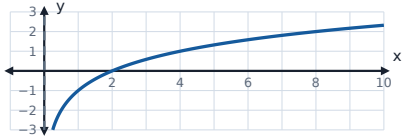
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q3/5

1867

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Down 1 from $y = \log_3 x$: asymptote $x = 0$, x-intercept (3, 0), through (1, −1) and (9, 1); domain $x > 0$. The others show $\log_3(x - 1)$, $\log_3 x + 1$, and $\log_2 x - 1$ (wrong base: it passes (2, 0)).

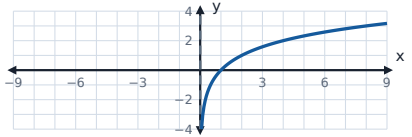
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q3/5

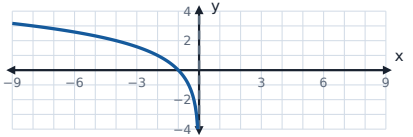
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = -\log_2 x$?

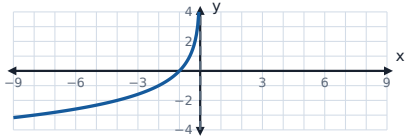
A



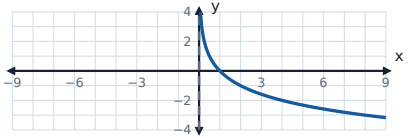
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q4/5

1868

QUESTION ID

S50-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Reflection of $y = \log_2 x$ over the x-axis: asymptote $x = 0$, x-intercept $(1, 0)$, through $(2, -1)$, $(4, -2)$, $(8, -3)$; decreasing; domain $x > 0$. The others show $\log_2 x$, $\log_2(-x)$, and $-\log_2(-x)$.

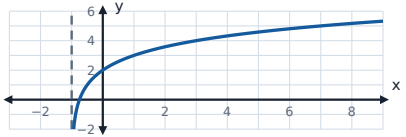
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q4/5

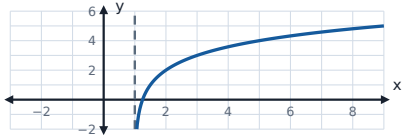
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_2(x - 1) + 2$?

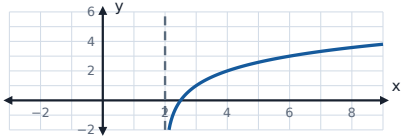
A



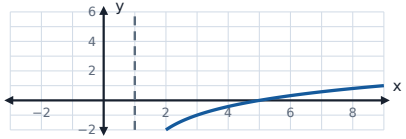
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q5/5

1869

QUESTION ID

S50-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Right 1 and up 2: asymptote $x = 1$, through (2, 2), (3, 3), (5, 4), x-intercept (1.25, 0); domain $x > 1$. The others show $\log_2(x + 1) + 2$, $\log_2(x - 2) + 1$, and $\log_2(x - 1) - 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • MAIN • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 1

Graph exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Find the y-values of $y = 2^x$ at x = 0, 1, and 2, in that order, to get three points on the graph.

y-values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q1/5

1871

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, 4

GRADING NOTE Form required: all three, in order: (0, 1), (1, 2), (2, 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the asymptote of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q2/5

1872

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q2/5

True or false?

$y = 3^x$ passes through (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q3/5

1873

QUESTION ID

S50-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^3 = 27$, not 1; the graph passes through (1, 3) and (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q3/5

$y = 2^x$ passes through

A. (2, 1)

B. (−1, −2)

C. (1/2, −1)

D. (−1, 1/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q4/5

1874

QUESTION ID

S50-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (−1, 1/2)

GRADING NOTE 2^{-1}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $y = 3^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q5/5

1875

QUESTION ID

S50-B1-Q5

TYPE

Short answer

CORRECT ANSWER

y > 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 2

Find inverses by swapping inputs and outputs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

(2, 4) is on $y = 2^x$. Which point is on its inverse?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

1877

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the inverse of $y = 3^x$?

A. $y = x^3$

B. $y = \log_3 x$

C. $y = x/3$

D. $y = 3^{-x}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

1878

QUESTION ID
S50-B2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. $y = \log_3 x$

GRADING NOTE Swapping x and y gives $x = 3^y$, so $y = \log_3 x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Swapping x and y in $y = 2^x$ gives $x = 2^y$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

1879

QUESTION ID

S50-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Inverse relation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on $y = \log_2 x$?

A. (3, 8)

B. (2, 4)

C. (8, 3)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

1880

QUESTION ID

S50-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (8, 3)

GRADING NOTE $2^3 = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\log_2 16$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

1881

QUESTION ID

S50-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-IF.C.7.e

BACKWARD 3

Shift and reflect graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = \log_2(x + 3)$ related to the graph of $y = \log_2 x$?

A. Shifted right 3

B. Shifted down 3

C. Shifted left 3

D. Shifted up 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q1/5

1883

QUESTION ID

S50-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Shifted left 3

GRADING NOTE Replacing x by x + 3 shifts left; the asymptote moves to x = −3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q1/5

How is the graph of $y = -\log_2 x$ related to the graph of $y = \log_2 x$?

A. Reflected over the line $y = x$

B. Reflected over the y-axis

C. Reflected over the x-axis

D. Shifted down 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q2/5

1884

QUESTION ID

S50-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Reflected over the x-axis

GRADING NOTE Negating the output reflects over the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q2/5

True or false?

A horizontal shift moves the vertical asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q3/5

1885

QUESTION ID

S50-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE x = 0 becomes x = h.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = \log_2(x - 1) + 2$ has asymptote

A. $x = -1$

B. $y = 2$

C. $x = 1$

D. $x = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q4/5

1886

QUESTION ID

S50-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x = 1

GRADING NOTE Right 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Where does (1, 0) of $y = \log_3 x$ move under $y = \log_3 x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q5/5

1887

QUESTION ID

S50-B3-Q5

TYPE

Short answer

CORRECT ANSWER

(1, −1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • HSF-IF.C.7.e

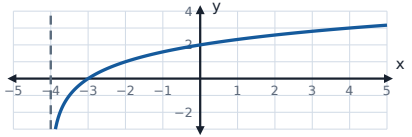
FORWARD 1

Graph logarithms with reflections that change the domain • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

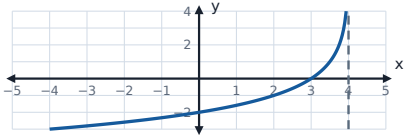
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows $y = \log_2(4 - x)$?

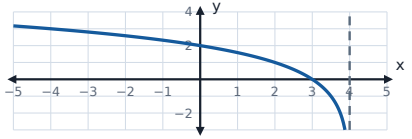
A



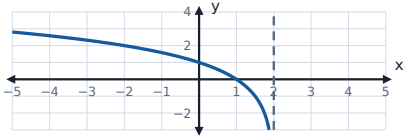
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q1/5

1889

QUESTION ID

S50-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE $4 - x > 0$ gives domain $x < 4$ and asymptote $x = 4$; x-intercept (3, 0); through (2, 1), (0, 2), (−4, 3); decreasing. The others show $\log_2(x + 4)$, $-\log_2(4 - x)$, and $\log_2(2 - x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q1/5

Why does the graph of $y = \log(x^2 - 1)$ have two separate branches?

A. Every logarithm graph has two branches, one on each side of its asymptote.

B. Its domain is $x < -1$ or $x > 1$: $x^2 - 1 > 0$ on two separate intervals.

C. $\log(x^2 - 1)$ is undefined at $x = 0$ only, which splits the graph in two.

D. Its domain is $-1 < x < 1$, an interval split in two at $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q2/5

1890

QUESTION ID

S50-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Its domain is $x < -1$ or $x > 1$: $x^2 - 1 > 0$ on two separate intervals.

GRADING NOTE A logarithm needs a positive input. The others reverse the domain or overgeneralize ($y = \log x$ has one branch).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q2/5

True or false?

$y = \log_2(2x)$ is $y = \log_2 x$ shifted up 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q3/5

1891

QUESTION ID

S50-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\log_2 2 + \log_2 x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The x-intercept of $y = \log_2(x + 3)$ is

A. -2

B. 1

C. -3

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q4/5

1892

QUESTION ID

S50-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. −2

GRADING NOTE $x + 3 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the x-intercept of $y = \log_2(x - 1) + 2$ exactly.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q5/5

1893

QUESTION ID

S50-F1-Q5

TYPE

Short answer

CORRECT ANSWER

5/4

GRADING NOTE Any equivalent form. $x - 1 = 2^{-2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-IF.C.7.e

FORWARD 2

Solve equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 12$. Give x to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

1895

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3.58

GRADING NOTE Rounding to 0.01. $x = \log_2 12 = \log 12/\log 2 \approx 3.585$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5 \cdot 10^x = 40$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

1896

QUESTION ID
S50-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.903

GRADING NOTE Rounding to 0.001. $10^x = 8$, so $x = \log 8 \approx 0.90309$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

True or false?

$\log_2 12$ is between 4 and 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

1897

QUESTION ID

S50-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^3 = 8 < 12 < 16 = 2^4$, so $\log_2 12$ is between 3 and 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The solution of $3^x = 7$ is

A. log 7/log 3

B. log 3/log 7

C. 7/3

D. log 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

1898

QUESTION ID

S50-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. log 7/log 3

GRADING NOTE Change of base.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a graphing calculator (or the logarithm key) to estimate $\log_2 6$ to the nearest tenth. A calculator may be used.

$\log_2 6 \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

1899

QUESTION ID

S50-F2-Q5

TYPE

Short answer

CORRECT ANSWER

2.6

GRADING NOTE Rounding to 0.1. $\log_2 6 = \log 6/\log 2 \approx 2.585$, and $2^2 = 4 < 6 < 8 = 2^3$ confirms it is between 2 and 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph trigonometric functions showing period, midline, and amplitude

MAIN	Algebra II main question	**HSF-IF.C.7.e**
BACKWARD 1	Unit-circle values at quarter turns	**HSF-TF.A.2**
BACKWARD 2	Read amplitude, period, and midline from a formula	**HSF-TF.B.5**
BACKWARD 3	Vertical stretches, shifts, and reflections	**HSF-BF.B.3**
FORWARD 1	Graph shifted sinusoids and the tangent function	**HSF-IF.C.7.e**
FORWARD 2	Horizontal stretches f(kx) of trigonometric graphs	**HSF-BF.B.3**

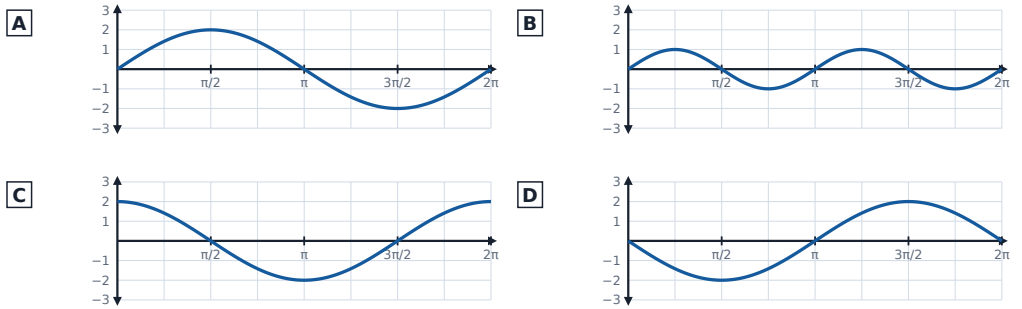
SET 51 • HSF-IF.C.7.e

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = 2sin x on 0 ≤ x ≤ 2π?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q1/5

1902

QUESTION ID

S51-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

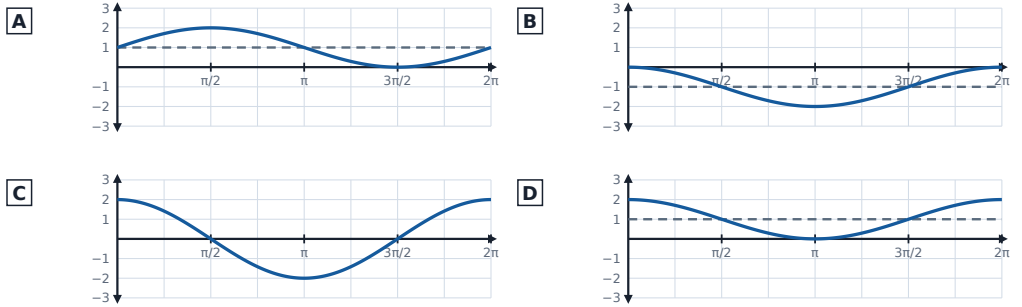
GRADING NOTE Amplitude 2, period 2π, midline y = 0: (0, 0), (π/2, 2), (π, 0), (3π/2, −2), (2π, 0). The others show sin(2x), 2cos x, and −2sin x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = cos x + 1 on 0 ≤ x ≤ 2π?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q2/5

1903

QUESTION ID

S51-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

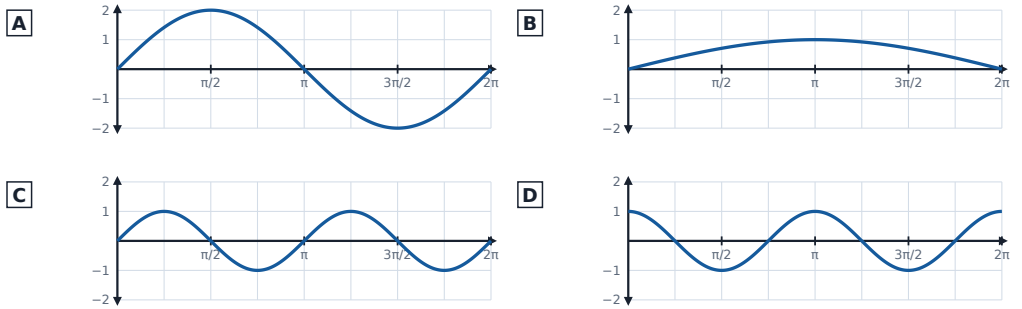
GRADING NOTE Midline y = 1, amplitude 1, period 2π: (0, 2), (π/2, 1), (π, 0), (3π/2, 1), (2π, 2). The others show sin x + 1, cos x − 1, and 2cos x (same maximum, wrong minimum).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = sin(2x) on 0 ≤ x ≤ 2π?



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q3/5

1904

QUESTION ID

S51-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Graph C

GRADING NOTE Period 2π/2 = π, amplitude 1, midline y = 0: two cycles on [0, 2π], starting at (0, 0) and rising to (π/4, 1). The others show 2sin x, sin(x/2), and cos(2x).

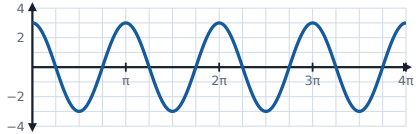
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q3/5

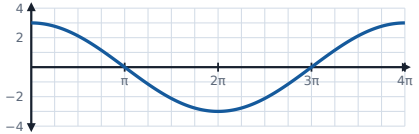
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = 3cos(x/2) on 0 ≤ x ≤ 4π?

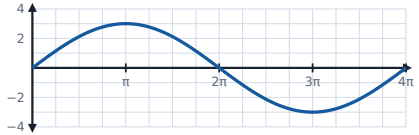
A



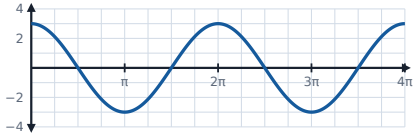
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q4/5

1905

QUESTION ID

S51-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Graph B

GRADING NOTE Period 2π/(1/2) = 4π, amplitude 3, midline y = 0: (0, 3), (π, 0), (2π, −3), (3π, 0), (4π, 3). The others show 3cos(2x), 3sin(x/2), and 3cos x.

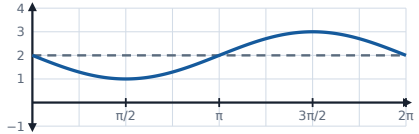
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q4/5

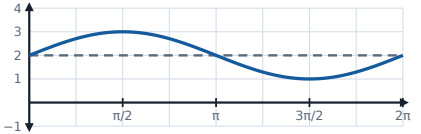
ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = −sin x + 2 on 0 ≤ x ≤ 2π?

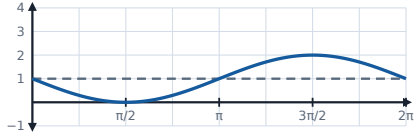
A



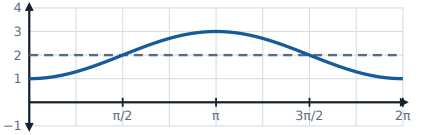
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q5/5

1906

QUESTION ID

S51-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Graph A

GRADING NOTE Midline y = 2, amplitude 1, reflected: (0, 2), (π/2, 1), (π, 2), (3π/2, 3), (2π, 2). The others show sin x + 2, −sin x + 1, and −cos x + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • MAIN • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 1

Unit-circle values at quarter turns • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Give sin x at x = 0, π/2, π, 3π/2, 2π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q1/5

1908

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 1, 0, −1, 0

GRADING NOTE Form required: all five.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give cos x at x = 0, π/2, π, 3π/2, 2π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q2/5

1909

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1, 0, −1, 0, 1

GRADING NOTE Form required: all five.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\sin(3\pi/2) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q3/5

1910

QUESTION ID

S51-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The point at $3\pi/2$ is $(0, -1)$, so $\sin(3\pi/2) = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

cos(π) equals

A. π

B. 1

C. −1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q4/5

1911

QUESTION ID

S51-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −1

GRADING NOTE Point (−1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give sin(π/4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q5/5

1912

QUESTION ID

S51-B1-Q5

TYPE

Short answer

CORRECT ANSWER

√2/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 2

Read amplitude, period, and midline from a formula • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the amplitude and the period of y = 3cos(x/2).

Amplitude: _____

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q1/5

1914

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3; 4π

GRADING NOTE Form required: both (the period exactly, in terms of π). 2π/(1/2) = 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the midline of $y = -\sin x + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q2/5

1915

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = sin(2x) has period 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q3/5

1916

QUESTION ID

S51-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The period is 2π/2 = π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

y = 2sin x has amplitude

A. 1

B. 2

C. π

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q4/5

1917

QUESTION ID

S51-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2

GRADING NOTE |A|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the period of y = cos(3x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q5/5

1918

QUESTION ID

S51-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2π/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSF-IF.C.7.e

BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How does the graph of y = cos x + 1 compare with the graph of y = cos x?

A. Shifted left 1

B. Stretched vertically by 1

C. Shifted down 1

D. Shifted up 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q1/5

1920

QUESTION ID

S51-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Shifted up 1

GRADING NOTE Adding 1 to the output shifts up. The others move left instead of up, call a factor of 1 a stretch, or shift the wrong way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q1/5

How does the graph of y = −sin x compare with the graph of y = sin x?

A. Reflected over the x-axis

B. Stretched vertically by 2

C. Shifted down 1

D. Unchanged

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q2/5

1921

QUESTION ID

S51-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Reflected over the x-axis

GRADING NOTE Negating the output reflects over the x-axis. The others shift, leave unchanged, or stretch.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = 2sin x has the same zeros as y = sin x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q3/5

1922

QUESTION ID

S51-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Vertical stretches keep zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The maximum of $y = -\sin x + 2$ is

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q4/5

1923

QUESTION ID

S51-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 3

GRADING NOTE $-(-1) + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the minimum of y = cos x + 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q5/5

1924

QUESTION ID

S51-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • HSF-IF.C.7.e

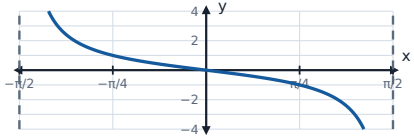
FORWARD 1

Graph shifted sinusoids and the tangent function • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Which graph shows y = tan x for −π/2 < x < π/2?

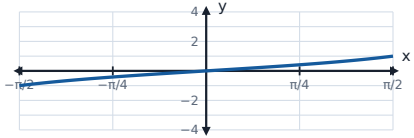
A



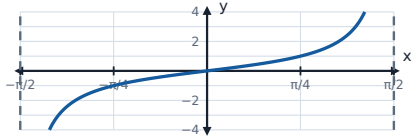
B



C



D



A. Graph A **B.** Graph B **C.** Graph C **D.** Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q1/5

1926

QUESTION ID

S51-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Graph D

GRADING NOTE Asymptotes x = ±π/2 (cos x = 0 there); increasing through (−π/4, −1), (0, 0), (π/4, 1). The others show −tan x, tan(2x), and tan(x/2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q1/5

How does y = sin(x − π/2) compare with y = sin x, and which simpler formula describes it?

A. It is shifted left π/2, and it equals cos x.

B. It is shifted right π/2, and it equals −cos x.

C. It is shifted down π/2, and it equals sin x − π/2.

D. It is shifted right π/2, and it equals cos x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q2/5

1927

QUESTION ID

S51-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. It is shifted right π/2, and it equals −cos x.

GRADING NOTE x − π/2 shifts right; sin(x − π/2) = −cos x (at x = 0 both are −1). The others shift the wrong way, shift vertically, or give the wrong sign (cos 0 = 1, but sin(−π/2) = −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The period of $y = \tan x$ is 2π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q3/5

1928

QUESTION ID

S51-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\tan(x + \pi) = \tan x$, so the period is π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The first maximum of $y = \cos(x - \pi/3)$ for $x \geq 0$ is at

A. $x = 2\pi/3$

B. $x = \pi/3$

C. $x = 0$

D. $x = -\pi/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q4/5

1929

QUESTION ID
S51-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. $x = \pi/3$

GRADING NOTE Shift right π/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the x-values of the maximum points of $y = \sin(x + \pi/4)$ on $0 \leq x < 2\pi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q5/5

1930

QUESTION ID

S51-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$x = \pi/4$

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-IF.C.7.e

FORWARD 2

Horizontal stretches f(kx) of trigonometric graphs • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

How is the graph of y = sin(2x) obtained from the graph of y = sin x?

A. Horizontal compression by a factor of 1/2

B. Horizontal stretch by a factor of 2

C. Shift left 2

D. Vertical stretch by a factor of 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q1/5

1932

QUESTION ID
S51-F2-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. Horizontal compression by a factor of 1/2

GRADING NOTE The period becomes π, half of 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q1/5

How is the graph of y = cos(x/2) obtained from the graph of y = cos x?

A. Shift right 2

B. Vertical compression by a factor of 1/2

C. Horizontal stretch by a factor of 2

D. Horizontal compression by a factor of 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q2/5

1933

QUESTION ID

S51-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Horizontal stretch by a factor of 2

GRADING NOTE The period becomes 4π, twice 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q2/5

True or false?

Replacing x by kx (k > 1) shortens the period by the factor k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q3/5

1934

QUESTION ID
S51-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Period 2π/k.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = \sin(-x)$ is the same as

A. $\cos x$

B. $\sin x$

C. $-\cos x$

D. $-\sin x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q4/5

1935

QUESTION ID

S51-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $-\sin x$

GRADING NOTE Sine is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For which k > 0 does y = cos(kx) have period 6π?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q5/5

1936

QUESTION ID

S51-F2-Q5

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE Form required: the value. The period is 2π/k = 6π, so k = 1/3. (Without the condition k > 0, k = −1/3 also works, since the period is 2π/|k|.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Rewrite exponential functions to reveal rates over other time units

MAIN	Algebra II main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth rates from exponential functions	**HSF-IF.C.8.b**
BACKWARD 2	Powers of powers and rational exponents	**HSN-RN.A.1**
BACKWARD 3	Percent change and growth factors	**7.RP.A.3**
FORWARD 1	Compare growth offers by rewriting to the same time unit	**HSF-IF.C.8.b**
FORWARD 2	Find doubling times and half-lives with logarithms	**HSF-LE.A.4**

SET 52 • HSF-IF.C.8.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

$y = 100(1.08)^t$ with t in years. Rewrite it as $y = 100(b)^{12t}$ with a monthly growth factor b. Give b to 6 decimal places and the monthly percent rate to the nearest hundredth of a percent. A calculator may be used.

b = _____

Monthly rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q1/5

1939

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

1.006434; 0.64%

GRADING NOTE Form required: both. $b = 1.08^{1/12} \approx 1.0064340$, about 0.64% per month.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q1/5

$y = 500(0.97)^t$ with t in years. Rewrite it as $y = 500(b)^{t/10}$ with a growth factor b per decade. Give b to 4 decimal places and the percent decrease per decade to the nearest hundredth of a percent. A calculator may be used.

b = _____

Percent decrease per decade: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q2/5

1940

QUESTION ID
S52-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.7374; 26.26%

GRADING NOTE Form required: both. $b = 0.97^{10} \approx 0.737424$; $1 - 0.737424 \approx 0.2626$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q2/5

$Q = 50(2)^{t/3}$ with t in hours. Rewrite it as $Q = 50(b)^{t}$ with an hourly growth factor b. Give b to 4 decimal places and the hourly percent rate to the nearest hundredth of a percent. A calculator may be used.

b = _____

Hourly rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S52-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q3/5

1941

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

1.2599; 25.99%

GRADING NOTE Form required: both. $b = 2^{1/3} \approx 1.259921$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-M-Q3**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$A = 1000(1.005)^{12t}$ with t in years. Rewrite it as $A = 1000(b)^{t}$ with an annual growth factor b. Give b to 6 decimal places and the annual percent rate to the nearest hundredth of a percent. A calculator may be used.

b = _____

Annual rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q4/5

1942

QUESTION ID
S52-M-Q4

TYPE
Short answer

CORRECT ANSWER
1.061678; 6.17%

GRADING NOTE Form required: both. $b = 1.005^{12} \approx 1.0616778$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q4/5

$N = 800(0.5)^{t/10}$ with t in years. Rewrite it as $N = 800(b)^t$ with a yearly factor b. Give b to 6 decimal places and the yearly percent decrease to the nearest hundredth of a percent. A calculator may be used.

b = _____

Yearly decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q5/5

1943

QUESTION ID

S52-M-Q5

TYPE

Short answer

CORRECT ANSWER

0.933033; 6.70%

GRADING NOTE Form required: both. $b = 0.5^{1/10} \approx 0.9330330$; $1 - 0.933033 \approx 0.0670$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • MAIN • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 1

Read growth rates from exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Is $y = 100(1.08)^t$ growth or decay, and by what percent per year? Answer Growth or Decay, then the percent.

Growth or Decay: _____

Percent per year: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q1/5

1945

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Growth; 8%

GRADING NOTE Form required: both. The factor 1.08 = 1 + 0.08.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $y = 500(0.97)^t$ growth or decay, and by what percent per year? Answer Growth or Decay, then the percent.

Growth or Decay: _____

Percent per year: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q2/5

1946

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Decay; 3%

GRADING NOTE Form required: both. The factor 0.97 = 1 − 0.03.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 2(1.25)^t$ grows 125% per unit of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q3/5

1947

QUESTION ID

S52-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The factor 1.25 means 25% growth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 40(0.9)^t$ decreases by

A. 9% per unit

B. 10% per unit

C. 90% per unit

D. 0.9% per unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q4/5

1948

QUESTION ID

S52-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 10% per unit

GRADING NOTE $1 - 0.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the growth factor for 6.17% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q5/5

1949

QUESTION ID

S52-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1.0617

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 2

Powers of powers and rational exponents • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite 1.08^{t} as a power of $1.08^{1/12}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q1/5

1951

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$(1.08^{1/12})^{12t}$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B2-Q1**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $2^{t/3}$ as $(2^{1/3})^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q2/5

1952

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(2^{1/3})^t$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S52-B2-Q2**

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(b^m)^n = b^{mn}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q3/5

1953

QUESTION ID

S52-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Power of a power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

1.005^{12t} equals

A. $(1.005^{12})^{t}$

B. $(1.005^{t})/12$

C. $12 \cdot 1.005^{t}$

D. $1.005^{12} + t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q4/5

1954

QUESTION ID

S52-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $(1.005^{12})^{t}$

GRADING NOTE Power of a power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write $8^{1/3}$ as a whole number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q5/5

1955

QUESTION ID

S52-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.1 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • HSF-IF.C.8.b

BACKWARD 3

Percent change and growth factors • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Write the growth factor for a 25.99% increase.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q1/5

1957

QUESTION ID

S52-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1.2599

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A factor 0.7374 means what percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q2/5

1958

QUESTION ID

S52-B3-Q2

TYPE

Short answer

CORRECT ANSWER

26.26%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A factor 1.006434 means about 0.64% growth.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q3/5

1959

QUESTION ID

S52-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtract 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A 3% decrease corresponds to the factor

A. 0.03

B. 0.97

C. 0.7

D. 1.03

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q4/5

1960

QUESTION ID

S52-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.97

GRADING NOTE 1 − 0.03.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

An amount is multiplied by 0.933. By what percent does it decrease?

Percent decrease: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q5/5

1961

QUESTION ID

S52-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6.7%

GRADING NOTE Form required: the percent. 1 − 0.933 = 0.067.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 52 • BACKWARD 3 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 1

Compare growth offers by rewriting to the same time unit • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Account A grows 0.5% per month. Account B grows 6.1% per year. Find A's annual growth factor (to 4 decimal places), and say which account grows faster (A or B). A calculator may be used.

A's annual factor: _____

Faster account (A or B): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q1/5

1963

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1.0617; A

GRADING NOTE Form required: both. $1.005^{12} \approx 1.061678$, about 6.17% per year, which is more than 6.1%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Culture 1 doubles every 3 hours; culture 2 triples every 5 hours. Find each hourly growth factor to 3 decimal places, and say which grows faster (1 or 2). A calculator may be used.

Hourly factor of culture 1: _____

Hourly factor of culture 2: _____

Faster culture (1 or 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q2/5

1964

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.260; 1.246; 1

GRADING NOTE Form required: all three. $2^{1/3} \approx 1.2599$ and $3^{1/5} \approx 1.2457$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q2/5

True or false?

Growing 1% per month is the same as growing 12% per year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q3/5

1965

QUESTION ID

S52-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1.01^{12} \approx 1.1268$, about 12.68%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the greatest annual growth?

A. 4% per half year

B. 0.6% per month

C. 8% per year

D. 2% per quarter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q4/5

1966

QUESTION ID

S52-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2% per quarter

GRADING NOTE $1.02^4 \approx 1.0824$; $1.006^{12} \approx 1.0744$; $1.04^2 = 1.0816$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A substance with a half-life of 10 years and a substance that decays 7% per year are compared. Which decays faster? A calculator may be used.

A. 7% per year: its yearly factor 0.93 is less than $0.5^{1/10} \approx 0.933$.

B. 7% per year: 7 is less than 10.

C. The half-life substance: its yearly factor 0.5 is less than 0.93.

D. They decay at the same rate, because $0.5^{1/10} \approx 0.93$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q5/5

1967

QUESTION ID
S52-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. 7% per year: its yearly factor 0.93 is less than $0.5^{1/10} \approx 0.933$.

GRADING NOTE Compare yearly factors: the smaller factor decays faster.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 52 • FORWARD 1 • Q5/5

SET 52 • HSF-IF.C.8.b

FORWARD 2

Find doubling times and half-lives with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the doubling time of $y = 100(1.08)^t$, in years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q1/5

1969

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

9.01

GRADING NOTE log 2/log 1.08 ≈ 9.006.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the half-life of $y = 500(0.97)^t$, in years to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q2/5

1970

QUESTION ID
S52-F2-Q2

TYPE
Short answer

CORRECT ANSWER
22.76

GRADING NOTE log 0.5/log 0.97 ≈ 22.757.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q2/5

True or false?

The doubling time of $y = 50(2)^{t/3}$ is 6 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q3/5

1971

QUESTION ID

S52-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The exponent t/3 increases by 1 every 3 hours, which doubles y.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q3/5

The doubling time of a quantity growing 6.17% per year is about

A. 16.2 years

B. 11.6 years

C. 6.17 years

D. 3.1 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q4/5

1972

QUESTION ID

S52-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 11.6 years

GRADING NOTE log 2/log 1.0617 ≈ 11.58.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 100(1.08)^t$ is rewritten as $y = 100 \cdot 2^{t/9.01}$. Why does the rewritten form show the doubling time?

A. Because 9.01 is the growth rate in percent per year, and doubling depends only on the rate.

B. Because 2 is the starting value, and the exponent t/9.01 counts how many times it is used.

C. Because y = 200 when t = 1, so y doubles from 100 to 200 in the first year and in each 9.01-year period.

D. The exponent t/9.01 increases by 1 every 9.01 years, and each such increase multiplies y by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q5/5

1973

QUESTION ID
S52-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. The exponent t/9.01 increases by 1 every 9.01 years, and each such increase multiplies y by 2.

GRADING NOTE Base 2 with exponent t/T doubles every T units. The others misread the parts (100 is the starting value; the rate is 8%).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 52 • FORWARD 2 • Q5/5

SET 53 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of Algebra II functions in different representations

MAIN	Algebra II main question	**HSF-IF.C.9**
BACKWARD 1	Compare linear, quadratic, and exponential functions	**HSF-IF.C.9**
BACKWARD 2	Recognize a pattern in a table	**8.F.A.3**
BACKWARD 3	Domains, asymptotes, and periods of Algebra II families	**HSF-IF.B.5**
FORWARD 1	Compare rates of change and long-run behavior across represent…	**HSF-IF.C.9**
FORWARD 2	Build a difference function to compare	**HSF-BF.A.1.b**

SET 53 • HSF-IF.C.9

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 3(1.1)^x$. The table shows the linear function g. Give the greater y-intercept, f(20) to the nearest hundredth, and g(20). A calculator may be used.

x	0	1	2	3	4
g(x)	10	15	20	25	30

Greater y-intercept: _____

f(20) = _____

g(20) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q1/5

1976

QUESTION ID

S53-M-Q1

TYPE

Short answer

CORRECT ANSWER

10; 20.18; 110

GRADING NOTE Form required: all three. f(0) = 3 and g(0) = 10. g is linear with slope 5, so g(x) = 10 + 5x and g(20) = 110; $f(20) = 3(1.1)^{20} \approx 20.18$, so g is greater at x = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

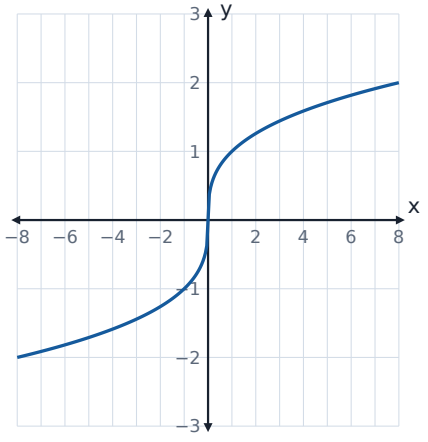
f(x) = √(x − 1). The graph of g is shown. Compare their domains.

A. f: x > 1; g: all real numbers

B. f: x ≥ 1; g: x ≥ 0

C. f: x ≥ 1; g: all real numbers

D. f: all real numbers; g: x ≥ 0



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q2/5

1977

QUESTION ID

S53-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. f: x ≥ 1; g: all real numbers

GRADING NOTE The radicand x − 1 must be nonnegative; the graph of g continues in both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

h(t) = 12 + 10sin(πt/15) gives one Ferris wheel's height (m, t in s). A second wheel has radius 8 m, its center 10 m high, and turns once every 40 s. Give the maximum height and the period of each wheel.

First wheel maximum (m): _____

First wheel period (s): _____

Second wheel maximum (m): _____

Second wheel period (s): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q3/5

1978

QUESTION ID

S53-M-Q3

TYPE

Short answer

CORRECT ANSWER

22; 30; 18; 40

GRADING NOTE Form required: all four. 12 + 10 = 22 and 2π/(π/15) = 30; 10 + 8 = 18. The first wheel is higher and faster.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \log_2 x$. The table shows g. Find f(9) to the nearest hundredth and g(9). A calculator may be used.

x	1	3	9	27
g(x)	0	1	2	3

f(9) = _____

g(9) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q4/5

1979

QUESTION ID

S53-M-Q4

TYPE

Short answer

CORRECT ANSWER

3.17; 2

GRADING NOTE Form required: both. $\log_2 9 \approx 3.170$, so f is greater at x = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S53-M-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

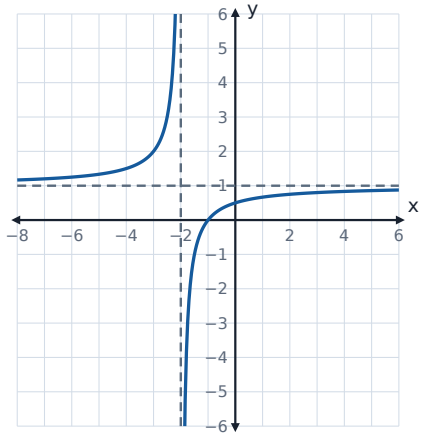
f(x) = (2x + 1)/(x − 3). The graph of g shows its asymptotes. Give the horizontal and vertical asymptotes of f and of g as equations.

f, horizontal: _____

f, vertical: _____

g, horizontal: _____

g, vertical: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q5/5

1980

QUESTION ID

S53-M-Q5

TYPE

Short answer

CORRECT ANSWER

y = 2; x = 3; y = 1; x = −2

GRADING NOTE Form required: all four. For f, the ratio of leading coefficients is 2 and the denominator is 0 at x = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • MAIN • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 1

Compare linear, quadratic, and exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = 2x + 1, and g has the table g(0) = 3, g(1) = 4. Which function has the greater y-intercept?

A. f, with y-intercept 2

B. f, with y-intercept 1

C. g, with y-intercept 3

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q1/5

1982

GRADING NOTE f(0) = 1 and g(0) = 3.

QUESTION ID

S53-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. g, with y-intercept 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = 2^x$. Find f(10) and g(10).

f(10) = _____

g(10) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q2/5

1983

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

100; 1024

GRADING NOTE Form required: both. g is greater at x = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q2/5

True or false?

An increasing exponential eventually exceeds any linear function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q3/5

1984

QUESTION ID

S53-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Algebra I result.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = -(x - 1)^2 + 5$ and g with maximum 4 in a graph. The greater maximum is

A. f: 5

B. cannot tell

C. equal

D. g: 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q4/5

1985

QUESTION ID

S53-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. f: 5

GRADING NOTE Vertex form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A table's outputs increase by 5 at each equal step of the input. What kind of function fits it?

A. Exponential

B. Linear

C. Quadratic

D. Logarithmic

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q5/5

1986

QUESTION ID

S53-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Linear

GRADING NOTE Equal differences over equal steps mean a constant rate of change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 2

Recognize a pattern in a table • Grade 8 • Functions • 8.F.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Is the table x: 0, 1, 2 and y: 10, 15, 20 linear?

A. No: it starts at 10, not 0, and a linear table must start at 0

B. Yes: every table of numbers can be joined by straight segments, so every table is linear

C. No: the outputs 10, 15, 20 are not all multiples of 10, so they are not linear

D. Yes: the outputs have a constant difference of 5 for equal steps in x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q1/5

1988

QUESTION ID
S53-B2-Q1

TYPE
Multiple choice

CORRECT ANSWER
D. Yes: the outputs have a constant difference of 5 for equal steps in x

GRADING NOTE Constant difference over equal input steps.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is the table x: 1, 3, 9, 27 and y: 0, 1, 2, 3 linear?

A. No: each output step of 1 needs the input multiplied by 3, so it is logarithmic ($y = \log_3 x$)

B. Yes: the inputs increase each time, and any table with increasing inputs is linear

C. No: it is exponential, because each input is 3 times the one before it ($y = 3^x$)

D. Yes: the outputs increase by 1 each time, so the table has a constant rate of change

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q2/5

1989

QUESTION ID

S53-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. No: each output step of 1 needs the input multiplied by 3, so it is logarithmic ($y = \log_3 x$)

GRADING NOTE The inputs are not equally spaced; equal output steps for multiplied inputs is logarithmic.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A table with constant ratios of outputs is exponential.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q3/5

1990

QUESTION ID

S53-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Repeated multiplication.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q3

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which table is exponential?

A. 1, 4, 9, 16

B. 1, 2, 4, 8

C. 2, 2, 2, 2

D. 1, 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q4/5

1991

GRADING NOTE Ratio 2.

QUESTION ID

S53-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1, 2, 4, 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q4

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the slope of a linear table with outputs 10, 15, 20 at x = 0, 1, 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q5/5

1992

QUESTION ID

S53-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 8 • Functions • 8.F.A.3 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • HSF-IF.C.9

BACKWARD 3

Domains, asymptotes, and periods of Algebra II families • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain of $\sqrt{(x - 1)}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q1/5

1994

QUESTION ID

S53-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 1$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the vertical and horizontal asymptotes of $y = (2x + 1)/(x - 3)$ as equations.

Vertical asymptote: _____

Horizontal asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q2/5

1995

QUESTION ID

S53-B3-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3; y = 2

GRADING NOTE Form required: both equations. The denominator is 0 at x = 3; the ratio of leading coefficients is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The period of $12 + 10\sin(\pi t/15)$ is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q3/5

1996

QUESTION ID

S53-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2\pi/(\pi/15) = 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $\log_2 x$ is

A. all reals

B. $x \geq 0$

C. $x > 0$

D. $x \neq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q4/5

1997

QUESTION ID

S53-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x > 0

GRADING NOTE Positive inputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A wheel of radius 8 m with center 10 m high has what maximum height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q5/5

1998

QUESTION ID

S53-B3-Q5

TYPE

Short answer

CORRECT ANSWER

18 m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • HSF-IF.C.9

FORWARD 1

Compare rates of change and long-run behavior across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 3(1.1)^x$ and $g(x) = 10 + 5x$. Estimate the first whole number x at which f exceeds g. A calculator may be used. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q1/5

2000

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

46

GRADING NOTE Any equivalent form. f(45) ≈ 218.7 < g(45) = 235; f(46) ≈ 240.6 > g(46) = 240.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \log_2 x$, and g has g(1) = 0, g(3) = 1, g(9) = 2. Find the average rates of change of f and of g on $1 \le x \le 9$, each to the nearest hundredth. A calculator may be used.

Rate of f: _____

Rate of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q2/5

2001

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.40; 0.25

GRADING NOTE Form required: both. f: $(\log_2 9 - 0)/8 \approx 3.170/8 \approx 0.396$; g: $2/8 = 0.25$. f has the greater rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A function with a larger y-intercept is always larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q3/5

2002

QUESTION ID

S53-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE g starts higher but f eventually passes it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which grows fastest for large x?

A. $\sqrt{(x - 1)}$

B. $3(1.1)^x$

C. $5x + 10$

D. $\log_2 x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q4/5

2003

QUESTION ID

S53-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $3(1.1)^x$

GRADING NOTE Exponential.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why does $g(x) = \log_3 x$ stay below $f(x) = \log_2 x$ for $x > 1$?

A. Because 3 > 2 makes every value of $\log_3 x$ larger, so g stays below f only for $0 < x < 1$.

B. Because both equal 0 at $x = 1$, and two graphs that share a point never cross after it.

C. Because $\log_3 x$ is negative for $x > 1$ while $\log_2 x$ is positive there, so g is below f.

D. $\log_3 x = \log_2 x / \log_2 3$, and $\log_2 3 > 1$, so for $x > 1$ (where $\log_2 x > 0$), g is a smaller positive multiple of f.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q5/5

2004

QUESTION ID
S53-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. $\log_3 x = \log_2 x / \log_2 3$, and $\log_2 3 > 1$, so for $x > 1$ (where $\log_2 x > 0$), g is a smaller positive multiple of f.

GRADING NOTE Change of base. The others reverse the comparison, use a single point, or state a false fact.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 53 • FORWARD 1 • Q5/5

2004

SET 53 • HSF-IF.C.9

FORWARD 2

Build a difference function to compare • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Write $d(x) = f(x) - g(x)$ for $f(x) = 3(1.1)^x$ and $g(x) = 5x + 10$.

$d(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q1/5

2006

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$3(1.1)^x - 5x - 10$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q1/5

2006

ANSWER KEY Reduced copy of the question page, with its answer

$d(x) = f(x) - g(x)$ is the difference of two functions. What does the sign of $d(x)$ tell you?

A. $d(x) > 0$ means f is increasing and g is decreasing at x.

B. $d(x) > 0$ means $f(x) > g(x)$; $d(x) < 0$ means $g(x) > f(x)$; $d(x) = 0$ means they are equal.

C. $d(x) > 0$ means both functions are positive at x; $d(x) < 0$ means both are negative.

D. The sign of $d(x)$ gives the slope of f: positive means f rises and negative means f falls.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q2/5

2007

QUESTION ID
S53-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
B. $d(x) > 0$ means $f(x) > g(x)$; $d(x) < 0$ means $g(x) > f(x)$; $d(x) = 0$ means they are equal.

GRADING NOTE $f(x) - g(x) > 0$ exactly when $f(x) > g(x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A zero of d(x) = f(x) − g(x) is an x where f(x) = g(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q3/5

2008

QUESTION ID
S53-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Equal outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$, $h_1(0) - h_2(0)$ is

A. 2

B. 22

C. −2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q4/5

2009

QUESTION ID

S53-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2

GRADING NOTE 12 − 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q4/5

2009

ANSWER KEY Reduced copy of the question page, with its answer

Two Ferris wheel heights are $h_1(t) = 12 + 10\sin(\pi t/15)$ and $h_2(t) = 10 + 8\sin(\pi t/20)$. Which is $h_1(t) - h_2(t)$?

A. $2 + 10\sin(\pi t/15) + 8\sin(\pi t/20)$

B. $2 + 2\sin(\pi t/60)$

C. $2 + 10\sin(\pi t/15) - 8\sin(\pi t/20)$

D. $22 + 10\sin(\pi t/15) + 8\sin(\pi t/20)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q5/5

2010

QUESTION ID

S53-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $2 + 10\sin(\pi t/15) - 8\sin(\pi t/20)$

GRADING NOTE Subtract term by term; sines with different periods do not combine.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 53 • FORWARD 2 • Q5/5

2010

Build models by combining Algebra II function types

MAIN	Algebra II main question	**HSF-BF.A.1.b**
BACKWARD 1	Combine linear and quadratic functions	**HSF-BF.A.1.b**
BACKWARD 2	Multiply and subtract polynomials	**HSA-APR.A.1**
BACKWARD 3	Evaluate exponential and trigonometric expressions	**HSF-IF.A.2**
FORWARD 1	Interpret the parts of a combined model	**HSF-BF.A.1.b**
FORWARD 2	Compose functions in context	**HSF-BF.A.1.c**

SET 54 • HSF-BF.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Revenue is $R(x) = x(50 - 0.5x)$ and cost is $C(x) = 200 + 10x$ dollars for x items. Write the profit $P(x) = R(x) - C(x)$ with its terms multiplied out, and find $P(30)$.

$P(x) =$ _____

$P(30) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q1/5

2013

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-0.5x^2 + 40x - 200$; 550

GRADING NOTE Form required: both. $50x - 0.5x^2 - 200 - 10x$; $P(30) = -450 + 1200 - 200 = 550$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The cost of x posters is C(x) = 1200 + 3x dollars. Write the average cost per poster A(x) = C(x)/x and find A(100).

A(x) = _____

A(100) ($) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q2/5

2014

QUESTION ID

S54-M-Q2

TYPE

Short answer

CORRECT ANSWER

(1200 + 3x)/x; 15

GRADING NOTE Form required: both. A(100) = 1500/100 = 15 dollars per poster.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A town has $P(t) = 5000(1.02)^t$ people and $F(t) = 6000 + 100t$ units of food after t years. Write the food per person $F(t)/P(t)$, and find it at $t = 10$ to the nearest hundredth. A calculator may be used.

Food per person = _____

At t = 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q3/5

2015

QUESTION ID
S54-M-Q3

TYPE
Short answer

CORRECT ANSWER
$(6000 + 100t)/(5000(1.02)^t)$; 1.15

GRADING NOTE Form required: both. $7000/(5000 \cdot 1.02^{10}) \approx 7000/6094.97 \approx 1.148$ units per person.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A spring starts at displacement +10 cm at t = 0. Its amplitude envelope is 10 cm and shrinks by 10% each second, and it oscillates once per second, so $y(t) = A(t)\cos(2\pi t)$. Write the envelope A(t) and find y(1).

A(t) = _____

y(1) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q4/5

2016

QUESTION ID

S54-M-Q4

TYPE

Short answer

CORRECT ANSWER

$10(0.9)^t$; 9

GRADING NOTE Form required: both. $A(t) = 10(0.9)^t$, so $y(t) = 10(0.9)^t \cos(2\pi t)$ ($y(0) = 10$), and $y(1) = 10 \cdot 0.9 \cdot \cos 2\pi = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Coffee starts 70°C above a 20°C room, and the difference shrinks 5% each minute. Write its temperature T(t) after t minutes as a sum of a constant and an exponential, and find T(10) to the nearest hundredth. A calculator may be used.

T(t) = _____

T(10) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q5/5

2017

QUESTION ID

S54-M-Q5

TYPE

Short answer

CORRECT ANSWER

$20 + 70(0.95)^t$; 61.91

GRADING NOTE Form required: both. $70 \cdot 0.95^{10} \approx 41.91$, so T(10) ≈ 61.91°C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • MAIN • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 1

Combine linear and quadratic functions • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

R(x) = 8x and C(x) = 3x + 50. Write P = R − C.

P(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q1/5

2019

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5x − 50

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = x + 2 and g(x) = x − 2. Write f · g.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q2/5

2020

QUESTION ID

S54-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 - 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Profit is revenue minus cost.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q3/5

2021

QUESTION ID

S54-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Average cost is

A. total cost times the number of items

B. total cost divided by the number of items

C. total cost minus total revenue

D. total revenue divided by total cost

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q4/5

2022

QUESTION ID
S54-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
B. total cost divided by the number of items

GRADING NOTE Per item.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q4/5

2022

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x$ and $g(x) = x^2 + 1$. Find $(f + g)(3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q5/5

2023

QUESTION ID

S54-B1-Q5

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • BACKWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 2

Multiply and subtract polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Expand x(50 − 0.5x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q1/5

2025

QUESTION ID

S54-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$50x - 0.5x^2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B2-Q1**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q1/5

2025

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(50x - 0.5x^2) - (200 + 10x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q2/5

2026

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$-0.5x^2 + 40x - 200$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B2-Q2**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q2/5

2026

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$-(200 + 10x) = -200 + 10x.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q3/5

2027

QUESTION ID

S54-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The minus sign applies to both terms: $-200 - 10x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

(3x + 1) − (x − 4) equals

A. 2x + 5

B. 4x − 3

C. 2x − 3

D. 2x + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q4/5

2028

QUESTION ID

S54-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2x + 5

GRADING NOTE Subtract each term.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Expand $2(x^2 - 3x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q5/5

2029

QUESTION ID
S54-B2-Q5

TYPE
Short answer

CORRECT ANSWER
$2x^2 - 6x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-B2-Q5**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • HSF-BF.A.1.b

BACKWARD 3

Evaluate exponential and trigonometric expressions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $70(0.95)^{10}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q1/5

2031

QUESTION ID

S54-B3-Q1

TYPE

Short answer

CORRECT ANSWER

41.91

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $5000(1.02)^{10}$ to the nearest whole number. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q2/5

2032

QUESTION ID
S54-B3-Q2

TYPE
Short answer

CORRECT ANSWER
6095

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\cos(2\pi) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q3/5

2033

QUESTION ID

S54-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Full turn.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q3/5

2033

ANSWER KEY Reduced copy of the question page, with its answer

$10(0.9)^2$ equals

A. 9

B. 18

C. 81

D. 8.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q4/5

2034

QUESTION ID

S54-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 8.1

GRADING NOTE $10 \cdot 0.81$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\cos(2\pi \cdot 0.5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q5/5

2035

QUESTION ID

S54-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 54 • BACKWARD 3 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 1

Interpret the parts of a combined model • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b

Coffee in a 20°C room has temperature $T(t) = 20 + 70(0.95)^t$ °C after t minutes. What do 20 and $70(0.95)^t$ represent?

A. 20 is the room temperature; $70(0.95)^t$ is how far the coffee is above room temperature, which shrinks 5% each minute.

B. 20 is the starting temperature of the coffee; $70(0.95)^t$ is the room temperature, which falls 5% each minute.

C. 20 is the number of minutes the coffee cools; $70(0.95)^t$ is the temperature of the coffee at time t.

D. 20 is the room temperature; $70(0.95)^t$ is the drop in the coffee's temperature during each minute.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q1/5

2037

QUESTION ID
S54-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
A. 20 is the room temperature; $70(0.95)^t$ is how far the coffee is above room temperature, which shrinks 5% each minute.

GRADING NOTE T approaches 20 as the second term decays. The others swap the parts (T(0) = 90), misread 20, or confuse the remaining difference with the drop.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q1/5

A spring's position is $y(t) = 10(0.9)^t \cos(2\pi t)$. Which factor controls the size of the swings and which controls their timing?

A. $10(0.9)^t$ controls the size (it shrinks 10% each second); $\cos(2\pi t)$ controls the timing (one swing per second).

B. $\cos(2\pi t)$ controls the size (it shrinks each second); $10(0.9)^t$ controls the timing (one swing per 0.9 seconds).

C. Both factors control only the size of the swings; the timing is set by the 10 in front.

D. 10 controls the timing (10 swings per second); 0.9 controls the size (it shrinks the swings).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q2/5

2038

QUESTION ID
S54-F1-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. $10(0.9)^t$ controls the size (it shrinks 10% each second); $\cos(2\pi t)$ controls the timing (one swing per second).

GRADING NOTE The exponential factor is the amplitude envelope; the cosine repeats with period 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q2/5

True or false?

In A(x) = (1200 + 3x)/x = 1200/x + 3, the term 1200/x is the fixed cost spread over x posters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q3/5

2039

QUESTION ID

S54-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Split the fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

As t grows, $T(t) = 20 + 70(0.95)^t$ approaches

A. 20

B. 0

C. 90

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q4/5

2040

QUESTION ID

S54-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 20

GRADING NOTE The exponential part → 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A town has $P(t) = 5000(1.02)^t$ people and $F(t) = 6000 + 100t$ units of food after t years. Why does the food per person $F(t)/P(t)$ eventually decrease?

A. $F(t)/P(t)$ is constant because both F and P increase every year, so their ratio does not change as time passes.

B. $P(t) = 5000(1.02)^t$ decreases over time, so there are fewer people for the same food.

C. $F(t) = 6000 + 100t$ decreases over time, so there is less food for the same number of people.

D. P grows exponentially (factor 1.02 each year) while F grows linearly, so P eventually grows much faster.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q5/5

2041

QUESTION ID
S54-F1-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. P grows exponentially (factor 1.02 each year) while F grows linearly, so P eventually grows much faster.

GRADING NOTE Exponential growth with a factor greater than 1 eventually outpaces any linear growth. The others contradict the formulas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.A.1.b • Set 54 • FORWARD 1 • Q5/5

SET 54 • HSF-BF.A.1.b

FORWARD 2

Compose functions in context • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

A balloon's radius is r(t) = 2t cm after t seconds and its volume is V(r) = (4/3)πr³. Write V(r(t)).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q1/5

2043

QUESTION ID
S54-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(32/3)πt³

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A store takes 20% off with d(p) = 0.8p and then adds a $5 fee with f(p) = p + 5. Write f(d(p)) and find it for p = 50.

f(d(p)) = _____

f(d(50)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q2/5

2044

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.8p + 5; 45

GRADING NOTE Form required: both. f(d(p)) = 0.8p + 5; 0.8 · 50 + 5 = 45 dollars.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q2**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q2/5

True or false?

f(g(x)) and g(f(x)) are always equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q3/5

2045

QUESTION ID

S54-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Order matters.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(x) = x^2$ and $g(x) = x + 1$, then $f(g(2)) =$

A. 4

B. 9

C. 3

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q4/5

2046

QUESTION ID

S54-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 9

GRADING NOTE f(3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q4/5

How does composing two functions differ from multiplying them?

A. Composition f(g(x)) adds the two outputs f(x) + g(x); multiplication f(x) · g(x) multiplies them.

B. Composition f(g(x)) uses the output of g as the input of f; multiplication f(x) · g(x) multiplies the two outputs.

C. Composition f(g(x)) multiplies the inputs of f and g; multiplication f(x) · g(x) multiplies the outputs.

D. They are the same operation written two ways: f(g(x)) and f(x) · g(x) are always equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q5/5

2047

QUESTION ID
S54-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. Composition f(g(x)) uses the output of g as the input of f; multiplication f(x) · g(x) multiplies the two outputs.

GRADING NOTE For f(x) = x + 1 and g(x) = 2x: f(g(x)) = 2x + 1, but f(x)g(x) = 2x² + 2x. The others say composition adds outputs or multiplies inputs, or claim the two operations are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.A.1.c • Set 54 • FORWARD 2 • Q5/5

Identify vertical shifts, vertical stretches, and horizontal shifts of Algebra II functions

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Shifts and stretches of linear and quadratic graphs	**HSF-BF.B.3**
BACKWARD 2	Translate points in the plane	**8.G.A.3**
BACKWARD 3	Key points of parent functions	**HSF-IF.C.7.e**
FORWARD 1	Write transformed functions from descriptions and order of transfo..	**HSF-BF.B.3**
FORWARD 2	Read sinusoid parameters as transformations	**HSF-TF.B.5**

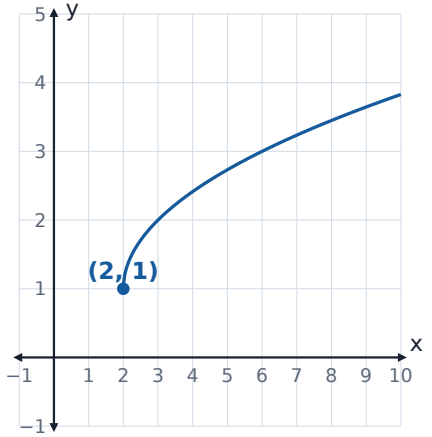
SET 55 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \sqrt{x}$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.



h = _____

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q1/5

2050

QUESTION ID

S55-M-Q1

TYPE

Short answer

CORRECT ANSWER

2; 1

GRADING NOTE Form required: both. The starting point moves from (0, 0) to (2, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

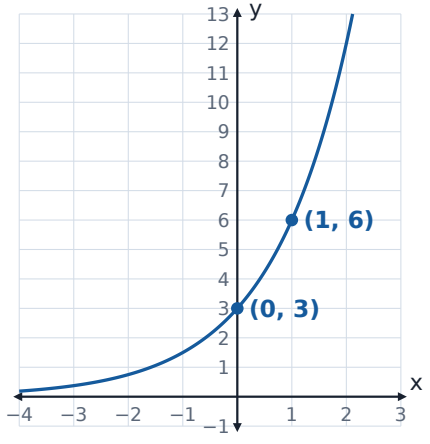
High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q1/5

2050

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2^x$. The graph of $g(x) = a \cdot f(x)$ is shown. Find a.

a = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q2/5

2051

QUESTION ID
S55-M-Q2

TYPE
Short answer

CORRECT ANSWER
3

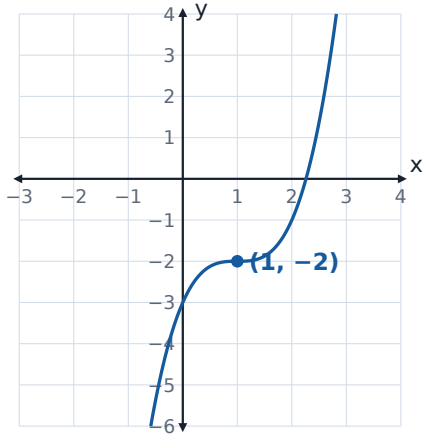
GRADING NOTE Any equivalent form. g(0) = a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^3$. The graph of $g(x) = f(x - h) + k$ is shown. Find h and k.



h = _____

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q3/5

2052

QUESTION ID

S55-M-Q3

TYPE

Short answer

CORRECT ANSWER

1; −2

GRADING NOTE Form required: both. The center moves from (0, 0) to (1, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q3/5

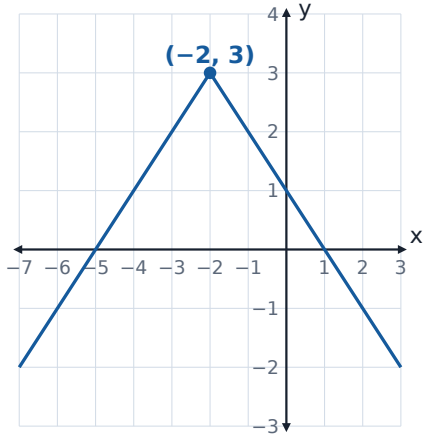
ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = |x|$. The graph of $g(x) = a \cdot f(x - h) + k$ is shown. Find a, h, and k.

a = _____

h = _____

k = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q4/5

2053

QUESTION ID

S55-M-Q4

TYPE

Short answer

CORRECT ANSWER

−1; −2; 3

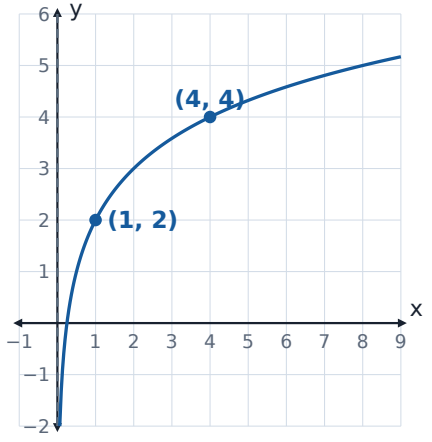
GRADING NOTE Form required: all three. The vertex is (−2, 3) and the slopes are ±1 with the V opening down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = \log_2 x$. The graph of $g(x) = f(x) + k$ is shown. Find k.



k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q5/5

2054

QUESTION ID
S55-M-Q5

TYPE
Short answer

CORRECT ANSWER
2

GRADING NOTE Any equivalent form. f(1) = 0 becomes 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • MAIN • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 1

Shifts and stretches of linear and quadratic graphs • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

How is the graph of $y = (x - 3)^2 + 1$ obtained from the graph of $y = x^2$?

A. Shift left 3 and up 1

B. Shift up 3 and right 1

C. Shift right 3 and up 1

D. Shift right 3 and down 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q1/5

2056

QUESTION ID

S55-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Shift right 3 and up 1

GRADING NOTE $x - 3$ shifts right 3; $+ 1$ shifts up 1. The vertex moves from (0, 0) to (3, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q1/5

How is the graph of $y = 2x^2$ obtained from the graph of $y = x^2$?

A. Shift up 2

B. Vertical stretch by 2

C. Vertical compression by 1/2

D. Horizontal stretch by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q2/5

2057

QUESTION ID

S55-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Vertical stretch by 2

GRADING NOTE Every output is doubled.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

y = f(x + 4) is the graph of f shifted right 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q3/5

2058

QUESTION ID

S55-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Left 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The vertex of $y = (x + 2)^2 - 5$ is

A. (2, 5)

B. (2, −5)

C. (−2, −5)

D. (−2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q4/5

2059

QUESTION ID

S55-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (−2, −5)

GRADING NOTE h = −2, k = −5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q4/5

2059

ANSWER KEY Reduced copy of the question page, with its answer

Write $y = x^2$ shifted left 1 and down 4.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q5/5

2060

QUESTION ID

S55-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(x + 1)^2 - 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 55 • BACKWARD 1 • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 2

Translate points in the plane • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Translate (0, 0) right 2 and up 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q1/5

2062

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(2, 1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Translate (1, 1) left 2 and up 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q2/5

2063

QUESTION ID

S55-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, 4)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q2/5

True or false?

Translating (0, 0) down 2 gives (−2, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q3/5

2064

QUESTION ID

S55-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Down 2 changes y: (0, −2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q3/5

$(x, y) \to (x - 2, y + 3)$ moves $(0, 0)$ to

A. $(-2, 3)$

B. $(-2, -3)$

C. $(2, 3)$

D. $(3, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q4/5

2065

QUESTION ID

S55-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (−2, 3)

GRADING NOTE Apply the rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which translation takes the point (0, 0) to (1, −2)?

A. Up 1 and left 2

B. Right 2 and down 1

C. Right 1 and down 2

D. Left 1 and up 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q5/5

2066

QUESTION ID

S55-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Right 1 and down 2

GRADING NOTE The x-coordinate increases by 1 and the y-coordinate decreases by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 55 • BACKWARD 2 • Q5/5

SET 55 • HSF-BF.B.3

BACKWARD 3

Key points of parent functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Give the point of $y = \sqrt{x}$ at $x = 0$ and at $x = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q1/5

2068

QUESTION ID

S55-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(0, 0) and (1, 1)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the y-intercept of $y = 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q2/5

2069

QUESTION ID

S55-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q2/5

True or false?

The graph of $y = \log_2 x$ passes through (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q3/5

2070

QUESTION ID

S55-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\log_2 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The vertex of y = |x| is

A. (1, 0)

B. (1, 1)

C. (0, 1)

D. (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q4/5

2071

QUESTION ID

S55-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 0)

GRADING NOTE Parent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the center (point of symmetry) of $y = x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q5/5

2072

QUESTION ID
S55-B3-Q5

TYPE
Short answer

CORRECT ANSWER
(0, 0)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 55 • BACKWARD 3 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 1

Write transformed functions from descriptions and order of transformations • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Write g, the graph of $f(x) = 2^x$ stretched vertically by 3, then shifted down 4.

$g(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q1/5

2074

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$3 \cdot 2^x - 4$

GRADING NOTE Any equivalent expression. Stretch: $3 \cdot 2^x$; then shift: $3 \cdot 2^x - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q1/5

Which is g, the graph of f(x) = √x shifted down 4 and then stretched vertically by 3?

A. g(x) = 3√(x − 4)

B. g(x) = 3√x − 4

C. g(x) = √(3x) − 4

D. g(x) = 3√x − 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q2/5

2075

QUESTION ID

S55-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. g(x) = 3√x − 12

GRADING NOTE Shift: √x − 4; then stretch: 3(√x − 4) = 3√x − 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q2/5

True or false?

A vertical stretch then a shift can give a different graph than the same shift then the stretch.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q3/5

2076

QUESTION ID
S55-F1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE See the two items above.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 2f(x) + 1$ moves the point (3, 4) of f to

A. (6, 5)

B. (4, 9)

C. (3, 9)

D. (3, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q4/5

2077

QUESTION ID

S55-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (3, 9)

GRADING NOTE $2 \cdot 4 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = f(x + 2) - 3$ and f has asymptote $x = 0$. Where is g's vertical asymptote?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q5/5

2078

QUESTION ID
S55-F1-Q5

TYPE
Short answer

CORRECT ANSWER
x = −2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 55 • FORWARD 1 • Q5/5

SET 55 • HSF-BF.B.3

FORWARD 2

Read sinusoid parameters as transformations • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

$y = 3\sin x + 2$ is a transformation of $y = \sin x$. Give the vertical stretch factor, the upward shift, and the maximum value.

Vertical stretch factor: _____

Upward shift: _____

Maximum value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q1/5

2080

QUESTION ID

S55-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3; 2; 5

GRADING NOTE Form required: all three. The maximum of sin x is 1, so the maximum is $3 \cdot 1 + 2 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q1/5

2080

ANSWER KEY Reduced copy of the question page, with its answer

The graph of y = cos x is stretched vertically by 4 and then shifted down 1. Write the result in the form y = A·cos x + D.

A = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q2/5

2081

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4; −1

GRADING NOTE Form required: both. Stretch: 4cos x; then shift: 4cos x − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In $y = A \sin x + D$, A is the midline.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q3/5

2082

QUESTION ID

S55-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE D gives the midline y = D; |A| is the amplitude.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = -2\cos x + 5$ has amplitude and midline

A. 5 and $y = 2$

B. -2 and $y = 5$

C. 2 and $y = -5$

D. 2 and $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q4/5

2083

QUESTION ID

S55-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2 and $y = 5$

GRADING NOTE |A|.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A model is y = A sin x + D with maximum 9 and minimum 1. Find A (A > 0) and D.

A = _____

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q5/5

2084

QUESTION ID

S55-F2-Q5

TYPE

Short answer

CORRECT ANSWER

4; 5

GRADING NOTE Form required: both. A = (9 − 1)/2 and D = (9 + 1)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 55 • FORWARD 2 • Q5/5

Horizontal stretches, compressions, and reflections f(kx)

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Reflections and dilations of points	**8.G.A.3**
BACKWARD 2	Solve kx = c for x	**6.EE.B.7**
BACKWARD 3	Period of y = sin(Bx)	**HSF-TF.B.5**
FORWARD 1	Combine horizontal stretches with shifts	**HSF-BF.B.3**
FORWARD 2	Graph exponential and trigonometric functions with horizontal cha…	**HSF-IF.C.7.e**

SET 56 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = √x passes through (4, 2). For g(x) = √(4x), by what number are the x-coordinates of points of f multiplied, and which point of g has y = 2?

x-coordinates are multiplied by: _____

Point of g with y = 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q1/5

2087

QUESTION ID

S56-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/4; (1, 2)

GRADING NOTE Form required: both. g is a horizontal compression of f by a factor of 1/4: √(4 · 1) = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = sin x has period 2π. For g(x) = sin(3x), by what number are the x-coordinates of points of f multiplied, and what is the period of g?

x-coordinates are multiplied by: _____

Period of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q2/5

2088

QUESTION ID

S56-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/3; 2π/3

GRADING NOTE Form required: both (the period exactly, in terms of π). A horizontal compression by a factor of 1/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2^x$ passes through (1, 2). For $g(x) = 2^{-x}$, by what number are the x-coordinates of points of f multiplied, and which point of g comes from (1, 2)?

x-coordinates are multiplied by: _____

Point of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q3/5

2089

QUESTION ID
S56-M-Q3

TYPE
Short answer

CORRECT ANSWER
−1; (−1, 2)

GRADING NOTE Form required: both. g is the reflection of f over the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

f passes through (−2, 1), (0, 3), and (4, −1). List the corresponding points of g(x) = f(2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q4/5

2090

QUESTION ID
S56-M-Q4

TYPE
Short answer

CORRECT ANSWER
(−1, 1), (0, 3), (2, −1)

GRADING NOTE Form required: all three.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

f passes through (2, 5). For g(x) = f(−x/2), by what number are the x-coordinates of points of f multiplied, and which point must be on the graph of g?

x-coordinates are multiplied by: _____

Point of g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q5/5

2091

QUESTION ID

S56-M-Q5

TYPE

Short answer

CORRECT ANSWER

−2; (−4, 5)

GRADING NOTE Form required: both. g(−4) = f(2) = 5: a horizontal stretch by 2 and a reflection over the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • MAIN • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 1

Reflections and dilations of points • Grade 8 • Geometry • 8.G.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Reflect (1, 2) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q1/5

2093

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(−1, 2)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply the x-coordinate of (4, −1) by 1/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q2/5

2094

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −1)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q2/5

True or false?

Reflecting over the y-axis changes the sign of the x-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S56-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q3/5

2095

QUESTION ID

S56-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x, y) → (−x, y).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S56-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

(x, y) → (x/3, y) applied to (6, 2) gives

A. (18, 2)

B. (6, 2/3)

C. (2, 2)

D. (3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q4/5

2096

QUESTION ID

S56-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 2)

GRADING NOTE Divide x by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Apply (x, y) → (−2x, y) to (2, 5).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q5/5

2097

QUESTION ID

S56-B1-Q5

TYPE

Short answer

CORRECT ANSWER

(−4, 5)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 2

Solve kx = c for x • Grade 6 • Expressions & Equations • 6.EE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x = 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q1/5

2099

QUESTION ID

S56-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-x/2 = 2$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q2/5

2100

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If 4x = 4, then x = 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q3/5

2101

QUESTION ID

S56-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Divide by 4: x = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3x = 2\pi$ gives $x =$

A. $2\pi/3$

B. $2\pi - 3$

C. $\pi/3$

D. 6π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q4/5

2102

QUESTION ID

S56-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2π/3

GRADING NOTE Divide by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-x = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q5/5

2103

QUESTION ID

S56-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • HSF-BF.B.3

BACKWARD 3

Period of y = sin(Bx) • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the period of y = sin(2x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q1/5

2105

QUESTION ID

S56-B3-Q1

TYPE

Short answer

CORRECT ANSWER

π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the period of y = cos(x/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q2/5

2106

QUESTION ID

S56-B3-Q2

TYPE

Short answer

CORRECT ANSWER

4π

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q2/5

True or false?

For y = sin(Bx) with B > 0, a larger B gives a longer period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q3/5

2107

QUESTION ID

S56-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The period 2π/B gets shorter as B grows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

y = sin(4x) has period

A. 8π

B. 4π

C. π/4

D. π/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q4/5

2108

QUESTION ID

S56-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. π/2

GRADING NOTE 2π/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q4/5

2108

ANSWER KEY Reduced copy of the question page, with its answer

For y = sin(Bx) with B > 0, which B gives period 6? Give B exactly, in terms of π.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q5/5

2109

QUESTION ID

S56-B3-Q5

TYPE

Short answer

CORRECT ANSWER

π/3

GRADING NOTE Form required: the exact value. The period is 2π/B = 6, so B = π/3. (Without B > 0, −π/3 would also give period 6.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 56 • BACKWARD 3 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 1

Combine horizontal stretches with shifts • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite g(x) = f(2x − 6) in the form f(k(x − h)). Give k and h.

k = _____

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q1/5

2111

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2; 3

GRADING NOTE Form required: both. 2x − 6 = 2(x − 3): compress horizontally by a factor of 1/2, then shift right 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q1/5

f passes through (4, 1). Which point is on $g(x) = f(2x - 6)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q2/5

2112

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 1)

GRADING NOTE Any equivalent answer. $2x - 6 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

f(2x − 6) is f(2x) shifted right 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q3/5

2113

QUESTION ID

S56-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is shifted right 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

y = √(9x) equals

A. √x + 9

B. 9√x

C. 3√x

D. √x/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q4/5

2114

QUESTION ID
S56-F1-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 3√x

GRADING NOTE Horizontal compression equals vertical stretch for √.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write g, the graph of $f(x) = 2^x$ reflected over the y-axis and then shifted right 1.

g(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q5/5

2115

QUESTION ID

S56-F1-Q5

TYPE

Short answer

CORRECT ANSWER

2^{1-x}

GRADING NOTE Any equivalent expression. Reflect: 2^{-x}; shift right 1: $2^{-(x-1)} = 2^{1-x}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F1-Q5**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 56 • FORWARD 1 • Q5/5

SET 56 • HSF-BF.B.3

FORWARD 2

Graph exponential and trigonometric functions with horizontal changes • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

For $y = 2^{-x} + 1$, give the y-intercept, the horizontal asymptote, and the y-values at $x = -1$ and at $x = 1$.

y-intercept: _____

Asymptote: $y =$ _____

y at $x = -1$: _____

y at $x = 1$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q1/5

2117

QUESTION ID
S56-F2-Q1

TYPE
Short answer

CORRECT ANSWER
2; 1; 3; 1.5

GRADING NOTE Form required: all four. $2^0 + 1 = 2$; $2^{-x} \to 0$ as x grows; $2^1 + 1 = 3$; $2^{-1} + 1 = 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

On $0 \le x \le 2\pi$, at which x-values does $y = \cos(2x)$ reach its maximum? List them separated by commas, exactly in terms of π.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q2/5

2118

QUESTION ID
S56-F2-Q2

TYPE
Short answer

CORRECT ANSWER
{0, π, 2π}

GRADING NOTE Form required: all three. cos(2x) = 1 when 2x = 0, 2π, 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = 3^{2x} = 9^{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q3/5

2119

QUESTION ID
S56-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Power of a power.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = \sin(x/2)$ completes one cycle on

A. $0 \le x \le \pi$

B. $0 \le x \le 2\pi$

C. $0 \le x \le \pi/2$

D. $0 \le x \le 4\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q4/5

2120

QUESTION ID
S56-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. $0 \le x \le 4\pi$

GRADING NOTE Period 4π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q4/5

Why is $y = 2^{2x}$ both a horizontal compression of $y = 2^x$ and an exponential function with base 4?

A. Because $2^{2x} = 2^x + 2^x$, which doubles each output, and doubling 2 gives the base 4.

B. Replacing x by 2x compresses the graph horizontally by a factor of 1/2, and $2^{2x} = (2^2)^x = 4^x$.

C. Because $2^{2x} = 2 \cdot 2^x$, a horizontal compression by 2, and $2 \cdot 2$ gives the base 4.

D. Replacing x by 2x stretches the graph vertically by a factor of 2, so the base 2 becomes $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q5/5

2121

QUESTION ID
S56-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. Replacing x by 2x compresses the graph horizontally by a factor of 1/2, and $2^{2x} = (2^2)^x = 4^x$.

GRADING NOTE Power of a power: $(2^2)^x = 4^x$. The others misuse exponent rules ($2 \cdot 2^x = 2^{x+1}$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 56 • FORWARD 2 • Q5/5

Recognize even and odd functions

MAIN	Algebra II main question	**HSF-BF.B.3**
BACKWARD 1	Signs of powers of negatives	**7.NS.A.2**
BACKWARD 2	Reflect points over the axes and the origin	**6.NS.C.6.b**
BACKWARD 3	Substitute -x into an expression	**HSF-IF.A.2**
FORWARD 1	Reason about sums and products of even and odd functions	**HSF-BF.B.3**
FORWARD 2	Explain trigonometric symmetry with the unit circle	**HSF-TF.A.4**

SET 57 • HSF-BF.B.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Is $f(x) = x^4 - 3x^2$ even, odd, or neither?

A. Even: $f(0) = 0$, and a graph through the origin is even.

B. Even: $f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$.

C. Odd: $f(-x) = -x^4 + 3x^2 = -f(x)$, because $-x$ makes each term negative.

D. Neither: $f(1) = -2$ and $f(-1) = 2$, so $f(-1) \neq f(1)$ and f is not even.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q1/5

2124

QUESTION ID
S57-M-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. Even: $f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$.

GRADING NOTE Even powers remove the sign. The others apply the sign to even powers, miscompute $f(-1)$ (it is -2), or use one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is $f(x) = x^3 - x$ even, odd, or neither?

A. Odd: $f(0) = 0$, and every function through the origin is odd.

B. Even: $f(-x) = x^3 - x = f(x)$, because the signs cancel in each term.

C. Odd: $f(-x) = (-x)^3 - (-x) = -x^3 + x = -f(x)$.

D. Neither: $f(-x) = -x^3 - x$, which is neither $f(x)$ nor $-f(x)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q2/5

2125

QUESTION ID
S57-M-Q2

TYPE
Multiple choice

CORRECT ANSWER
C. Odd: $f(-x) = (-x)^3 - (-x) = -x^3 + x = -f(x)$.

GRADING NOTE Odd powers keep the sign. The others drop signs or rely on one point ($f(0) = 0$ alone does not show oddness).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q2/5

Is $f(x) = x^2 + x$ even, odd, or neither?

A. Even: $f(0) = 0$, and a graph through the origin is symmetric about the y-axis.

B. Even: x^2 is even, and the term of highest degree decides whether f is even.

C. Odd: x is odd, and adding an odd term to x^2 makes the whole function odd.

D. Neither: $f(1) = 2$ and $f(-1) = 0$, so $f(-1) \neq f(1)$ and $f(-1) \neq -f(1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q3/5

2126

QUESTION ID
S57-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. Neither: $f(1) = 2$ and $f(-1) = 0$, so $f(-1) \neq f(1)$ and $f(-1) \neq -f(1)$.

GRADING NOTE One input that fails both tests shows neither. The others judge from one term or one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

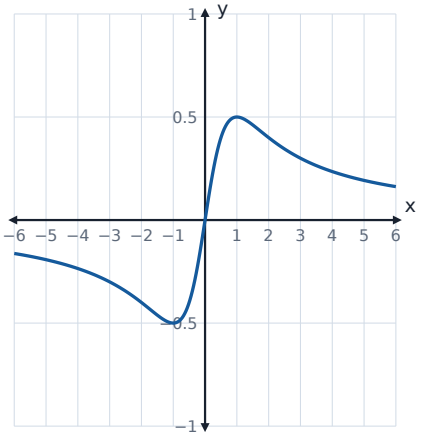
The graph of $f(x) = x/(x^2 + 1)$ is shown. Is f even, odd, or neither?

A. Even: the graph is symmetric about the y-axis (reflecting it over the y-axis gives the same graph).

B. Odd: the graph passes through the origin, and every graph through the origin is odd.

C. Odd: the graph is symmetric about the origin (rotating it 180° about the origin gives the same graph).

D. Neither: the graph has a peak on one side and a low point on the other, so it has no symmetry.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q4/5

2127

QUESTION ID
S57-M-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. Odd: the graph is symmetric about the origin (rotating it 180° about the origin gives the same graph).

GRADING NOTE Origin symmetry: $f(-x) = -x/(x^2 + 1) = -f(x)$. The others misread the symmetry or rely on one point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q4/5

Is $f(x) = \cos x + x^2$ even, odd, or neither?

A. Odd: cos is odd.

B. Even: $f(-x) = \cos(-x) + (-x)^2 = \cos x + x^2 = f(x)$.

C. Odd: $f(-x) = -\cos x + x^2$.

D. Neither: cos x and x^2 are different kinds of functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q5/5

2128

QUESTION ID

S57-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Even: $f(-x) = \cos(-x) + (-x)^2 = \cos x + x^2 = f(x)$.

GRADING NOTE cos is even and x^2 is even, so the sum is even. The others misstate cos(−x) or give a non-reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • MAIN • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 1

Signs of powers of negatives • Grade 7 • The Number System • 7.NS.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(-x)^4$ and $(-x)^3$.

$(-x)^4 =$ _____

$(-x)^3 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q1/5

2130

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x^4; $-x^3$

GRADING NOTE Form required: both. An even power removes the sign; an odd power keeps it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q1/5

2130

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $(-x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q2/5

2131

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x^2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(-x)^5 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q3/5

2132

QUESTION ID

S57-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An odd power keeps the sign: $(-x)^5 = -x^5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(-2)^2 - (-2)$ equals

A. -6

B. 2

C. 6

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q4/5

2133

GRADING NOTE 4 + 2.

QUESTION ID

S57-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Simplify $-(-x)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q5/5

2134

QUESTION ID

S57-B1-Q5

TYPE

Short answer

CORRECT ANSWER

x^3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 2

Reflect points over the axes and the origin • Grade 6 • The Number System • 6.NS.C.6.b

ANSWER KEY Reduced copy of the question page, with its answer

Reflect (2, 3) over the y-axis.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q1/5

2136

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−2, 3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rotate (2, 3) 180° about the origin.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q2/5

2137

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −3)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q2/5

True or false?

A graph symmetric about the y-axis contains (−a, b) whenever it contains (a, b).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q3/5

2138

QUESTION ID

S57-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Even symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Symmetry about the origin pairs (a, b) with

A. (−a, b)

B. (−a, −b)

C. (a, −b)

D. (b, a)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q4/5

2139

QUESTION ID

S57-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (−a, −b)

GRADING NOTE 180° rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph is symmetric about the origin and contains (1, 0.5). Name another point.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q5/5

2140

QUESTION ID

S57-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(−1, −0.5)

GRADING NOTE Any equivalent answer.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 57 • BACKWARD 2 • Q5/5

SET 57 • HSF-BF.B.3

BACKWARD 3

Substitute −x into an expression • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x^2 + x$, find $f(-x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q1/5

2142

QUESTION ID

S57-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 - x$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = x^3 - x$, find $f(-2)$ and $f(2)$.

$f(-2) =$ _____

$f(2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q2/5

2143

QUESTION ID

S57-B3-Q2

TYPE

Short answer

CORRECT ANSWER

−6; 6

GRADING NOTE Form required: both. $-8 + 2 = -6$ and $8 - 2 = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q2/5

True or false?

For f(x) = 3x, f(−x) = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q3/5

2144

QUESTION ID

S57-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(−x) = 3(−x) = −3x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q3/5

For $f(x) = x^4 - 3x^2$, $f(-1)$ is

A. 2

B. 4

C. -2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q4/5

2145

QUESTION ID

S57-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. −2

GRADING NOTE 1 − 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

For f(x) = 1/x, find f(−x).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q5/5

2146

QUESTION ID

S57-B3-Q5

TYPE

Short answer

CORRECT ANSWER

−1/x

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 1

Reason about sums and products of even and odd functions • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Which argument proves that the sum of two odd functions f and g is odd?

A. x^3 and x^5 are odd and $x^3 + x^5$ is odd, so every sum of two odd functions is odd.

B. $(f + g)(-x) = f(-x) + g(-x) = -f(x) - g(x) = -(f(x) + g(x)) = -(f + g)(x)$.

C. $(f + g)(-x) = f(x) + g(x) = (f + g)(x)$, so the sum is odd.

D. $(f + g)(-x) = -f(-x) - g(-x) = -(f + g)(-x)$, so the sum is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q1/5

2148

QUESTION ID
S57-F1-Q1

TYPE
Multiple choice

CORRECT ANSWER
B. $(f + g)(-x) = f(-x) + g(-x) = -f(x) - g(x) = -(f(x) + g(x)) = -(f + g)(x)$.

GRADING NOTE Use $f(-x) = -f(x)$ and $g(-x) = -g(x)$, then factor out -1. The others use the even-function rule (and would show even), check one example, or misapply the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q1/5

Which argument proves that the product of two odd functions f and g is even?

A. $(fg)(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) = (fg)(x)$.

B. $x \cdot x^3 = x^4$ is even, so every product of two odd functions is even.

C. Odd times odd is odd for whole numbers, so the product of two odd functions is odd.

D. $(fg)(-x) = f(-x)g(-x) = -f(x)g(x)$, so the product is odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q2/5

2149

QUESTION ID

S57-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A.

$(fg)(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) = (fg)(x)$.

GRADING NOTE The two negative signs multiply to +1. The others drop one negative sign, check one example, or confuse odd functions with odd numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q2/5

True or false?

The only function that is both even and odd is f(x) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q3/5

2150

QUESTION ID

S57-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(x) = f(−x) = −f(x) gives f(x) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = |x| + x^3$ is

A. even

B. both

C. neither

D. odd

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q4/5

2151

QUESTION ID

S57-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. neither

GRADING NOTE $f(1) = 2$, $f(-1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Is f(x) = x sin x even, odd, or neither?

A. Odd: f(−x) = (−x) sin x = −x sin x = −f(x), since only the first factor changes sign.

B. Odd: sin x is odd, and multiplying an odd function by x keeps it odd, so x sin x is odd.

C. Even: x and sin x are both odd, and a product of two odd functions is even.

D. Neither: x is not a trigonometric function, so the rules for even and odd functions do not apply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q5/5

2152

QUESTION ID

S57-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Even: x and sin x are both odd, and a product of two odd functions is even.

GRADING NOTE f(−x) = (−x)sin(−x) = (−x)(−sin x) = x sin x. The others drop one sign or give a non-reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 57 • FORWARD 1 • Q5/5

SET 57 • HSF-BF.B.3

FORWARD 2

Explain trigonometric symmetry with the unit circle • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Why is cos(−x) = cos x, using the unit circle?

A. The points at angles x and −x are reflections of each other over the x-axis, which keeps the x-coordinate (the cosine).

B. The points at angles x and −x are reflections of each other over the y-axis, which keeps the x-coordinate (the cosine).

C. The points at angles x and −x are the same point on the unit circle, so they have the same x-coordinate.

D. Because cos is never negative, so cos(−x) and cos x always have the same value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q1/5

2154

QUESTION ID

S57-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. The points at angles x and −x are reflections of each other over the x-axis, which keeps the x-coordinate (the cosine).

GRADING NOTE Reflection over the x-axis keeps x. The others name the wrong reflection, confuse the points, or are false (cos π = −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q1/5

Is tan even, odd, or neither?

A. Odd: tan(−x) = sin(−x)/cos(−x) = −sin x/cos x = −tan x.

B. Even: cos is even, so tan(−x) = sin x/cos(−x) = sin x/cos x = tan x.

C. Even: tan(−x) = sin x/cos x = tan x.

D. Neither: tan has vertical asymptotes, so it cannot be even or odd.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q2/5

2155

QUESTION ID
S57-F2-Q2

TYPE
Multiple choice

CORRECT ANSWER
A. Odd: tan(−x) = sin(−x)/cos(−x) = −sin x/cos x = −tan x.

GRADING NOTE sin is odd and cos is even, so their quotient is odd. The others drop the sign or give a non-reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q2/5

True or false?

sin is an even function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q3/5

2156

QUESTION ID

S57-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sin(-x) = -\sin x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q3/5

Which is an even function?

A. cos x

B. tan x

C. sin x + cos x

D. sin x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q4/5

2157

QUESTION ID

S57-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. cos x

GRADING NOTE Reflection over the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Use symmetry to find $\sin(-\pi/6)$ from $\sin(\pi/6) = 1/2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q5/5

2158

QUESTION ID

S57-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$-1/2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q5**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 57 • FORWARD 2 • Q5/5

Write inverses of radical, rational, cubic, exponential, and logarithmic functions

MAIN	Algebra II main question	**HSF-BF.B.4.a**
BACKWARD 1	Inverses of linear functions	**HSF-BF.B.4.a**
BACKWARD 2	Rearrange formulas with roots, powers, and fractions	**HSA-CED.A.4**
BACKWARD 3	Domain and range	**HSF-IF.B.5**
FORWARD 1	Inverses that are their own inverse, and checking by points	**HSF-BF.B.4.a**
FORWARD 2	Verify inverses by composition	**HSF-BF.B.4.b**

SET 58 • HSF-BF.B.4.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = (x - 1)^3 + 2$ (all real x). It has the form $f^{-1}(x) = \sqrt[3]{}(x - a) + b$; give a and b.

a = _____

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q1/5

2161

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

2; 1

GRADING NOTE Any equivalent expression. $y - 2 = (x - 1)^3$ gives $x = \sqrt[3]{}(y - 2) + 1$; the domain is all real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = 2x/(x - 3)$, $x \neq 3$, and the value excluded from its domain.

$f^{-1}(x) =$ _____

Excluded value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q2/5

2162

QUESTION ID

S58-M-Q2

TYPE

Short answer

CORRECT ANSWER

3x/(x − 2); {2}

GRADING NOTE Form required: both. $y(x - 3) = 2x$ gives $x(y - 2) = 3y$, $x = 3y/(y - 2)$; $y = 2$ is not in the range of f.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of $f(x) = \sqrt{(x + 4)}$, $x \geq -4$, and its domain.

$f^{-1}(x) =$ _____

Domain of f^{-1}: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q3/5

2163

QUESTION ID

S58-M-Q3

TYPE

Short answer

CORRECT ANSWER

$x^2 - 4; x \geq 0$

GRADING NOTE Form required: both. $y^2 = x + 4$ gives $x = y^2 - 4$; the domain of f^{-1} is the range of f, $y \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q3/5

Which gives the inverse of $f(x) = 5 \cdot 2^x$ and its domain?

A. $f^{-1}(x) = \log_5(x/2)$, $x > 0$

B. $f^{-1}(x) = \log_2(x) - 5$, $x > 0$

C. $f^{-1}(x) = \log_2(x/5)$, all real x

D. $f^{-1}(x) = \log_2(x/5)$, $x > 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q4/5

2164

QUESTION ID

S58-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $f^{-1}(x) = \log_2(x/5)$, $x > 0$

GRADING NOTE $y/5 = 2^x$ gives $x = \log_2(y/5)$; the domain of f^{-1} is the range of f, $y > 0$. The others subtract instead of dividing, swap the base, or give the wrong domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q4/5

Which gives the inverse of $f(x) = \log_3(x - 1)$ and its domain?

A. $f^{-1}(x) = 3^x + 1$, $x > 1$

B. $f^{-1}(x) = 3^x + 1$, all real x

C. $f^{-1}(x) = 3^{x+1}$, all real x

D. $f^{-1}(x) = x^3 + 1$, all real x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q5/5

2165

QUESTION ID

S58-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. $f^{-1}(x) = 3^x + 1$, all real x

GRADING NOTE $y = \log_3(x - 1)$ means $x - 1 = 3^y$. The domain of f^{-1} is the range of f, all real numbers. The others misplace the 1, give the wrong domain, or use a power instead of an exponential.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • MAIN • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 1

Inverses of linear functions • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Find the inverse of f(x) = 3x − 6.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

2167

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 6)/3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 5 = y for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

2168

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(y − 5)/2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

True or false?

The inverse undoes the function: $f^{-1}(f(x)) = x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

2169

QUESTION ID

S58-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The inverse of f(x) = x/4 + 1 is

A. 4(x − 1)

B. (x − 1)/4

C. 4x − 1

D. x/4 − 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

2170

QUESTION ID

S58-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4(x − 1)

GRADING NOTE Undo in reverse order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

If $f(2) = 7$, what is $f^{-1}(7)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

2171

QUESTION ID

S58-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 2

Rearrange formulas with roots, powers, and fractions • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = \sqrt{(x + 4)}$ for x ($y \geq 0$).

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q1/5

2173

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$y^2 - 4$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $y = 2x/(x - 3)$ for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q2/5

2174

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3y/(y − 2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving $y = (x - 1)^3 + 2$ for x gives $x = \sqrt[3]{(y - 2)} + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q3/5

2175

QUESTION ID

S58-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Undo each step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Solving $y = x^3$ for x gives

A. y/3

B. $\sqrt[3]{y}$

C. $\sqrt{y}$

D. y^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q4/5

2176

QUESTION ID

S58-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $\sqrt[3]{y}$

GRADING NOTE Cube root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve y = x/(x + 1) for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q5/5

2177

QUESTION ID

S58-B2-Q5

TYPE

Short answer

CORRECT ANSWER

y/(1 − y)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • HSF-BF.B.4.a

BACKWARD 3

Domain and range • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of f(x) = √(x + 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q1/5

2179

QUESTION ID

S58-B3-Q1

TYPE

Short answer

CORRECT ANSWER

y ≥ 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the range of $f(x) = 5 \cdot 2^x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q2/5

2180

QUESTION ID

S58-B3-Q2

TYPE

Short answer

CORRECT ANSWER

y > 0

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q2/5

True or false?

The range of f(x) = 2x/(x − 3) excludes y = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q3/5

2181

QUESTION ID

S58-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(9) = 18/6 = 3; the excluded value is y = 2 (the horizontal asymptote).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The domain of $\log_3(x - 1)$ is

A. $x > 1$

B. all reals

C. $x > 0$

D. $x \geq 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q4/5

2182

QUESTION ID

S58-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. x > 1

GRADING NOTE $x - 1 > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of $f(x) = \log_3(x - 1)$?

A. $y > 0$

B. $y > 1$

C. All real numbers

D. $y \geq 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q5/5

2183

QUESTION ID

S58-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. All real numbers

GRADING NOTE A logarithm takes every real value (it is the inverse of an exponential, whose domain is all real numbers).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 58 • BACKWARD 3 • Q5/5

SET 58 • HSF-BF.B.4.a

FORWARD 1

Inverses that are their own inverse, and checking by points • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x + 1)/(x - 1)$, $x \neq 1$. Find $f^{-1}(x)$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q1/5

2185

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x + 1)/(x - 1)$

GRADING NOTE Any equivalent expression. $y(x - 1) = x + 1$ gives $x = (y + 1)/(y - 1)$: f is its own inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 5 \cdot 2^x$ passes through (1, 10), and $f^{-1}(x) = \log_2(x/5)$. Check the inverse with this point: find $f^{-1}(10)$.

$f^{-1}(10) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q2/5

2186

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Form required: the value. $\log_2(10/5) = \log_2 2 = 1$, matching the point (1, 10) on f.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q2/5

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q3/5

2187

QUESTION ID

S58-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only one-to-one functions; $f(x) = x^2$ on all reals does not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is its own inverse?

A. $f(x) = x^2$

B. $f(x) = x + 1$

C. $f(x) = 2x$

D. $f(x) = 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q4/5

2188

QUESTION ID

S58-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. f(x) = 1/x

GRADING NOTE $1/(1/x) = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q4/5

Use the horizontal line test on the graph of $f(x) = x^3 - x$. Does f have an inverse function?

A. No: the horizontal line y = 0 meets the graph three times (x = −1, 0, 1), so f is not one-to-one.

B. Yes: f is a polynomial, and every polynomial passes the horizontal line test, so f has an inverse function.

C. Yes: every cubic passes the horizontal line test, because its graph rises from left to right, so f is one-to-one.

D. No: the vertical line x = 0 meets the graph three times, so f fails the vertical line test and is not one-to-one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q5/5

2189

QUESTION ID

S58-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. No: the horizontal line y = 0 meets the graph three times (x = −1, 0, 1), so f is not one-to-one.

GRADING NOTE One horizontal line meeting the graph more than once is enough. The others overgeneralize or confuse the tests (f is a function, so it passes the vertical line test).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.4.a • Set 58 • FORWARD 1 • Q5/5

SET 58 • HSF-BF.B.4.a

FORWARD 2

Verify inverses by composition • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b

$f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2)} + 1$. Which computation of $f(g(x))$ is correct?

A. $f(g(x)) = f(x) \cdot g(x) = ((x - 1)^3 + 2)(\sqrt[3]{(x - 2)} + 1) = x$

B. $f(g(x)) = ((\sqrt[3]{(x - 2)} + 1) - 1)^3 + 1 = (\sqrt[3]{(x - 2)})^3 + 1 = x - 1$

C. $f(g(x)) = ((\sqrt[3]{(x - 2)} + 1) - 1)^3 + 2 = (\sqrt[3]{(x - 2)})^3 + 2 = x - 2 + 2 = x$

D. $f(g(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2)} + 1 = \sqrt[3]{((x - 1)^3)} + 1 = x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q1/5

2191

QUESTION ID

S58-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. $f(g(x)) = ((\sqrt[3]{(x - 2)} + 1) - 1)^3 + 2 = (\sqrt[3]{(x - 2)})^3 + 2 = x - 2 + 2 = x$

GRADING NOTE Substitute g(x) for x in f. The others compute g(f(x)) instead, drop a step, or multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q1/5

$f(x) = (x - 1)^3 + 2$ and $g(x) = \sqrt[3]{(x - 2) + 1}$. Which computation of $g(f(x))$ is correct?

A. $g(f(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2) + 1} = \sqrt[3]{((x - 1)^3)} + 1 = x - 1 + 1 = x$

B. $g(f(x)) = g(x) \cdot f(x) = (\sqrt[3]{(x - 2) + 1})((x - 1)^3 + 2) = x$

C. $g(f(x)) = ((\sqrt[3]{(x - 2) + 1}) - 1)^3 + 2 = (\sqrt[3]{(x - 2)})^3 + 2 = x$

D. $g(f(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2) + 2} = \sqrt[3]{((x - 1)^3)} + 2 = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q2/5

2192

QUESTION ID

S58-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A.

$g(f(x)) = \sqrt[3]{(((x - 1)^3 + 2) - 2) + 1} = \sqrt[3]{((x - 1)^3)} + 1 = x - 1 + 1 = x$

GRADING NOTE Substitute f(x) for x in g. The others compute f(g(x)), mix up the constants, or multiply.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q2/5

True or false?

To verify inverses, both f(g(x)) = x and g(f(x)) = x must hold on the domains.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q3/5

2193

QUESTION ID
S58-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Two-sided.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For $f(x) = \sqrt{(x + 4)}$ and $g(x) = x^2 - 4$ $(x \geq 0)$, $g(f(x)) =$

A. x for all x

B. $|x|$

C. x for $x \geq -4$

D. $x^2 + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q4/5

2194

QUESTION ID

S58-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. x for x ≥ −4

GRADING NOTE $(\sqrt{(x + 4)})^2 - 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q4/5

Why is $g(x) = x^2 - 4$, without a domain restriction, not the inverse of $f(x) = \sqrt{(x + 4)}$?

A. Because $f(g(2)) \neq 2$, so the composition fails even for positive inputs.

B. $f(g(x)) = \sqrt{(x^2)} = |x|$, which is not x when $x < 0$; g is the inverse only on $x \geq 0$ (the range of f).

C. Because $g(x) = x^2 - 4$ is not a function, since it gives two outputs for each input.

D. Because $g(f(x)) = (\sqrt{(x + 4)})^2 - 4$ is undefined for every x, so the composition test cannot be done.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q5/5

2195

QUESTION ID

S58-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. $f(g(x)) = \sqrt{(x^2)} = |x|$, which is not x when $x < 0$; g is the inverse only on $x \geq 0$ (the range of f).

GRADING NOTE $\sqrt{(x^2)} = |x|$. The others are false: $g(f(x)) = x$ for $x \geq -4$, g is a function, and $f(g(2)) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.4.b • Set 58 • FORWARD 2 • Q5/5

Write exact logarithmic solutions of exponential equations

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Isolate the exponential expression	**HSA-REI.B.3**
BACKWARD 2	Powers of 2, 3, and 10	**8.EE.A.1**
BACKWARD 3	Logarithms as the inverse of exponentials	**HSF-BF.B.4.a**
FORWARD 1	Solve equations with different bases or variables on both sides	**HSF-LE.A.4**
FORWARD 2	Use the inverse relationship to solve logarithmic equations	**HSF-BF.B.5**

SET 59 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3 \cdot 2^{5t} = 48$ exactly.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q1/5

2198

QUESTION ID

S59-M-Q1

TYPE

Short answer

CORRECT ANSWER

4/5

GRADING NOTE Any equivalent form. $2^{5t} = 16 = 2^4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the exact solution of $2 \cdot 10^{3t} = 9$?

A. t = log(9)/6

B. t = (log 9 − 2)/3

C. t = 3 log 4.5

D. t = log(4.5)/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q2/5

2199

QUESTION ID

S59-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. t = log(4.5)/3

GRADING NOTE $10^{3t} = 4.5$, so 3t = log 4.5 (≈ 0.2177). The others divide by 2 after taking the log, subtract 2 instead of dividing, or multiply by 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the exact solution of $5e^{2t} = 30$?

A. $t = (\ln 6) \cdot 2$

B. $t = (\ln 6)/2$

C. $t = \ln 3$

D. $t = \ln 30/10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q3/5

2200

QUESTION ID

S59-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. t = (ln 6)/2

GRADING NOTE $e^{2t} = 6$, so $2t = \ln 6$ ($t \approx 0.896$). The others divide the logarithm wrongly, multiply by 2, or halve 6 before taking the log.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the exact solution of $100(1.05)^t = 300$?

A. t = 3/1.05

B. t = log 3 − log 1.05

C. t = log 3/log 1.05

D. t = log 300/log 1.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q4/5

2201

QUESTION ID

S59-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. t = log 3/log 1.05

GRADING NOTE $(1.05)^t = 3$, so t = log 3/log 1.05 ≈ 22.52. The others forget to divide by 100, subtract the logs, or skip the logarithm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-M-Q4**

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the exact solution of $4 \cdot 3^{t/2} = 20$?

A. $t = 2 \log_3 5$

B. $t = \log_3 10$

C. $t = 2 \log_3 20$

D. $t = \log_3 5/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q5/5

2202

QUESTION ID

S59-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $t = 2 \log_3 5$

GRADING NOTE $3^{t/2} = 5$, so $t/2 = \log_3 5$ ($t \approx 2.93$). The others divide instead of multiplying by 2, forget to divide by 4, or halve 20 first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • MAIN • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 1

Isolate the exponential expression • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3y = 48 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q1/5

2204

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 5y = 30 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q2/5

2205

QUESTION ID

S59-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q2/5

True or false?

From $2 \cdot 10^{3t} = 9$, we get $10^{3t} = 4.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q3/5

2206

QUESTION ID

S59-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Divide by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

From $4 \cdot 3^{t/2} = 20$, $3^{t/2}$ equals

A. 80

B. 5

C. 1/5

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q4/5

2207

QUESTION ID

S59-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5

GRADING NOTE Divide by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 100y = 300 for y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q5/5

2208

QUESTION ID

S59-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 59 • BACKWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 2

Powers of 2, 3, and 10 • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Write 16 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q1/5

2210

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

2^4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 1000 as a power of 10.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q2/5

2211

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10^3

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$3^4 = 12$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q3/5

2212

QUESTION ID

S59-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^4 = 3 \cdot 3 \cdot 3 \cdot 3 = 81$; 12 is $3 \cdot 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$2^{5t} = 2^4$ gives

A. t = 4

B. 5t = 4

C. 5t = 16

D. t = 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q4/5

2213

QUESTION ID

S59-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5t = 4

GRADING NOTE Equal exponents.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 1/8 as a power of 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q5/5

2214

QUESTION ID

S59-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2^{-3}

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • HSF-LE.A.4

BACKWARD 3

Logarithms as the inverse of exponentials • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $10^x = 4.5$ as a logarithm.

A. $x = 10^{4.5}$

B. $x = 4.5/10$

C. $x = \log 10/4.5$

D. $x = \log 4.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q1/5

2216

QUESTION ID

S59-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. x = log 4.5

GRADING NOTE log is the inverse of 10^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $e^u = 6$ as a logarithm.

A. $u = \log 6$

B. $u = 6/e$

C. $u = \ln 6$

D. $u = e^6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q2/5

2217

QUESTION ID

S59-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. u = ln 6

GRADING NOTE In is the inverse of e^u; log 6 is base 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log_2 8 = 4$ because $2^4 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q3/5

2218

QUESTION ID

S59-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^4 = 16$; $\log_2 8 = 3$ because $2^3 = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$3^y = 5$ means y =

A. $\log_3 5$

B. 5/3

C. $\log_5 3$

D. 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q4/5

2219

QUESTION ID
S59-B3-Q4

TYPE
Multiple choice

CORRECT ANSWER
A. $\log_3 5$

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $\log_2 32$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q5/5

2220

QUESTION ID
S59-B3-Q5

TYPE
Short answer

CORRECT ANSWER
5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 59 • BACKWARD 3 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 1

Solve equations with different bases or variables on both sides • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2^x = 3^{x-1}$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q1/5

2222

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2.710

GRADING NOTE Rounding to 0.001. $x \ln 2 = (x - 1) \ln 3$ gives $x = \ln 3/(\ln 3 - \ln 2) \approx 2.7095$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5^{2x} = 7$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q2/5

2223

QUESTION ID
S59-F1-Q2

TYPE
Short answer

CORRECT ANSWER
0.605

GRADING NOTE Rounding to 0.001. x = log 7/(2 log 5) ≈ 0.6045.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

log(ab) = log a + log b for positive a and b.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q3/5

2224

QUESTION ID

S59-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Product property.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The exact solution of $e^{x+1} = 10$ is

A. 10/e

B. ln 10/1

C. ln 9

D. ln 10 − 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q4/5

2225

QUESTION ID

S59-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. ln 10 − 1

GRADING NOTE x + 1 = ln 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3 \cdot 4^x = 2 \cdot 5^x$. Give x to the nearest thousandth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q5/5

2226

QUESTION ID
S59-F1-Q5

TYPE
Short answer

CORRECT ANSWER
1.817

GRADING NOTE Rounding to 0.001. $(5/4)^x = 3/2$, so $x = \log(3/2)/\log(5/4) \approx 1.8171$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 59 • FORWARD 1 • Q5/5

SET 59 • HSF-LE.A.4

FORWARD 2

Use the inverse relationship to solve logarithmic equations • High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\log_2(x + 3) = 5$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q1/5

2228

QUESTION ID

S59-F2-Q1

TYPE

Short answer

CORRECT ANSWER

29

GRADING NOTE Any equivalent form. x + 3 = 32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve log x + log(x − 3) = 1. Give every value found after combining the logarithms and solving, and the actual solutions (enter none if there are none).

Values found: _____

Actual solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q2/5

2229

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

{5, −2}; {5}

GRADING NOTE Form required: both. log(x(x − 3)) = 1 gives $x^2 - 3x - 10 = 0$, x = 5 or x = −2; log(−2) is undefined, so −2 is extraneous.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\log_b(b^x) = x$ for every real x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q3/5

2230

QUESTION ID

S59-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Inverse functions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$\ln(e^7)$ equals

A. ln 7

B. 1

C. 7

D. e^7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q4/5

2231

QUESTION ID
S59-F2-Q4

TYPE
Multiple choice

CORRECT ANSWER
C. 7

GRADING NOTE Inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve ln(2x) = 3 exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q5/5

2232

QUESTION ID

S59-F2-Q5

TYPE

Short answer

CORRECT ANSWER

$e^3/2$

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Building Functions • HSF-BF.B.5 • Set 59 • FORWARD 2 • Q5/5

Evaluate logarithmic solutions of exponential models with technology

MAIN	Algebra II main question	**HSF-LE.A.4**
BACKWARD 1	Write exact logarithmic solutions	**HSF-LE.A.4**
BACKWARD 2	Evaluate logarithms with a calculator	**HSF-IF.A.2**
BACKWARD 3	Choose an appropriate level of accuracy	**HSN-Q.A.3**
FORWARD 1	Analyze how parameters change the solution	**HSF-LE.A.4**
FORWARD 2	Rewrite models to show doubling time or half-life	**HSF-IF.C.8.b**

SET 60 • HSF-LE.A.4

MAIN

Algebra II main question • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$2,500 earns 3.5% interest, credited once at the end of each year. Solve $2500(1.035)^t = 4000$ for t (to the nearest hundredth of a year), and give the number of yearly credits after which the balance first reaches at least $4,000. A calculator may be used.

t (years): _____

Number of yearly credits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q1/5

2235

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

13.66; 14

GRADING NOTE Form required: both. t = log 1.6/log 1.035 ≈ 13.662; after 13 credits the balance is $3,909.89 and after 14 it is $4,046.74.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Carbon-14 has a half-life of 5730 years. A sample has 30 of its original 80 units of carbon-14. Solve $80(0.5)^{t/5730} = 30$ to estimate its age to the nearest year. A calculator may be used.

Age (years): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q2/5

2236

QUESTION ID
S60-M-Q2

TYPE
Short answer

CORRECT ANSWER
8108

GRADING NOTE Rounding to the nearest year. $t = 5730 \cdot \log(3/8)/\log 0.5 \approx 8108.2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Coffee cools by $T(t) = 20 + 70(0.95)^t$ °C after t minutes. When is it 50°C? Give minutes to the nearest hundredth. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q3/5

2237

QUESTION ID
S60-M-Q3

TYPE
Short answer

CORRECT ANSWER
16.52

GRADING NOTE $0.95^t = 3/7$, so $t = \ln(3/7)/\ln 0.95 \approx 16.52$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A city of 1.2 million grows by $P(t) = 1.2e^{0.015t}$ million people. When does it reach 1.5 million? Give years to the nearest hundredth. A calculator may be used.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q4/5

2238

QUESTION ID
S60-M-Q4

TYPE
Short answer

CORRECT ANSWER
14.88

GRADING NOTE ln 1.25/0.015 ≈ 14.88.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A dose leaves $200(0.8)^t$ mg of a drug after t hours. At which whole hour is the amount first below 50 mg? A calculator may be used.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q5/5

2239

QUESTION ID

S60-M-Q5

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE $0.8^t < 0.25$ gives $t > \ln 0.25/\ln 0.8 \approx 6.21$: $200(0.8)^6 \approx 52.4$ and $200(0.8)^7 \approx 41.9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • MAIN • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 1

Write exact logarithmic solutions • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Which is the exact solution of $1.035^t = 1.6$?

A. t = log(1.6 − 1.035)

B. t = log 1.6/log 1.035

C. t = log 1.035/log 1.6

D. t = 1.6/1.035

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q1/5

2241

QUESTION ID

S60-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. t = log 1.6/log 1.035

GRADING NOTE t log 1.035 = log 1.6 (t ≈ 13.66). The others invert the quotient, skip the logarithm, or use a false rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q1/5

Which is the exact solution of $e^{0.015t} = 1.25$?

A. t = ln 1.25/0.015

B. t = 1.25/0.015

C. t = ln(1.25/0.015)

D. t = 0.015 ln 1.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q2/5

2242

QUESTION ID

S60-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. t = ln 1.25/0.015

GRADING NOTE 0.015t = ln 1.25 (t ≈ 14.88). The others multiply, divide inside the log, or skip the logarithm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$0.8^t = 0.25$ gives t = log 0.25/log 0.8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q3/5

2243

QUESTION ID
S60-B1-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Take logs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$(0.5)^{t/5730} = 0.375$ gives t =

A. 0.375 · 5730

B. 5730/log 0.375

C. log 0.375/5730

D. 5730 log 0.375/log 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q4/5

2244

QUESTION ID
S60-B1-Q4

TYPE
Multiple choice

CORRECT ANSWER
D. 5730 log 0.375/log 0.5

GRADING NOTE Multiply by 5730.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the exact solution of $0.95^t = 3/7$?

A. t = log(3/7) − log 0.95

B. t = (3/7)/0.95

C. t = log 0.95/log(3/7)

D. t = log(3/7)/log 0.95

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q5/5

2245

QUESTION ID
S60-B1-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. t = log(3/7)/log 0.95

GRADING NOTE t log 0.95 = log(3/7) (t ≈ 16.52). The others invert, skip the logarithm, or subtract the logs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 2

Evaluate logarithms with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate log 1.6 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q1/5

2247

QUESTION ID
S60-B2-Q1

TYPE
Short answer

CORRECT ANSWER
0.2041

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate ln 1.25 to four decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q2/5

2248

QUESTION ID
S60-B2-Q2

TYPE
Short answer

CORRECT ANSWER
0.2231

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q2/5

2248

True or false?

log 0.25 is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q3/5

2249

QUESTION ID

S60-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 0.25 < 1, so log 0.25 < 0 (about −0.602).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

log 1.6/log 1.035 is about

A. 0.20

B. 46.0

C. 1.55

D. 13.66

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q4/5

2250

QUESTION ID

S60-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 13.66

GRADING NOTE 0.2041/0.01494.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q4/5

2250

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate log 0.25/log 0.8 to two decimal places. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q5/5

2251

QUESTION ID

S60-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6.21

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • HSF-LE.A.4

BACKWARD 3

Choose an appropriate level of accuracy • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Interest is added once a year. A model says a balance reaches its goal at t = 13.66 years. After how many whole years is the goal first reached?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q1/5

2253

QUESTION ID

S60-B3-Q1

TYPE

Short answer

CORRECT ANSWER

14

GRADING NOTE The balance changes only at whole years, so round up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q1/5

2253

ANSWER KEY Reduced copy of the question page, with its answer

Round 8108.16 years to the nearest hundred years.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q2/5

2254

QUESTION ID

S60-B3-Q2

TYPE

Short answer

CORRECT ANSWER

8100

GRADING NOTE 8108.16 is between 8100 and 8200, closer to 8100.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q2/5

True or false?

A time found from a model should be reported with more decimal places than the data justify.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q3/5

2255

QUESTION ID

S60-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Precision should match the context.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q3/5

2255

ANSWER KEY Reduced copy of the question page, with its answer

A drug amount first drops below 50 mg at t = 6.21 h. If it is checked every hour, it is first below 50 mg at

A. 6.21 h

B. 6 h

C. 7 h

D. 5 h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q4/5

2256

QUESTION ID

S60-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 7 h

GRADING NOTE Next whole hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A population model gives t = 14.88 years. Why might this be reported as "about 15 years"?

A. Because rounding a model's answer up always gives the exact answer for the population.

B. Because 14.88 is not a real number of years, so it must be replaced by a whole number.

C. The model and its data are approximate, so reporting more precision than they justify would be misleading.

D. Because answers about years must always be whole numbers, whatever the model shows.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q5/5

2257

QUESTION ID

S60-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The model and its data are approximate, so reporting more precision than they justify would be misleading.

GRADING NOTE Precision should match the context and the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • HSF-LE.A.4

FORWARD 1

Analyze how parameters change the solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$2,500 is invested at 5% interest, credited once at the end of each year. Using the model $2500(1.05)^t$, find the time t when the model first equals $4,000 (to the nearest hundredth of a year), and the number of yearly credits after which the account first holds at least $4,000. A calculator may be used.

Model time t (years): _____

Number of yearly credits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q1/5

2259

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9.63; 10

GRADING NOTE Form required: both. t = log 1.6/log 1.05 ≈ 9.633; after 9 credits the balance is $3,878.32 and after 10 it is $4,072.24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q1/5

2259

ANSWER KEY Reduced copy of the question page, with its answer

Money earns 6% interest per year. Using the model $(1.06)^t = 2$, find the doubling time to the nearest hundredth of a year, the Rule of 72 estimate 72/6, and the number of yearly credits after which the money is first at least doubled when interest is credited once a year. A calculator may be used.

Model doubling time (years): _____

Rule of 72 estimate: _____

Number of yearly credits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q2/5

2260

QUESTION ID

S60-F1-Q2

TYPE

Short answer

CORRECT ANSWER

11.90; 12; 12

GRADING NOTE Form required: all three. $\log 2/\log 1.06 \approx 11.896$; $1.06^{11} \approx 1.898 < 2 \leq 1.06^{12} \approx 2.012$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q2/5

True or false?

Doubling the initial amount in $a(b)^t = d$ halves the time t needed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q3/5

2261

QUESTION ID

S60-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It reduces t by one doubling time, not by half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

In $200(0.8)^t = 50$, replacing 200 by 400 changes t by

A. doubling it

B. halving it

C. adding log 2/log(1/0.8) ≈ 3.11 hours

D. nothing

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q4/5

2262

QUESTION ID

S60-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. adding log 2/log(1/0.8) ≈ 3.11 hours

GRADING NOTE $400(0.8)^t = 50$ needs $0.8^t = 1/8$ instead of 1/4: t grows by log(1/2)/log 0.8 ≈ 3.11.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

At 3.5% yearly interest, why does $2,500 take as long to grow to $4,000 as $5,000 takes to grow to $8,000?

A. Because $5,000 is twice $2,500, and twice the money grows twice as fast, which exactly cancels the doubled target.

B. Both need the growth factor 1.6, and t = log 1.6/log 1.035 depends only on that factor, not on the starting amount.

C. Because both amounts earn the same interest in dollars each year, and both need to gain the same number of dollars.

D. They do not take the same time: the larger amount earns more interest each year, so it reaches its target sooner.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q5/5

2263

QUESTION ID

S60-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Both need the growth factor 1.6, and t = log 1.6/log 1.035 depends only on that factor, not on the starting amount.

GRADING NOTE 4000/2500 = 8000/5000 = 1.6. The others give an incomplete reason or a false claim (the larger amount earns more dollars, not a larger factor).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 60 • FORWARD 1 • Q5/5

SET 60 • HSF-LE.A.4

FORWARD 2

Rewrite models to show doubling time or half-life • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $2500(1.035)^t$ in the form $2500 \cdot 2^{t/D}$ and give D, in years to the nearest hundredth. A calculator may be used.

D ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q1/5

2265

QUESTION ID
S60-F2-Q1

TYPE
Short answer

CORRECT ANSWER
20.15

GRADING NOTE D = log 2/log 1.035 ≈ 20.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q1**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rewrite $200(0.8)^t$ in the form $200(0.5)^{t/H}$ and give H, in hours to the nearest hundredth. A calculator may be used.

H ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q2/5

2266

QUESTION ID
S60-F2-Q2

TYPE
Short answer

CORRECT ANSWER
3.11

GRADING NOTE H = log 0.5/log 0.8 ≈ 3.106.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$80(0.5)^{t/5730}$ shows the half-life 5730 directly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q3/5

2267

QUESTION ID
S60-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Exponent t/5730.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$1.2e^{0.015t}$ has doubling time about

A. 46.2 years

B. 66.7 years

C. 15 years

D. 0.015 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q4/5

2268

QUESTION ID

S60-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 46.2 years

GRADING NOTE ln 2/0.015.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A substance decays by $A = A_0(0.5)^{t/h}$, where h is its half-life. Why does this form make it easy to find how much remains after 3 half-lives?

A. After 3 half-lives t/h = 3, so $A = A_0(0.5)^3 = A_0/8$.

B. Because the substance is gone after 3 half-lives.

C. Because 3 half-lives leave 3/8 of the amount.

D. Because $A = A_0 - 3h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q5/5

2269

QUESTION ID
S60-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
A. After 3 half-lives t/h = 3, so $A = A_0(0.5)^3 = A_0/8$.

GRADING NOTE Each half-life multiplies by 0.5. The others are false (decay never reaches 0; it is not linear; $(0.5)^3 = 1/8$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 60 • FORWARD 2 • Q5/5

SET 61 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Estimate population percentages with a normal distribution

MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Interpret mean and standard deviation	**HSS-ID.A.2**
BACKWARD 2	Describe the shape of a distribution	**HSS-ID.A.1**
BACKWARD 3	Percents of a whole	**6.RP.A.3.c**
FORWARD 1	Find values from percentiles of a normal model	**HSS-ID.A.4**
FORWARD 2	Use a normal model for a margin of error	**HSS-IC.B.4**

SET 61 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent between 59 and 69 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q1/5

2272

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

95%

GRADING NOTE Any equivalent form. 59 and 69 are 2 standard deviations from the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q1**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Adult women's heights are approximately normal with mean 64 in and standard deviation 2.5 in. Estimate the percent of women taller than 66.5 inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q2/5

2273

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

16%

GRADING NOTE Any equivalent form. (100 − 68)/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q2**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Test scores are approximately normal with mean 500 and standard deviation 100. Estimate the percent of scores between 400 and 700. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q3/5

2274

QUESTION ID

S61-M-Q3

TYPE

Short answer

CORRECT ANSWER

81.5%

GRADING NOTE Any equivalent form. 34 + 34 + 13.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q3**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Battery life is approximately normal with mean 40 h and standard deviation 4 h. Use the table to estimate the percent of batteries lasting less than 46 hours.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q4/5

2275

QUESTION ID

S61-M-Q4

TYPE

Short answer

CORRECT ANSWER

93.32%

GRADING NOTE Any equivalent form. z = 1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q4/5

Cereal box weights are approximately normal with mean 16.0 oz and standard deviation 0.2 oz. Use the table to estimate the percent between 15.8 and 16.3 oz.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q5/5

2276

QUESTION ID

S61-M-Q5

TYPE

Short answer

CORRECT ANSWER

77.45%

GRADING NOTE Any equivalent form. z from −1 to 1.5: 0.9332 − 0.1587.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-M-Q5**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • MAIN • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 1

Interpret mean and standard deviation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A data set has mean 64 and standard deviation 2.5.
What value is one standard deviation above the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q1/5

2278

QUESTION ID
S61-B1-Q1

TYPE
Short answer

CORRECT ANSWER
66.5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B1-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value is two standard deviations below a mean of 500 with standard deviation 100?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q2/5

2279

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

300

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q2/5

True or false?

A larger standard deviation means the data are more spread out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q3/5

2280

QUESTION ID

S61-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q3/5

2280

ANSWER KEY Reduced copy of the question page, with its answer

46 is how many standard deviations above a mean of 40 when the standard deviation is 4?

A. 0.67

B. 4

C. 6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q4/5

2281

QUESTION ID

S61-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1.5

GRADING NOTE (46 − 40)/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B1-Q4**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Compute (15.8 − 16.0)/0.2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q5/5

2282

QUESTION ID

S61-B1-Q5

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 2

Describe the shape of a distribution • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Which describes a histogram that is highest in the middle and falls off evenly on both sides?

A. skewed right

B. uniform

C. bimodal

D. symmetric and mound-shaped

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q1/5

2284

QUESTION ID

S61-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. symmetric and mound-shaped

GRADING NOTE Bell-shaped.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B2-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a symmetric distribution, how do the mean and the median compare?

A. The median is much larger than the mean.

B. The mean is much larger than the median.

C. They are about equal, both at the center.

D. They are at opposite ends of the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q2/5

2285

QUESTION ID

S61-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. They are about equal, both at the center.

GRADING NOTE Symmetry puts both measures at the center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B2-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q2/5

True or false?

A normal curve is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q3/5

2286

QUESTION ID

S61-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Bell shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A bell-shaped histogram is called

A. skewed right

B. uniform

C. mound-shaped and symmetric

D. bimodal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q4/5

2287

QUESTION ID

S61-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. mound-shaped and symmetric

GRADING NOTE Shape vocabulary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A continuous distribution is symmetric about its median. What percent of the distribution lies below the median? Give a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q5/5

2288

QUESTION ID

S61-B2-Q5

TYPE

Short answer

CORRECT ANSWER

50%

GRADING NOTE The median splits the area in half.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-B2-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 61 • BACKWARD 2 • Q5/5

SET 61 • HSS-ID.A.4

BACKWARD 3

Percents of a whole • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Find 95% of 2,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q1/5

2290

QUESTION ID

S61-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1900

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 34% + 13.5%?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q2/5

2291

QUESTION ID

S61-B3-Q2

TYPE

Short answer

CORRECT ANSWER

47.5%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Half of 68% is 36%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q3/5

2292

QUESTION ID

S61-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 68/2 = 34.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

100% − 95% split equally into two tails gives each tail

A. 5%

B. 47.5%

C. 0.5%

D. 2.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q4/5

2293

QUESTION ID

S61-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 2.5%

GRADING NOTE 5/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 0.7745 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q5/5

2294

QUESTION ID

S61-B3-Q5

TYPE

Short answer

CORRECT ANSWER

77.45%

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 61 • BACKWARD 3 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 1

Find values from percentiles of a normal model • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Women's heights are approximately normal with mean 64 inches and standard deviation 2.5 inches. Above what height are the tallest 2.5% of women? Give the height in inches. Use the 68-95-99.7 rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

2296

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

69

GRADING NOTE 2.5% lies above mean + 2 SD = 64 + 5 = 69.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Battery life is approximately normal with mean 40 hours and standard deviation 4 hours. Use the table to find the battery life at the 84th percentile, in hours.

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

2297

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

44

GRADING NOTE The table gives area 0.8413 at z = 1, so 40 + 4 = 44.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q2**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q2/5

True or false?

In a normal model, the median equals the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

2298

QUESTION ID

S61-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Symmetry.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

The 16th percentile of scores with mean 500 and SD 100 is

A. 484

B. 600

C. 340

D. 400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

2299

QUESTION ID

S61-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 400

GRADING NOTE One SD below.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A cereal box below 15.7 oz is rejected. Use the table to estimate the percent rejected (mean 16.0, SD 0.2).

z	−1.5	−1	−0.5	0	0.5	1	1.5	2
Area to the left	0.0668	0.1587	0.3085	0.5000	0.6915	0.8413	0.9332	0.9772

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

2300

QUESTION ID

S61-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6.68%

GRADING NOTE Any equivalent form. z = −1.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F1-Q5**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 61 • FORWARD 1 • Q5/5

SET 61 • HSS-ID.A.4

FORWARD 2

Use a normal model for a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Simulated sample proportions are approximately normal with mean 0.58 and standard deviation 0.049. Using the 68-95-99.7 rule, within what interval do about 95% of them fall?

Lower end: _____

Upper end: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q1/5

2302

QUESTION ID

S61-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.482; 0.678

GRADING NOTE Form required: both. 0.58 ± 2(0.049).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S61-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Simulated sample proportions from random samples of 100 have standard deviation 0.049. What margin of error does this give for the population proportion? Give a decimal. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q2/5

2303

QUESTION ID
S61-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.098

GRADING NOTE $2 \times 0.049 = 0.098$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q2/5

True or false?

When a simulated statistic is approximately normal and the margin of error is defined as the half-width of a 95% range, the margin of error is about two standard deviations of the simulated statistic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q3/5

2304

QUESTION ID

S61-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE About 95% of a normal distribution is within 2 SD of the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q3/5

Simulated sample means are approximately normal with standard deviation 0.2 hours. Using a margin of error of 2 standard deviations (the 95% convention), the margin of error is

A. 0.1 hours

B. 0.4 hours

C. 2 hours

D. 0.2 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q4/5

2305

QUESTION ID

S61-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.4 hours

GRADING NOTE $2 \times 0.2 = 0.4$. The others use 1 SD, half an SD, or 10 SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can the 68-95-99.7 rule be used to find a 95% range for simulated sample proportions from large random samples?

A. The population values must be normal, even though the sample proportions themselves are not normal.

B. For large enough samples their distribution is approximately normal, so about 95% fall within 2 SD of the mean.

C. Sample proportions are exactly normal for every sample size, so the rule needs no conditions at all.

D. The 68-95-99.7 rule applies to every histogram, whatever its shape, so no check is needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q5/5

2306

QUESTION ID

S61-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. For large enough samples their distribution is approximately normal, so about 95% fall within 2 SD of the mean.

GRADING NOTE The normal model is an approximation that works when the simulated distribution is mound-shaped and symmetric. The others require the population to be normal, claim exact normality for every sample size, or apply the rule to any shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 61 • FORWARD 2 • Q5/5

SET 62 • HSS-ID.A.4 • STATISTICS, INFERENCE, AND DECISIONS

Recognize data sets for which a normal model is not appropriate

MAIN	Algebra II main question	**HSS-ID.A.4**
BACKWARD 1	Describe skewed, symmetric, bimodal, and uniform shapes	**HSS-ID.A.3**
BACKWARD 2	Measures of center and spread for a data set	**6.SP.A.2**
BACKWARD 3	Groups combined in one data set	**HSS-ID.A.1**
FORWARD 1	Check a normal model with the 68-95-99.7 rule	**HSS-ID.A.4**
FORWARD 2	Decide whether a probability model fits observed data	**HSS-IC.A.2**

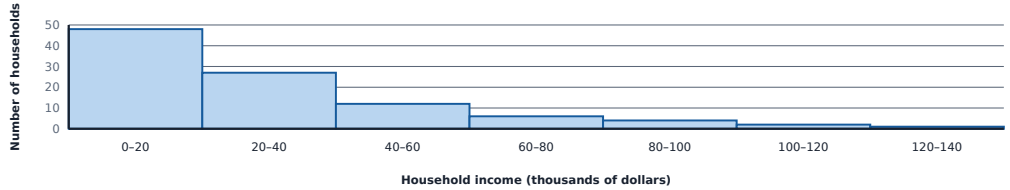
SET 62 • HSS-ID.A.4

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows household incomes in a town. Is a normal model appropriate?



A. No: the distribution is strongly skewed right, not symmetric and mound-shaped.

B. No: it has two separate peaks, so it is bimodal rather than normal.

C. Yes: incomes are measured in dollars, and measurements are always normal.

D. Yes: it has a single peak, and any single-peaked distribution is normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q1/5

2309

QUESTION ID

S62-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. No: the distribution is strongly skewed right, not symmetric and mound-shaped.

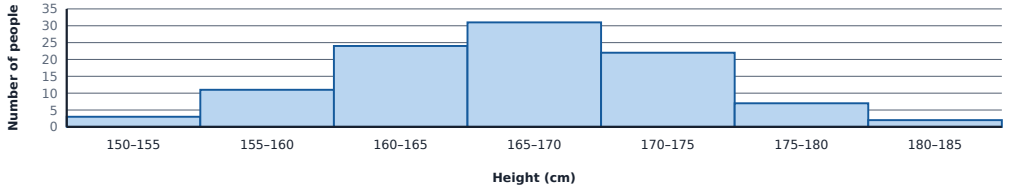
GRADING NOTE The long right tail rules out a normal model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows heights of adults in a club. Is a normal model appropriate?



A. No: it has two separate peaks, one for each group in the club.

B. No: it is strongly skewed right with a long tail to the right.

C. Yes: it is roughly symmetric and mound-shaped with a single peak.

D. No: the bars are not exactly equal on both sides of the peak.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q2/5

2310

QUESTION ID

S62-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Yes: it is roughly symmetric and mound-shaped with a single peak.

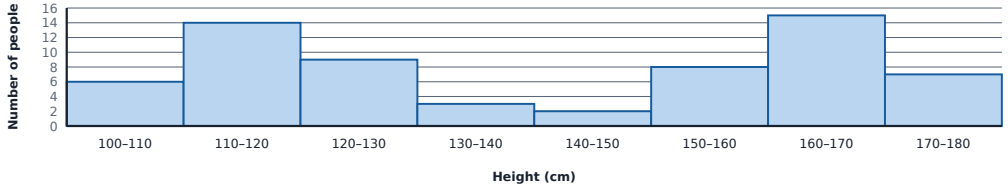
GRADING NOTE Approximate symmetry and one peak support a normal model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows heights of all people at a school event (children and adults). Is a normal model appropriate?



A. No: it is uniform, with all of the bars the same height.

B. No: it has two peaks, from two different groups combined.

C. Yes: heights of people are always approximately normal.

D. Yes: it is roughly symmetric about its middle, so it is normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q3/5

2311

QUESTION ID

S62-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. No: it has two peaks, from two different groups combined.

GRADING NOTE A bimodal shape is not normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Waiting times at a clinic have mean 4 minutes, standard deviation 5 minutes, and minimum 0. Is a normal model appropriate?

A. No: a normal model requires the standard deviation to be smaller than 1 for any data set.

B. Yes: a normal model can describe any data set once its mean and standard deviation are known.

C. Yes: the mean and the standard deviation are both positive, so a normal model fits the times.

D. No: a normal model would put about 16% of times below $4 - 5 = -1$ minute, which is impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q4/5

2312

QUESTION ID

S62-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. No: a normal model would put about 16% of times below $4 - 5 = -1$ minute, which is impossible.

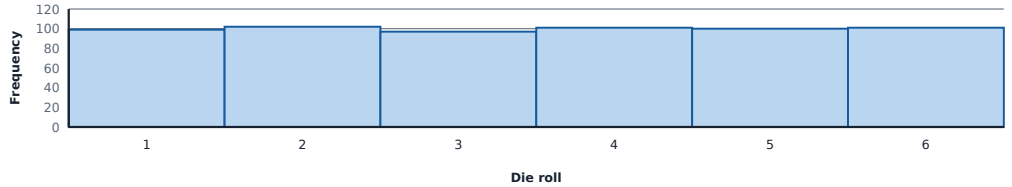
GRADING NOTE About 16% of a normal model lies below mean − 1 SD. The data must be skewed right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-M-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

The histogram shows the results of 600 rolls of a fair die. Is a normal model appropriate?



A. Yes: it is symmetric about 3.5, so it is normal.

B. No: it is strongly skewed left with a long tail.

C. No: it is roughly uniform (flat), not mound-shaped.

D. Yes: results of dice are always approximately normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q5/5

2313

QUESTION ID

S62-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. No: it is roughly uniform (flat), not mound-shaped.

GRADING NOTE Symmetric is not enough; the shape is flat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-M-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • MAIN • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 1

Describe skewed, symmetric, bimodal, and uniform shapes • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A histogram has a long tail to the right. Which describes its shape?

A. skewed right

B. symmetric

C. uniform

D. skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q1/5

2315

QUESTION ID

S62-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. skewed right

GRADING NOTE The tail names the skew.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B1-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q1/5

A histogram has two separate peaks. Which describes its shape?

A. symmetric with one peak

B. uniform

C. skewed left

D. bimodal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q2/5

2316

QUESTION ID

S62-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. bimodal

GRADING NOTE Two modes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q2/5

True or false?

In a right-skewed distribution, the mean is usually less than the median.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q3/5

2317

QUESTION ID

S62-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The long right tail pulls the mean above the median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A histogram with bars all about the same height is

A. skewed left

B. uniform

C. bimodal

D. normal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q4/5

2318

QUESTION ID

S62-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. uniform

GRADING NOTE Flat.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Which measure of center is more resistant to a long tail?

A. the maximum

B. the median

C. the mean

D. both equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q5/5

2319

QUESTION ID

S62-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. the median

GRADING NOTE The median depends only on the middle values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 62 • BACKWARD 1 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 2

Measures of center and spread for a data set • Grade 6 • Statistics & Probability • 6.SP.A.2

What does the standard deviation of a data set describe?

A. how spread out the values are around the mean

B. the middle value when the data are in order

C. the number of values in the data set

D. the most common value in the data set

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q1/5

2321

QUESTION ID

S62-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. how spread out the values are around the mean

GRADING NOTE Spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A data set has all values at least 0 and mean 4. Some value in the data set can be more than 4 below the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q2/5

2322

QUESTION ID

S62-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A value more than 4 below 4 would be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A distribution can be described by its center, spread, and shape.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q3/5

2323

QUESTION ID

S62-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Three features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a measure of center?

A. median

B. range

C. standard deviation

D. IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q4/5

2324

QUESTION ID

S62-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. median

GRADING NOTE Center.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Give one value two standard deviations below a mean of 4 with SD 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q5/5

2325

QUESTION ID

S62-B2-Q5

TYPE

Short answer

CORRECT ANSWER

−6

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • HSS-ID.A.4

BACKWARD 3

Groups combined in one data set • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Two groups with very different centers are combined into one data set. What will the histogram of the combined data most likely show?

A. a flat shape

B. a long right tail only

C. two peaks

D. one peak exactly in the middle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q1/5

2327

QUESTION ID

S62-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. two peaks

GRADING NOTE Each group forms its own peak.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B3-Q1**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q1/5

Which data set is most likely to have a histogram with two peaks?

A. heights of the parents only

B. heights of 5-year-olds and their parents together

C. heights of the 5-year-olds only

D. ages of students in one 10th-grade class

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q2/5

2328

QUESTION ID

S62-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. heights of 5-year-olds and their parents together

GRADING NOTE Two groups with different centers are combined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B3-Q2**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q2/5

True or false?

A histogram of one group can look different from a histogram of two groups combined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q3/5

2329

QUESTION ID
S62-B3-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Mixtures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A bimodal histogram suggests

A. an error in counting

B. a small standard deviation

C. two groups in the data

D. a normal model

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q4/5

2330

QUESTION ID

S62-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. two groups in the data

GRADING NOTE Two populations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q4/5

Heights of all people at a school event (young children and adults) form a two-peaked histogram. Which change is most likely to give mound-shaped distributions?

A. Add more children to the combined data set.

B. Round every height in the combined data to the nearest centimeter.

C. Remove the tallest person from the combined data.

D. Make separate histograms for the children and for the adults.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q5/5

2331

QUESTION ID

S62-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Make separate histograms for the children and for the adults.

GRADING NOTE Separating the groups removes the second peak.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-B3-Q5**

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1 • Set 62 • BACKWARD 3 • Q5/5

SET 62 • HSS-ID.A.4

FORWARD 1

Check a normal model with the 68-95-99.7 rule • High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

ANSWER KEY Reduced copy of the question page, with its answer

In 1,000 measurements, 70% are within 1 SD of the mean and 96% are within 2 SDs. Is a normal model reasonable?

A. No: the percents must be exactly 68% and 95%.

B. Yes: any data set with 1,000 values is normal.

C. Yes: 70% and 96% are close to the normal 68% and 95%.

D. No: 96% is more than 95%, so the data cannot be normal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q1

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q1/5

2333

QUESTION ID

S62-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Yes: 70% and 96% are close to the normal 68% and 95%.

GRADING NOTE Approximate agreement with 68-95 supports the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q1**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A data set has 85% of its values within 1 SD of the mean and 91% within 2 SDs. Is a normal model reasonable?

A. Yes: 91% is close to 95%, so 85% does not matter.

B. Yes: both percents are above 50%.

C. No: a normal model needs 100% within 2 SDs.

D. No: 85% is far above 68% and 91% is below 95%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q2

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q2/5

2334

QUESTION ID

S62-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. No: 85% is far above 68% and 91% is below 95%.

GRADING NOTE Both percents disagree with the normal model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q2**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q2/5

True or false?

If a data set has an extreme outlier, the normal model may still fit the rest but the mean and SD are distorted.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q3/5

2335

QUESTION ID

S62-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outliers affect mean and SD.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q3

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

About what percent of normal data lies more than 3 SDs from the mean?

A. 5%

B. 3%

C. 0.3%

D. 32%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q4

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q4/5

2336

QUESTION ID

S62-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.3%

GRADING NOTE 100 − 99.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q4**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A data set of 200 values has 25 values more than 2 SDs from the mean. What percent of the values is that, and about what percent does the 68-95-99.7 rule predict for a normal model?

Observed percent: _____

Normal model percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F1-Q5

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q5/5

2337

QUESTION ID

S62-F1-Q5

TYPE

Short answer

CORRECT ANSWER

12.5%; 5%

GRADING NOTE Form required: both. 25/200 = 12.5%, far more than 5%, so a normal model does not fit.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S62-F1-Q5**

High School — Algebra II • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 62 • FORWARD 1 • Q5/5

SET 62 • HSS-ID.A.4

FORWARD 2

Decide whether a probability model fits observed data • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A spinner is claimed to land on each of 4 colors equally often. In 400 spins the counts are 98, 105, 97, 100. Are the results consistent with the claim?

A. Yes: each count is close to the expected 100.

B. No: the counts are not all exactly 100.

C. No: 105 proves the spinner favors one color.

D. Yes: any 4 counts adding to 400 are consistent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q1/5

2339

QUESTION ID

S62-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Yes: each count is close to the expected 100.

GRADING NOTE Small chance variation around 100 is expected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q1/5

A model says a coin is fair. In 50 flips there were 34 heads. In 200 simulated runs of 50 fair flips, 3 runs had 34 or more heads. Use 5% as the cutoff for unusual. What do you conclude?

A. 1.5% is not unusual, so the data show that the fair-coin model fits.

B. 1.5% < 5%: unusual for a fair coin, so evidence against the model.

C. No simulated run had exactly 34 heads, so a fair coin has probability 0 of giving 34.

D. 1.5% < 5%: this proves the coin is unfair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q2/5

2340

QUESTION ID

S62-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 1.5% < 5%: unusual for a fair coin, so evidence against the model.

GRADING NOTE 3/200 = 1.5%. Evidence is not proof, and a rare result is not impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q2/5

True or false?

A model is rejected only if the observed result would be unusual under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q3/5

2341

QUESTION ID

S62-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Logic of the test.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q3/5

Under a model, a simulated result at least as extreme as the observed one occurs in 34 of 200 runs. Use 5% as the cutoff for unusual. The model is

A. proved true

B. rejected

C. consistent with the data

D. impossible

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q4/5

2342

QUESTION ID

S62-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. consistent with the data

GRADING NOTE 17% is not unusual; consistency is not proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q4/5

How does judging a normal model for a data set differ from judging a probability model with a simulation?

A. Both check whether the data form a bell curve; the simulation just repeats the same comparison many more times.

B. Both compare the sample mean with the population mean and ignore the shape of the data and the simulation.

C. The first uses many simulated runs of the model; the second only compares the shape of the histogram with a bell curve.

D. The first compares the shape of the data with a normal curve; the second asks whether an observed result is unusual among simulated results.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q5/5

2343

QUESTION ID

S62-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The first compares the shape of the data with a normal curve; the second asks whether an observed result is unusual among simulated results.

GRADING NOTE Shape check versus how-unusual check. The others swap the two methods, describe both as the same shape check, or describe a comparison of means that neither method makes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S62-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 62 • FORWARD 2 • Q5/5

SET 63 • HSS-IC.A.1 • STATISTICS, INFERENCE, AND DECISIONS

Distinguish population, sample, parameter, and statistic

MAIN	Algebra II main question	**HSS-IC.A.1**
BACKWARD 1	Samples and populations	**7.SP.A.1**
BACKWARD 2	Draw inferences from random samples	**7.SP.A.2**
BACKWARD 3	Compute sample proportions and means	**6.SP.B.5.c**
FORWARD 1	Describe sampling variability	**HSS-IC.A.1**
FORWARD 2	Quantify variability with a margin of error	**HSS-IC.B.4**

SET 63 • HSS-IC.A.1

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

A city surveys 400 randomly chosen residents, and 62% of them support a new park. Which identifies the population, sample, parameter, and statistic correctly?

A. Population: all residents; sample: the 62% who support it; parameter: 400; statistic: 62%.

B. Population: all residents; sample: the 400 surveyed; parameter: percent of all residents in support; statistic: 62%.

C. Population: the 400 surveyed; sample: all residents; parameter: 62%; statistic: percent of all residents in support.

D. Population: all residents; sample: the 400 surveyed; parameter: 62%; statistic: percent of all residents in support.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q1/5

2346

QUESTION ID

S63-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Population: all residents; sample: the 400 surveyed; parameter: percent of all residents in support; statistic: 62%.

GRADING NOTE A parameter describes the population; a statistic describes the sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q1/5

A factory measures 50 randomly selected bolts from a day's production; their mean length is 5.02 cm. Which identifies the parameter and the statistic?

A. Parameter: 5.02 cm, the mean of the 50 bolts; statistic: the mean length of all bolts made that day.

B. Parameter: the number of bolts measured, 50; statistic: the mean length of those bolts, 5.02 cm.

C. Parameter: the mean length of all bolts made that day; statistic: 5.02 cm, the mean of the 50 bolts.

D. Parameter: the 50 bolts that were measured; statistic: all of the bolts that were made that day.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q2/5

2347

QUESTION ID

S63-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Parameter: the mean length of all bolts made that day; statistic: 5.02 cm, the mean of the 50 bolts.

GRADING NOTE Parameter = population value; statistic = sample value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q2/5

A news website posts a poll; 80% of the 2,000 visitors who answered want a later school start. Can this result be generalized to all students?

A. No: it is a voluntary-response sample, so people with strong opinions are over-represented.

B. Yes: 80% is so far above 50% that any bias could not change the overall conclusion.

C. No: a poll can never be used to estimate a population percent, even with random sampling.

D. Yes: 2,000 responses form a large sample, so the result must be accurate for all students.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q3/5

2348

QUESTION ID
S63-M-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. No: it is a voluntary-response sample, so people with strong opinions are over-represented.

GRADING NOTE Voluntary response is biased regardless of size.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q3/5

Why does a random sample allow conclusions about the whole population?

A. A random sample always includes every member of the population, so nothing is left out of it.

B. Chance decides who is chosen, so the sample tends to represent the population without systematic bias.

C. A random sample is always larger than a voluntary sample, so its result is always more exact.

D. A random sample always has exactly the same statistic as the population parameter it estimates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q4/5

2349

QUESTION ID

S63-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Chance decides who is chosen, so the sample tends to represent the population without systematic bias.

GRADING NOTE Random selection removes selection bias; chance variation remains. The others are false: a sample does not include everyone, is not always larger, and does not match the parameter exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A school uses the first 30 students arriving in the morning to estimate the students' average hours of sleep. Which statement is correct?

A. It is a random sample, because the times the students arrive vary by chance.

B. It is not random, but 30 students is enough to remove any bias from the estimate.

C. It is a random sample, because no one chose the students by name or by any other rule.

D. It is not a random sample: early arrivers may sleep differently, so the estimate may be biased.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q5/5

2350

QUESTION ID
S63-M-Q5

TYPE
Multiple choice

CORRECT ANSWER
D. It is not a random sample: early arrivers may sleep differently, so the estimate may be biased.

GRADING NOTE Convenience sample; early arrivers may sleep less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • MAIN • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 1

Samples and populations • Grade 7 • Statistics & Probability • 7.SP.A.1

ANSWER KEY Reduced copy of the question page, with its answer

In statistics, what is a population?

A. the number of people in a city

B. the part of the group that is measured

C. the entire group a study wants to describe

D. a number computed from a sample

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

2352

QUESTION ID

S63-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. the entire group a study wants to describe

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In statistics, what is a sample?

A. the part of the population that is studied

B. the entire group of interest

C. a number describing the whole population

D. a list of every possible outcome

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

2353

GRADING NOTE Definition.

QUESTION ID

S63-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. the part of the population that is studied

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A random sample gives every member of the population a chance to be chosen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

2354

QUESTION ID

S63-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Surveying every student in a school is

A. a statistic

B. an experiment

C. a sample

D. a census

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

2355

QUESTION ID

S63-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. a census

GRADING NOTE Whole population.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why do studies often use a sample instead of the whole population?

A. Samples have no chance variation at all.

B. Samples always give the exact population value.

C. A population cannot be described by a number.

D. Studying everyone may be too costly, slow, or impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

2356

QUESTION ID

S63-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Studying everyone may be too costly, slow, or impossible.

GRADING NOTE Practical limits. The others are false: samples do vary by chance, do not give the exact population value, and populations are described by numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 63 • BACKWARD 1 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 2

Draw inferences from random samples • Grade 7 • Statistics & Probability • 7.SP.A.2

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 50 students, 20 walk to school. Estimate how many of 600 students walk.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q1/5

2358

QUESTION ID

S63-B2-Q1

TYPE

Short answer

CORRECT ANSWER

240

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two random samples of 500 voters from the same city give 40% and 44% support for a plan. What is the best explanation for the difference?

A. The population percent changed between the two samples.

B. Random samples vary by chance, so a 4-point difference is not surprising.

C. One of the two samples must have been chosen incorrectly.

D. Random samples from one population always give the same percent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q2/5

2359

QUESTION ID

S63-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Random samples vary by chance, so a 4-point difference is not surprising.

GRADING NOTE For n = 500 the sample percent varies by about 2 points (1 SD), so 4 points apart is common.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q2/5

True or false?

Different random samples from the same population usually give slightly different results.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q3/5

2360

QUESTION ID

S63-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

From a random sample, 30% of 200 voters prefer a candidate. The best estimate for all voters is

A. 30%

B. 200

C. cannot estimate

D. 60%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q4/5

2361

QUESTION ID

S63-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 30%

GRADING NOTE Statistic estimates parameter.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q4/5

Why is a larger random sample usually better for estimating a population value?

A. Its statistic varies less from sample to sample, so the estimate is more precise.

B. It always gives exactly the population value, with no chance variation.

C. It makes the population value larger, so the estimate is closer to it.

D. It removes all bias, even when the sample is not chosen at random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q5/5

2362

QUESTION ID

S63-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Its statistic varies less from sample to sample, so the estimate is more precise.

GRADING NOTE Less sampling variability. The others are false: no sample gives the exact value, size does not change the population, and size does not remove bias.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • HSS-IC.A.1

BACKWARD 3

Compute sample proportions and means • Grade 6 • Statistics & Probability • 6.SP.B.5.c

ANSWER KEY Reduced copy of the question page, with its answer

248 of 400 people say yes. What proportion is that?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q1/5

2364

QUESTION ID

S63-B3-Q1

TYPE

Short answer

CORRECT ANSWER

0.62

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the mean of 5.0, 5.1, 4.9, 5.08.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q2/5

2365

QUESTION ID

S63-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5.02

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A proportion can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q3/5

2366

QUESTION ID

S63-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A part of a whole is between 0 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

1,600 of 2,000 is

A. 16%

B. 1.25%

C. 80%

D. 20%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q4/5

2367

QUESTION ID

S63-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 80%

GRADING NOTE 1600/2000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Write 62% as a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q5/5

2368

QUESTION ID

S63-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.62

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-B3-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 63 • BACKWARD 3 • Q5/5

SET 63 • HSS-IC.A.1

FORWARD 1

Describe sampling variability • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Two random samples of 400 residents of the same city give 62% and 59% support for a park. Which statement is correct?

A. One sample must be biased, since random samples agree exactly.

B. Each sample estimates a different population percent.

C. The samples differ by chance; both estimate the same population percent.

D. The population percent is exactly their average, 60.5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q1/5

2370

QUESTION ID

S63-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. The samples differ by chance; both estimate the same population percent.

GRADING NOTE Sampling variability.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q1/5

2370

Would random samples of 40 or random samples of 400 from the same population vary more from sample to sample?

A. Samples of 40: small samples are always biased.

B. Samples of 400: more people means more variation.

C. Samples of 40: smaller samples are affected more by chance.

D. They vary the same amount, since both are random.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q2/5

2371

QUESTION ID

S63-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Samples of 40: smaller samples are affected more by chance.

GRADING NOTE Variability decreases as n increases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q2/5

True or false?

A parameter changes from sample to sample.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q3/5

2372

QUESTION ID

S63-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The parameter is fixed; the statistic varies.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is a statistic?

A. the population size

B. the mean height of 30 sampled students

C. the mean height of all students

D. the population proportion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q4/5

2373

QUESTION ID

S63-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. the mean height of 30 sampled students

GRADING NOTE From a sample.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

Why can a large voluntary-response sample be worse than a small random sample?

A. Voluntary samples are random, but a large one has too many answers to analyze well.

B. Large samples always have a larger margin of error than small random samples.

C. A large size does not remove bias; volunteers are systematically unrepresentative.

D. Small samples have no chance variation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q5/5

2374

QUESTION ID

S63-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. A large size does not remove bias; volunteers are systematically unrepresentative.

GRADING NOTE Bias is not reduced by size. The others are false: voluntary samples are not random, larger samples have smaller margins, and small samples do vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F1-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 63 • FORWARD 1 • Q5/5

2374

SET 63 • HSS-IC.A.1

FORWARD 2

Quantify variability with a margin of error • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A city surveys a random sample of 400 residents, and 62% support a new park. Simulated sample proportions (samples of 400 with p = 0.62) have standard deviation 0.024. What is the margin of error? Give a decimal. Use the convention that the margin of error is 2 standard deviations of the simulated statistic (a 95% range).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q1/5

2376

QUESTION ID

S63-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.048

GRADING NOTE 2 × 0.024 = 0.048, or 4.8 percentage points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S63-F2-Q1**

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a random sample of 400 residents, 62% support a new park, and the margin of error is 4.8 percentage points. Give the interval estimate for the percent of all residents who support the park.

Lower end (%): _____

Upper end (%): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q2/5

2377

QUESTION ID

S63-F2-Q2

TYPE

Short answer

CORRECT ANSWER

57.2; 66.8

GRADING NOTE Form required: both. 62 ± 4.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q2/5

True or false?

A margin of error describes chance variation, not bias.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q3/5

2378

QUESTION ID
S63-F2-Q3

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Bias is not measured.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q3/5

If the sample size is multiplied by 4, the margin of error is about

A. unchanged

B. halved

C. doubled

D. divided by 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q4/5

2379

QUESTION ID

S63-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. halved

GRADING NOTE Square-root law.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A random sample of 400 residents gives 62% support for a park, with the 95% interval 57.2% to 66.8%. What does the interval say?

A. Exactly 62% of all of the residents of the city support the park.

B. The percent of all residents who support the park is plausibly between 57.2% and 66.8%.

C. 95% of all of the residents of the city support the park.

D. Between 57.2% and 66.8% of the 400 surveyed residents support the park.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q5/5

2380

QUESTION ID
S63-F2-Q5

TYPE
Multiple choice

CORRECT ANSWER
B. The percent of all residents who support the park is plausibly between 57.2% and 66.8%.

GRADING NOTE The interval estimates the population percent; the sample percent is exactly 62%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S63-F2-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 63 • FORWARD 2 • Q5/5

2380

SET 64 • HSS-IC.A.2 • STATISTICS, INFERENCE, AND DECISIONS

Decide whether a model is consistent with observed results using a simulation

MAIN	Algebra II main question	**HSS-IC.A.2**
BACKWARD 1	Probability as long-run relative frequency	**7.SP.C.6**
BACKWARD 2	Use simulations of compound events	**7.SP.C.8.c**
BACKWARD 3	Relative frequency of a group of outcomes	**6.RP.A.3.c**
FORWARD 1	Design a simulation to test a model	**HSS-IC.A.2**
FORWARD 2	Compare two treatments with a re-randomization simulation	**HSS-IC.B.5**

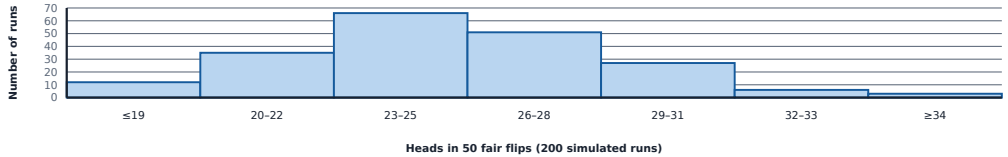
SET 64 • HSS-IC.A.2

MAIN

Algebra II main question • High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A coin is claimed to be fair. It landed heads 34 times in 50 flips. The histogram shows 200 simulated runs of 50 flips of a fair coin. Count runs with 34 or more heads. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. 3 of 200 runs (1.5%) had 34 or more heads; 1.5% < 5%, so the data are evidence against a fair coin.

B. 3 of 200 runs (1.5%) had 34 or more heads, so the coin is proved to be unfair.

C. A fair coin can never land heads 34 times in 50 flips, so the claim is false.

D. 36 of 200 runs (18%) had 29 or more heads, so 34 heads is not unusual for a fair coin.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q1/5

2383

QUESTION ID

S64-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

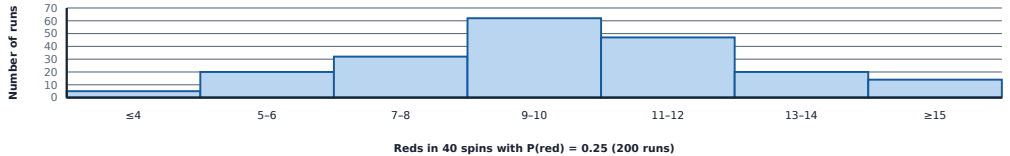
A. 3 of 200 runs (1.5%) had 34 or more heads; 1.5% < 5%, so the data are evidence against a fair coin.

GRADING NOTE Only the ≥34 bar counts: 3/200. Evidence is not proof, and the event is rare, not impossible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q1

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q1/5

A spinner is claimed to land on red 25% of the time. It landed on red 13 times in 40 spins. The histogram shows 200 simulated runs of 40 spins under the claim. Count runs with 13 or more reds. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. The expected count is 10, and 13 is not 10, so the claim is false.

B. 34 of 200 runs (17%) had 13 or more reds, so the claim is proved true.

C. Only runs with exactly 13 reds count, which is rare, so the claim is rejected.

D. 34 of 200 runs (17%) had 13 or more reds; 17% > 5%, so the result is consistent with the claim.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q2/5

2384

QUESTION ID

S64-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 34 of 200 runs (17%) had 13 or more reds; 17% > 5%, so the result is consistent with the claim.

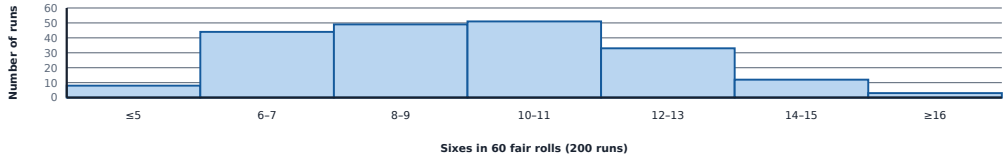
GRADING NOTE 20 + 14 = 34 runs. Consistent with the model does not prove it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-M-Q2

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A die is claimed to be fair. It showed 16 sixes in 60 rolls. The histogram shows 200 simulated runs of 60 rolls of a fair die. Count runs with 16 or more sixes. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. 15 of 200 runs (7.5%) had 14 or more sixes, so 16 sixes is not unusual.

B. 3 of 200 runs (1.5%) had 16 or more sixes; 1.5% < 5%, so the data are evidence against a fair die.

C. 3 of 200 runs (1.5%) had 16 or more sixes, so the die is proved to be unfair.

D. A fair die can never show 16 sixes in 60 rolls, so the claim is false.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q3/5

2385

QUESTION ID

S64-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

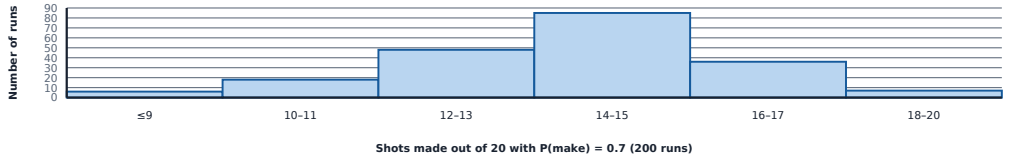
B. 3 of 200 runs (1.5%) had 16 or more sixes; 1.5% < 5%, so the data are evidence against a fair die.

GRADING NOTE Only the ≥16 bar counts: 3/200.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-M-Q3

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q3/5

A player claims to make 70% of her free throws. She made 11 of 20. The histogram shows 200 simulated runs of 20 shots under the claim. Count runs with 11 or fewer makes. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. She made 11, not the expected 14, so the claim is false.

B. 24 of 200 runs (12%) had 11 or fewer makes, so the 70% claim is proved true.

C. 24 of 200 runs (12%) had 11 or fewer makes; 12% > 5%, so the result is consistent with the claim.

D. 6 of 200 runs (3%) had 9 or fewer makes; 3% < 5%, so 11 makes is unusual.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q4/5

2386

QUESTION ID

S64-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 24 of 200 runs (12%) had 11 or fewer makes; 12% > 5%, so the result is consistent with the claim.

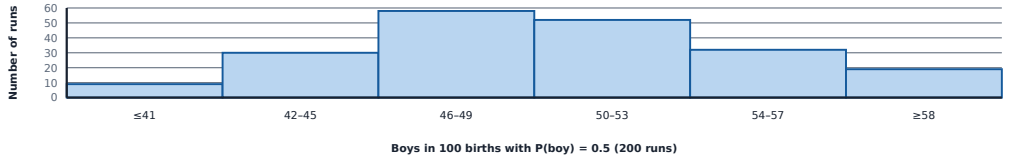
GRADING NOTE 6 + 18 = 24 runs. Not unusual is not proof.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S64-M-Q4

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A model says boys and girls are equally likely. A hospital had 58 boys in 100 births. The histogram shows 200 simulated runs of 100 births under the model. Count runs with 58 or more boys. Use 5% as the cutoff for unusual. Which conclusion is justified?



A. 19 of 200 runs (9.5%) had 58 or more boys, so the model is proved true.

B. 19 runs had 58 or more boys, and 19 is more than 5, so the result is unusual.

C. There were 58 boys, not the expected 50, so the model is false.

D. 19 of 200 runs (9.5%) had 58 or more boys; 9.5% > 5%, so the result is consistent with the model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q5/5

2387

QUESTION ID

S64-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 19 of 200 runs (9.5%) had 58 or more boys; 9.5% > 5%, so the result is consistent with the model.

GRADING NOTE 19/200 = 9.5%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-M-Q5

High School — Algebra II • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 64 • MAIN • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 1

Probability as long-run relative frequency • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped 50 times. About how many heads do you expect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q1/5

2389

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner has P(red) = 0.25. About how many reds in 40 spins?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q2/5

2390

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Getting exactly the expected number every time is likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q3/5

2391

QUESTION ID

S64-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Results vary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q3/5

A fair die is rolled 60 times. Expected sixes:

A. 10

B. 16

C. 6

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q4/5

2392

QUESTION ID

S64-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 10

GRADING NOTE 60/6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

A fair coin is flipped 50 times and lands heads 26 times. Why is this not surprising?

A. 26 is close to the expected 25, and results vary by chance.

B. Every number of heads from 0 to 50 is equally likely.

C. A fair coin always gives exactly 26 heads in 50 flips.

D. 26 is more than half, which proves the coin is fair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q5/5

2393

QUESTION ID

S64-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 26 is close to the expected 25, and results vary by chance.

GRADING NOTE Chance variation around 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B1-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 2

Use simulations of compound events • Grade 7 • Statistics & Probability • 7.SP.C.8.c

ANSWER KEY Reduced copy of the question page, with its answer

Which design simulates 50 flips of a fair coin with random digits 0-9?

A. Use 50 random digits; only the digit 5 = heads.

B. Use 50 random digits; even = heads, odd = tails.

C. Use 50 random digits; 0 = heads, 1-9 = tails.

D. Use 1 random digit and multiply it by 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q1/5

2395

QUESTION ID

S64-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Use 50 random digits; even = heads, odd = tails.

GRADING NOTE Half of the digits must represent heads.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which design simulates one spin of a spinner with P(red) = 0.25 using two-digit random numbers 00-99?

A. 00-25 = red; 26-99 = not red

B. 25 = red; all other numbers = not red

C. 00-49 = red; 50-99 = not red

D. 00-24 = red; 25-99 = not red

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q2/5

2396

QUESTION ID

S64-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 00-24 = red; 25-99 = not red

GRADING NOTE 25 of the 100 numbers must mean red; 00-25 has 26.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q2/5

True or false?

Each run of a simulation should copy the real process under the model.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q3/5

2397

QUESTION ID

S64-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q3/5

A simulation has 200 runs. Each run represents

A. the final answer

B. one repetition of the whole experiment

C. one histogram bar

D. one flip

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q4/5

2398

QUESTION ID

S64-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. one repetition of the whole experiment

GRADING NOTE Run = experiment.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q4/5

Why does a simulation use many runs instead of just one?

A. Because a single run always gives the wrong answer.

B. Because more runs change the model's probabilities.

C. To see the pattern of results the model produces by chance.

D. To make the model match the observed data exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q5/5

2399

QUESTION ID

S64-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. To see the pattern of results the model produces by chance.

GRADING NOTE Many runs show the distribution under the model. The others are false: one run is not wrong, more runs do not change the model, and runs are not fitted to the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S64-B2-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • HSS-IC.A.2

BACKWARD 3

Relative frequency of a group of outcomes • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c